

Drilling Supervisor Study Guide Day 1

DS1.1 **Porosity can be defined by which of the following statements?**

Answer

- B**
- (A) When KMW is at the shoe
 - (B) The amount of void space in the rock, expressed as a percent.
 - (C) The measure of ease or ability of a rock to transmit a one-phase fluid under conditions of laminar flow. The unit of porosity is the Darcy.
 - (D) Pore throat pressure

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DS1.2 **The drilling crew has successfully spaced out, shut down and shut-in the well. Which pressure do we want to keep the Bottom Hole Pressure (BHP) at least equal to?**

Answer

- B**
- (A) At least equal to the drillpipe pressure
 - (B) At least equal to the formation fluid pressure
 - (C) Initial circulating pressure
 - (D) At least equal to the casing pressure

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DS1.3 **Which of the following factors can change the value of Shut-in Casing Pressure (SICP) after we shut-in a well during a kick?**

Answer

- C**
- (A) Choke line length
 - (B) SCRCP pressure
 - (C) The formation fluid pressure (pore pressure)
 - (D) Low surface temperatures

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DS1.4 **In order to kill a well, which statement is correct?**

Answer

- D**
- (A) Keep BHP 100 psi over Formation Pressure
 - (B) Keep BHP 100 psi under formation pressure
 - (C) Keep BHP equal to Casing Shoe Pressure
 - (D) Keep BHP at least equal to formation pressure

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DS1.5 **Which of the following equipment is used to secure a well while the drilling line is cut and the drillstring is in the well?**

Answer

- B**
- (A) Inside Blowout Preventer
 - (B) Full Opening Safety Valve or Non-Return Valve
 - (C) Check valve
 - (D) Secure hanging wire rope in derrick

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DS1.6 **The operator's drilling engineer has calculated a kick tolerance of 35 bbls based on a 1 ppg kick intensity. If we recalculate for a kick intensity of 1.5 ppg, what will happen to the maximum allowable kick volume?**

Answer

- B**
- (A) Maximum allowable kick volume will increase
 - (B) Maximum allowable kick volume will decrease
 - (C) Maximum allowable kick volume would not be affected
 - (D) Both A and C

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DS1.7 **What can raise the risk of going over the Maximum Allowable Annulus Surface Pressure (MAASP) during a kill operation?**

Answer

- A**
- (A) Long open hole section
 - (B) Short open hole section
 - (C) Weak casing shoe
 - (D) No open hole section

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DS1.8 **While circulating the gas influx inside the casing, if the casing pressure gets close to the Maximum Allowable Annulus Surface Pressure (MAASP), what should we do?**

Answer

- D**
- (A) Continue with the kill procedure, but open the choke as needed to maintain casing pressure equal to MAASP, thereby risking additional influx.
 - (B) Shut-in well, use Volumetric Well Control Method after mobilizing a Well Control provider.
 - (C) Shut-in well, use the Lube and Bleed Method
 - (D) Continue with the correct kill procedure, the increasing MAASP will not cause formation breakdown.

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DS1.9 **During a well kill operation, which of the following is LEAST likely to cause us to exceed the Maximum Anticipated Annular Surface Pressure (MAASP)?**

Answer

- C**
- (A) Increase mud weight
 - (B) Small difference between formation fracture pressure and drilling fluid hydrostatic pressure.
 - (C) Short open hole section
 - (D) None of the above

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DS1.10 **At what point during a kill operation could the choke pressure exceed the Maximum Anticipated Annular Surface Pressure (MAASP) without causing the formation at the shoe to break down?**

Answer

- C**
- (A) When performing the pump start-up procedure
 - (B) When the influx is in the open hole section
 - (C) When the influx is above the casing shoe.
 - (D) When the influx is at the choke

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DS1.11 **The greatest pressure at the casing shoe occurs at what point in a well kill circulation?**

Answer

- B**
- (A) During the pump start-up
 - (B) Either at initial shut-in or when influx reaches the shoe.
 - (C) Only when the top of the influx reaches the shoe.
 - (D) When gas reaches the choke

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DS1.12 **Which operation affects the Maximum Allowable Annular Surface Pressure (MAASP) and requires that it be recalculated?**

Answer

- A** (A) After mud weight change
- (B) Every 300 to 500 feet drilled
- (C) After changing out a swab on the mud pump.
- (D) When alternate choke is used

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DS1.13 **Most problems in well control can be recognized early. For a Mud Gas Separator, what indicators would suggest that blow through was about to occur?**

Answer

- D** (A) Increasing trend in hydrostatic seal pressure
- (B) Loss of mud leg or dip tube
- (C) Reduction in flow coming from vent line
- (D) Increasing trend in vent line pressure

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DS1.14 **After noting an increase in flow, the crew spaced out, shut down and successfully shut-in the well. Which of the following statement best describes a crewmember responsibility?**

Answer

- D** (A) Communicate kill plans and procedures to crewmembers
- (B) Calculate the correct kill mud density
- (C) Choose which kill method to use
- (D) Check for leaks at the mud pumps, pipework and pit areas

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DS1.15 **Which statement best describes when a flow check is needed?**

Answer

- C** (A) Immediately after a 5% decrease in the weight on bit
- (B) Immediately after a 5% increase in the rate of penetration
- (C) Before the BHA is pulled up through the BOP stack
- (D) After the BHA is pulled up to the surface

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DS1.16 Which of the following well is NOT a wellbore barrier test?

Answer

- A** (A) Leak-off Test
- (B) Positive Test
- (C) Negative Test
- (D) Cumulative Pressure Test

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DS1.17 What is a definition of a negative barrier test?

Answer

- A** (A) A test on a barrier where pressure on the formation side of the barrier is greater than the pressure on the surface side of the barrier.
- (B) A test on a barrier where pressure on the surface side of the barrier is greater than the formation side of the barrier..
- (C) A test to see if the safety valve is holding
- (D) Opposite of a positive barrier test

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DS1.18 What is the proper definition of a "negative" pressure test?

Answer

- B** (A) A test where the barrier is made to fail.
- (B) A test on a barrier where pressure is reduced to a value lower than the other side of the barrier
- (C) A test on a barrier where pressure is increased to a value higher than the other side of the barrier.
- (D) The opposite of a positive test

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DS1.19 Which of the following statements is a proper example of a "physical barrier"?

Answer

- A** (A) A cement plug
- (B) Proper BOP shut-in procedure
- (C) Monitoring the trip tank
- (D) Monitoring DP and CP at the remote choke

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DS1.20 **During tripping operations, if an influx is swabbed into the well, what primary type of barrier failed?**

Answer

- B** (A) Permanent mechanical barrier
- (B) Operational barrier
- (C) Temporary mechanical barrier
- (D) Retrievable mechanical barrier

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DS1.21 **"Procedural barriers" can be defined by which of the following statements**

Answer

- A** (A) Monitoring well for gains and losses
- (B) Calculating KMW
- (C) Mixing mud
- (D) Procedural barriers are those barriers with written operating procedures

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DS1.22 **Procedural barrier can be defined as?**

Answer

- B** (A) Mud pumped into well
- (B) Monitoring the well for gains or losses.
- (C) Running casing in well
- (D) Tripping pipe from well

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DS1.23 **Concerning most rig operations, the API recommends at least two independent barriers to be in place throughout operations. Which pair of barriers are NOT independent of each other?**

Answer

- C** (A) Drilling fluid and an annular preventer
- (B) A tubing plug set downhole and a backpressure valve installed in the hanger
- (C) An annular and a pipe ram on a BOP stack
- (D) Drilling fluid and a pipe ram

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DS1.24 **For most operations, B.S.S.E. always recommends installation of two independent barriers. Which pair of barriers are NOT independent of each other?**

Answer

- C**
- (A) Viscosity and mud weight
 - (B) Kill mud in the well and the BOP
 - (C) Annular and a Ram on the BOP stack
 - (D) Drillpipe Safety Valve and Kill Mud

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DS1.25 **A 9.7 ppg brine has been circulated above a plug set from 7500 ft to 7850 ft across a casing shoe. The highest-pressure zone below the plug was drilled with a 10.5 ppg mud. What would happen if there is communication across the plug?**

Answer

- C**
- (A) Well will remain static.
 - (B) Well will lose circulation
 - (C) Well will flow
 - (D) The well will remain static until the brine begins to increase in temperature

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DS1.26 **If a safety margin is added to the mud weight what is the result?**

Answer

- A**
- (A) Cannot be removed easily
 - (B) Increases MAASP at the shoe
 - (C) Decreases casing shoe pressure
 - (D) Decreases MAASP at the shoe

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DS1.27 **What is the definition of a positive test?**

Answer

- C**
- (A) A test on a barrier where the pressure on the formation side of the barrier is equal to the pressure on the surface side of the barrier.
 - (B) A test on a barrier where the pressure applied to the formation side of the barrier is greater than the pressure on the surface side of the barrier.
 - (C) A test on a barrier where the pressure on the formation side of the barrier is less than the pressure on the surface side of the barrier.
 - (D) A test performed on a barrier where gas is injected downhole to test the integrity of the barrier

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DS1.28 **Which of the following statements best describes a "procedural barrier"?**

Answer

- C** (A) Writing handover notes for your relief
- (B) Setting a packer assembly
- (C) Actively monitoring the well for gains or losses
- (D) Decreasing the rate of penetration

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DS1.29 **Which type of barrier has failed when a kick is swabbed into the wellbore during tripping?**

Answer

- B** (A) Secondary Hydrostatic barrier
- (B) Primary Hydrostatic barrier
- (C) Tertiary Hydrostatic barrier
- (D) Primary Hydrostatic Pressure

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DS1.30 **Which type of barrier has failed when a kick is swabbed into the wellbore during tripping?**

Answer

- A** (A) Primary hydrostatic barrier
- (B) Temporary mechanical barrier
- (C) Permanent mechanical barrier
- (D) Secondary barrier

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DS1.31 **Which statement is NOT a wellbore barrier verification method?**

Answer

- A** (A) Leak off test
- (B) Weight set confirmation (Tag top of cement)
- (C) Inflow test (Negative test)
- (D) Measuring the mud density

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DS1.32 **Which statement best describes a positive test?**

Answer

- C**
- (A) A test performed on a barrier where the pressure applied to the formation side of the barrier is equal to the pressure on the surface side of the barrier
 - (B) A test performed on a barrier where the pressure applied to the formation side of the barrier is greater than the pressure on the surface side of the barrier
 - (C) A test performed on a barrier where the pressure on the formation side of the barrier is less than the pressure on the surface side of the barrier
 - (D) A good pressure test on surface equipment

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DS1.33 **Which statement best describes a "Barrier System" (Envelope)?**

Answer

- D**
- (A) A set of processes and procedures that prevent the well from kicking
 - (B) The BOP only
 - (C) Drilling fluid and barite
 - (D) More than one barrier that act together to prevent flow from the formation

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DS1.34 **Which statement best describes an item that is NOT classified as a barrier?**

Answer

- B**
- (A) The mud hydrostatic pressure
 - (B) A drill string float valve
 - (C) The pipe rams
 - (D) A cement plug

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DS1.35 **We are drilling a development well within an established producing field. Which statement best describes the pressure test normally used to determine formation strength at the casing shoe?**

Answer

- B**
- (A) Leak Off Test
 - (B) Formation Integrity Test
 - (C) Drill Stem Test
 - (D) BOP Test

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DS1.36 **Please select the best statement which describes what is needed to determine the casing shoe formation strength?**

Answer

- A** (A) The exact TVD of the casing shoe
- (B) Accurate hole capacity
- (C) Accurate pump stroke counters
- (D) Kill weight mud

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DS1.37 **Which of the following statements may NOT warn of an increase in formation pressure?**

Answer

- C** (A) WOB display or gauge
- (B) ROP display or gauge
- (C) Accumulator Pressure display or gauge
- (D) Torque display or gauge

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DS1.38 **Which statement describes the best definition of Maximum Allowable Annular Surface Pressure (MAASP)?**

Answer

- C** (A) The pressure which will cause losses at the bottom of the hole
- (B) The pressure which will cause flow at the casing
- (C) The pressure which will cause losses at the casing shoe minus the mud hydrostatic at the casing shoe
- (D) The pressure which will cause pit gains when bit is on bottom

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DS1.39 **Which statement defines the percentage of void space in a formation?**

Answer

- A** (A) Porosity
- (B) Permeability
- (C) Viscosity
- (D) Mud density

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DS1.40 **When ballooning occurs, Bottomhole Pressure (BHP) is slightly increased more than the formation fracture pressure. Which statement best describes what causes an increase?**

Answer

- C**
- (A) Mud hydrostatic pressure in the well
 - (B) Abnormal formation pore pressure
 - (C) Annular friction pressure while circulating
 - (D) Subnormal formation pore pressure

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DS1.41 **The crew has spaced out, shut down and successfully shut-in on a kick. The stabilized Shut-in Drillpipe Pressure (SIDPP) reads 300 psi and the Shut-in Casing Pressure (SICP) reads 450 psi. If SIDPP is held at 300 psi, what's the BHP?**

Answer

- C**
- (A) Increase
 - (B) Decrease
 - (C) Stay the same
 - (D) None of the above

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DS1.42 **Which statement best describes the amount of pressure a formation can withstand before fracturing?**

Answer

- B**
- (A) Mud Hydrostatic pressure
 - (B) Formation strength
 - (C) Mud type
 - (D) Annular pressure loss while circulating

42

DS1.43 **What circumstances require a flow check?**

Answer

- D**
- (A) Before the drill bit is run-in past the BOP
 - (B) Before the drill bit is run-in past the casing shoe
 - (C) After pulling the entire drill string out of hole
 - (D) Before beginning to pull out of the hole

43

DS1.44 **Which of the following statements describes the LEAST likely event which will cause pressures to exceed Maximum Allowable Annular Surface Pressures (MAASP)?**

Answer

- D**
- (A) A small pit gain
 - (B) A large pit gain
 - (C) A long open hole section
 - (D) A short open hole section

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DS1.45 **Which statement best describes Maximum Allowable Annular Surface Pressure (MAASP)?**

Answer

- B**
- (A) Total pressure applied at the shoe that will cause fluid losses
 - (B) Total pressure that will cause fluid losses at the casing shoe minus the mud hydrostatic at the casing shoe
 - (C) The difference in pressure between the annulus hydrostatic pressure and formation pressure
 - (D) Minimum pump pressure required to start a well kill operation

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DS1.46 **As a gas influx is circulated to surface within an oil-based mud, which statement best describes the cause of gas to break out of solution?**

Answer

- C**
- (A) The large inside diameter of the riser causes gas to break out of solution
 - (B) The temperature change of the mud as it gets cooler near surface
 - (C) The decrease in mud hydrostatic pressure above the gas when it gets close to surface
 - (D) The increasing mud hydrostatic pressure in the annulus

46

DS1.47 **While drilling ahead at a constant rate of penetration, returns on the flowline become excessive. What is the safest next step?**

Answer

- A**
- (A) Check for flow – if there is none, circulate bottoms up at a reduced pump rate so that the shale shakers can handle the cuttings volume
 - (B) Reduce the mud pump rate until the shakers can handle the amount of cuttings in the returns and continue drilling
 - (C) By pass the shale shakers until the problem is fixed
 - (D) Increase pump rate to the maximum pump speed

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DS1.48 **The crew has spaced out, shut down and successfully shut-in on a kick. The stabilized Shut-in Drillpipe Pressure (SIDPP) reads 300 psi and the Shut-in Casing Pressure (SICP) reads 450 psi. What is occurring in the shut-in well?**

Answer

B

- (A) The well is not balanced, bottom hole pressure on the annulus side is higher than on the drill string side of the U-tube
- (B) The well is currently balanced, pressures at the bottom are balanced on both sides of the well
- (C) The well is not balanced, bottom hole pressure on the annulus side is lower due to the lighter density of formation fluids
- (D) The well is overbalanced causing loss of returns

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DS1.49 **Which statement best describes Shut-in Casing Pressure (SICP)?**

Answer

- D**
- (A) Total pressure applied at the casing shoe likely to cause losses
 - (B) Total pressure inside the drill string minus the bottom hole pressure
 - (C) Total pressure applied below the casing shoe likely to cause losses
 - (D) The difference between the hydrostatic pressure in the annulus and the formation pressure

49

DS1.50 **Which of the following statements best describes factors needed to determine drillpipe pressure?**

Answer

- D**
- (A) Influx gradient psi/ft
 - (B) Type of influx in the well
 - (C) Mud density in the annulus
 - (D) Mud density inside the drill string

50

DS1.51 **During well control, Kill Weight Mud (KMW) has been pumped to the bit. If the pumps were correctly staged down and the well shut-in, what would the Shut-in Drillpipe Pressure (SIDPP) reading be? JUST THEORY, DO NOT ATTEMPT IN REAL WORLD**

Answer

- D**
- (A) The drill pipe pressure gauge will read the same pressure as the original SIDPP
 - (B) The drill pipe pressure gauge will read the same pressure as the original SICP
 - (C) The drill pipe pressure gauge will read greater than the new SICP value
 - (D) The drill pipe pressure gauge should read 0 psi due to the string being full of kill mud

51

DS1.52 **Why is Staging Up to Speed (SPUTS) so important in well control?**

Answer

- D** (A) To maintain a pressure safety margin of 100 psi while bringing the pump up to the kill rate
- (B) To allow the choke operator to reach the calculated initial circulating pressure
- (C) To reduce the time it takes for the hydraulic delay
- (D) To ensure the correct bottom hole pressure is set for the kill operation

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DS1.53 **What is the lagging indicator of an increase in formation pressure?**

Answer

- D** (A) Increasing ROP
- (B) Decreasing WOB
- (C) Decrease in background gas
- (D) Change in background gas

53

DS1.54 **During a well kill, the pump rate is increased while keeping the drill pipe pressure constant. If we ignore any effects from the Equivalent Circulating Density (ECD), how will the increase in pump rate affect Bottom Hole Pressure (BHP)?**

Answer

- D** (A) BHP will increase risking formation damage.
- (B) BHP will remain constant, risking no further influx or damage to the formation.
- (C) BHP will decrease slowly and increase dramatically, risking further influx.
- (D) None of the above

54

DS1.55 **A well has been shut-in after a kick. The stabilized SIDPP = 300 psi and SICP = 500 psi. If the DP stays at 300 psi, what happens to the BHP?**

Answer

- B** (A) Increase.
- (B) Stay the same
- (C) Decrease
- (D) None of the above

55

DS1.56 **What is the purpose of a Staging Pumps Up to Speed (SPUTS) procedure on a surface stack rig?**

Answer

- B** (A) To ensure correct drillpipe pressure is maintained.
- (B) To ensure correct bottomhole pressure is maintained.
- (C) To ensure correct casing pressure is maintained.
- (D) To ensure the driller can respond to orders

56

DS1.57 **What causes the increased Bottom Hole Pressure when ballooning formations occur and the BHP is slightly higher than the formation fracture pressure?**

Answer

- D** (A) Failure to fill up when tripping out.
- (B) Abnormal formation pressure
- (C) Drillpipe friction causes bottomhole pressure to decrease
- (D) Annular friction causes bottomhole pressure to exceed formation fracture pressure.

57

DS1.58 **How will reducing the pump rate while keeping drillpipe pressure constant during a well kill on a surface BOP affect Bottom Hole Pressure (BHP)?**

Answer

- C** (A) BHP will drop dramatically
- (B) There is no way to know what will happen to BHP
- (C) BHP will increase
- (D) Both B and C

58

DS1.59 **A well is shut-in after taking a 20 bbl kick with a Shut-in Drillpipe Pressure (SIDPP) reads 400 psi and a Shut-in Casing Pressure (SICP) reads 500 psi. If the well had been shut-in on a 10 bbl kick, how would the shut-in pressure change?**

Answer

- A** (A) SICP would be lower
- (B) SIDPP would be lower
- (C) SICP would be higher
- (D) Shut-in pressures are not affected by influx size

59

DS1.60 **Both drillpipe and casing pressures are rising because of gas migration. If we keep the casing pressure constant at 500 psi, what will happen to the Bottom Hole Pressure (BHP)?**

Answer

- A** (A) Decrease
- (B) Increase
- (C) BHP will not be affected
- (D) Slowly increase

60

DS1.61 **During the drilling a high temperature well, our Measurement While Drilling (MWD) shows the downhole temperature is significantly higher than the surface mud temperature. What effect will this temperature increase have on downhole hydrostatic pressure?**

Answer

- A** (A) It will decrease the effective downhole mud weight and cause a lower hydrostatic pressure.
- (B) It will increase the effective downhole mud weight and cause a higher hydrostatic pressure.
- (C) Increased temperature has no effect on downhole hydrostatic pressure.
- (D) All the above

61

DS1.62 **During the 1st Circulation of the Driller's Method, the influx is currently being circulated out the hole. If the gas influx is not allowed to expand, what will happen to the Bottom Hole Pressure (BHP)?**

Answer

- D** (A) Stay the same
- (B) It will decrease
- (C) Both A and B
- (D) It will increase

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DS1.63 **When we properly circulate out a kick, what is the main goal of well control?**

Answer

- B** (A) To have an overbalance pressure as close to the fracture pressure as possible.
- (B) To maintain a BHP at least equal to formation
- (C) To use the highest pump rate possible.
- (D) To ensure the kick does not lead to a blowout during the 2nd Circulation of the Driller's Method

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DS1.64 **When soft shales break down in the mud and increase mud viscosity, what happens to Bottom Hole Pressure (BHP)?**

Answer

- C** (A) BHP will decrease.
- (B) BHP will stay constant
- (C) BHP will increase.
- (D) All the above

64

DS1.65 **If a gas influx is not allowed to expand while being displaced up the annulus, what will happen to Bottom Hole Pressure (BHP)?**

Answer

- D** (A) Decrease
- (B) Stay the same
- (C) Both B and C
- (D) Increase

65

DS1.66 **The rig crew has successfully spaced out, shut down and shut-in. The stabilized SIDPP =350 psi and SICP = 450 psi. If gas begins migrating and the pressure begin to rise, what would be the effect on BHP if the SIDPP was maintained at 350 psi?**

Answer

- A** (A) It will stay the same
- (B) It will decrease
- (C) It will increase
- (D) Both A and B

66

DS1.67 **What does the Bottom Hole Pressure do if the drillpipe pressure is kept constant as kill weight mud is pumped down the drillstring?**

Answer

- A** (A) Increase
- (B) Stay the same
- (C) Decrease
- (D) Circulating CP will be variable

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DS1.68 **What will happen to Bottom Hole Pressure (BHP) if the Final Circulating Pressure (FCP) is held constant as Kill Weight Mud (KMW) is circulated up the annulus?**

Answer

- A** (A) Stay the same
- (B) Decrease
- (C) Increase
- (D) Both A and B

68

DS1.69 **On a shut-in stabilized well, Shut-in Drillpipe Pressure (SIDPP) reads 325 psi and Shut-in Casing Pressure (SICP) reads 540 psi. If CP is held constant at 540 psi during gas migration, what will happen to BHP?**

Answer

- B** (A) Increase
- (B) Decrease
- (C) Stay the same
- (D) All the above

69

DS1.70 **If the choke is closed to raise drillpipe pressure while circulating at constant SPM, what happens to bottom-hole pressure?**

Answer

- C** (A) It will stay the same
- (B) It will decrease
- (C) It will increase
- (D) All the above

70

DS1.71 **What is an uncontrolled flow of formation fluids to the surface called?**

Answer

- B** (A) Kick
- (B) Blowout
- (C) Porosity
- (D) Underground flow

71

DS1.72 **The Equivalent Circulating Density (ECD) determines the actual Bottom Hole Pressure (BHP) while circulating down the drillstring and up the annulus. Which part of the system's pressure losses is used to calculate the ECD?**

Answer

- D**
- (A) The pressure loss across the nozzles
 - (B) The pressure loss in the drillstring
 - (C) The pressure loss in surface lines
 - (D) The pressure loss in the annulus

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DS1.73 **A loss of Equivalent Circulating Density (ECD) may show that the well has become underbalanced. For our operations, what downhole tool would be the best option to accurately measure ECD while drilling?**

Answer

- A**
- (A) A pressure while drilling (PWD) tool
 - (B) A rotary steerable tool
 - (C) An adjustable gauge stabilizer (AGS) tool
 - (D) A cement bond log tool

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DS1.74 **The Pressure While Drilling (PWD) tool in the bottomhole assembly (BHA) gives us data which shows a drop in Equivalent Circulating Density (ECD) while drilling. What could lead to this drop in ECD during drilling?**

Answer

- A**
- (A) A loss of overbalance with formation fluids contaminating the mud in the annulus.
 - (B) A change in the ROP
 - (C) Increasing the RPM on the drillstring.
 - (D) A washout in the drillstring above the BHA

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DS1.75 **Shut-in Casing Pressure (SICP) value is affected by which condition?**

Answer

- A**
- (A) Annulus capacity per foot
 - (B) Slow circulating rate pressure
 - (C) Length of choke line
 - (D) Pump rate

75

DS1.76 **Concerning shallow hazards, which statement best describes a procedure which could result in an influx into the wellbore?**

Answer

- B**
- (A) Pumping out of the hole
 - (B) Not filling the hole properly while tripping out of the hole
 - (C) Reducing the pump rate
 - (D) Performing a flow check

76

DS1.77 **Which statement can increase risk for exceeded the Maximum Allowable Annular Surface Pressure (MAASP) during a kill operation?**

Answer

- B**
- (A) Short open hole section
 - (B) Small difference between formation fracture pressure and mud hydrostatic pressure
 - (C) Large difference between formation fracture pressure and mud hydrostatic pressure
 - (D) Long casing section

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DS1.78 **What happens to bottom-hole pressure if the pumps are speeded up while holding drillpipe pressure constant?**

Answer

- A**
- (A) It will decrease
 - (B) It will increase
 - (C) It will stay the same
 - (D) All the above

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DS1.79 **During the second circulation of the Driller's Method, the final circulating pressure is 350 psi at 150 gallons per minute. What would happen to the casing pressure if circulation was stopped and the choke closed once KWM enters the annulus?**

Answer

- B**
- (A) Decreases
 - (B) Increases
 - (C) Stays the same.
 - (D) All the above

79

DS1.80 **What are the changes in Drillpipe Pressure which can be expected if the mud density is decreased?**

Answer

- D** (A) Pump pressure will increase
- (B) Pump pressure will stay the same
- (C) Pump pressure will slowly decline to zero
- (D) Pump pressure will decrease

80

DS1.81 **Which of the following statements best describes needed action for updating and recording a new Slow Circulating Rate Pressure (SCR)?**

Answer

- C** (A) Every time 50 ft of new formation is drilled
- (B) Only after recharging the mud pump pulsation dampener
- (C) When you drill long sections of hole rapidly
- (D) Only at the beginning of a new hole section

81

DS1.82 **A rig drilled 2000 feet since the Slow Circulating Rate Pressure (SCR) was recorded. A kick occurred and was successfully shut in. The plan is to circulate the influx out using the Driller's method, and then staged pumps up to speed (SPUTS) was performed**

Answer

- C** (A) ICP actual = ICP calculated
- (B) ICP actual < ICP calculated
- (C) ICP actual > ICP calculated
- (D) ICP actual = FCP

82

DS1.83 **When circulating out an influx from a non-aqueous (oil-based) mud, what happens to the choke pressure as gas is liberated from the non-aqueous (oil-based) mud?**

Answer

- C** (A) Stay the same.
- (B) Decrease rapidly
- (C) Increase rapidly
- (D) None of the above

83

DS1.84 **Which of the following properties will affect the value of the Shut-In Casing Pressure (SICP)?**

Answer

- A** (A) The formation fluid pressure (pore pressure)
- (B) Permeability of formation
- (C) Porosity of formation.
- (D) Type of formation fluids

84

DS1.85 **If a kick is shut-in and the Slow Circulating Rate Pressure (SCRPP) is not known, what procedure should be used to get the correct Initial Circulating Pressure (ICP)?**

Answer

- A** (A) Follow the correct start-up procedure. When the kill rate has been reached, the drillpipe pressure minus any overbalance or safety factor will be the ICP
- (B) Contact mud logger and request the best possible calculation of ICP
- (C) Check the records and chose the kill rate circulating pressure taken with the last BHA in the hole nearest to the depth where the kick took place. Add 100 psi as a safety margin.
- (D) Stage pumps up to speed holding drillpipe pressure constant. When kill rate has been reached, subtract off any overbalance or safety factor will be the ICP

85

DS1.86 **SICP value is affected by which factor?**

Answer

- C** (A) The ID of the BHA and drillpipe.
- (B) Drillstring capacity per foot
- (C) Annulus capacity per foot
- (D) Hole diameter

86

DS1.87 **If a drillstring 750 ft off bottom is stripped into an influx which is not migrating and is at the bottom of the hole, what would be expected to happen to the SICP?**

Answer

- C** (A) Stay the same
- (B) Decrease
- (C) Increase
- (D) All the above

87

DS1.88	A salt-water kick is circulated out using the Driller's Method. The shut-in pressures and influx gradient calculation determine the influx has a greater density than current mud in the well. When will the surface casing pressure be at its minimum valve?
Answer	
C	(A) When the influx is at the choke. (B) When the annular geometry causes the influx to have minimum height. (C) When the annular geometry causes the influx to have the maximum height (D) At shut-in conditions only
88	
DS1.89	After the crew has successfully spaced out, shut down and shut-in. The pressures are allowed to stabilize. The remote choke panel Shut-in Drillpipe Pressure (SIDPP) reads 300 psi. The standpipe SICP reads 750 psi. What is the Driller's safest next step?
Answer	
A	(A) Investigate the reason for the difference (B) Record the pressure from the pressure gauge on the choke panel. (C) Record the pressure from the pressure gauges on the choke manifold (D) Assume the remote panel gauges are inaccurate and need to be replaced
89	
DS1.90	Which of these methods could be used to find the Shut-In Drillpipe Pressure (SIDPP) if a kick occurs with a non-ported float valve in the drillstring?
Answer	
D	(A) Pump slowly down the drillpipe until the float is observed opening, then shut down the pumps. Record the pressure when the float opened as the SIDPP. (B) Pump slowly down the casing, until the SIDPP begins to rise, then stop pumping. The SIDPP is the SICP minus any trapped pressure. (C) Pump quickly down the drillpipe until the SICP starts to increase, then increase pumping until 500 psi worth of trapped pressure is built. The SIDPP is the drillpipe pressure minus 500 psi. (D) Pump slowly down the drillpipe until the SICP starts to increase, then stop pumping. The SIDPP is the drillpipe pressure minus current trapped pressure.
90	
DS1.91	A well has been shut-in after a kick, and the drillpipe pressure is zero because there is a Non-Return Float Valve in the string. How should the operator obtain the Shut-In Drill Pipe Pressure (SIDPP)?
Answer	
C	(A) Pump very slowly into the drillstring with the well shut-in and continue pumping until the casing pressure reaches 100 psi over SICP. Stop the pumps. Record current SICP and subtract 100 psi to achieve SIDPP. (B) Bring the pump up to kill rate holding the casing pressure constant by opening the choke. The pressure shown when the pump is at kill rate is the Shut-In Drill Pipe Pressure (C) Pump very slowly into the drill string with the well shut-in and as casing pressure starts to rise, stop the pump, read the pressure, and subtract any overbalance. (D) None of the above
91	

DS1.92 **After observing a well flowing, space out, shut down and observe flow, then shut-in the BOP, what key data set must the Driller record?**

Answer

- C** (A) Rotary Torque
- (B) WOB
- (C) SIDPP
- (D) SICP and ROP

92

DS1.93 **How is the density of oil-based mud affected due to increasing wellbore temperature?**

Answer

- C** (A) Increases mud density
- (B) Mud Density remains the same
- (C) Decreases mud density
- (D) None of the above

93

DS1.94 **What effect does Annular Pressure Loss (APL) have on Bottom Hole Pressure (BHP)?**

Answer

- A** (A) It causes BHP to be greater than the hydrostatic pressure
- (B) It causes BHP to be less than the hydrostatic pressure
- (C) It causes BHP to be equal to the hydrostatic pressure
- (D) None of the above

94

DS1.95 **When would a flowcheck not be necessary?**

Answer

- B** (A) After tripping back to bottom.
- (B) After driller has increased the WOB
- (C) After seeing a pit level increase
- (D) When you are below the BOP stack during a trip out of the hole

95

DS1.96 **Why are well control drills are conducted?**

Answer

- D** (A) To meet governmental regulations and policies surrounding operations of well control
- (B) To train crew members to be able to have the fastest documented times to close in the well
- (C) To familiarize the rig crews with techniques that will be implemented in the event of a kick
- (D) All the above

96

DS1.97 **What is the function of pit drills?**

Answer

- C** (A) To ensure that the crew can fill out a kill sheet
- (B) To ensure that the BOPs are functioning properly
- (C) To ensure that the crew can effectively recognize and react to a kick.
- (D) To ensure rig crews can effectively clean pits

97

DS1.98 **On the BOP Closing Unit, the By-Pass Lever is used to perform which operation?**

Answer

- D** (A) Increase the accumulator pressure to 3000 psi
- (B) Decrease the accumulator pressure to 1500 psi
- (C) Increase air pressure for the Remote BOP panel
- (D) Increase the manifold pressure to the current accumulator pressure

98

DS1.99 **From the BOP Remote panel after the air lever handle has been engaged and the Upper Pipe Ram is closed, what has happened if the open light goes out, the close light does illuminates and the pressure gauges do not decrease?**

Answer

- B** (A) Faulty pressure switch
- (B) Close line is most likely blocked
- (C) Close line is most likely leaking
- (D) Faulty hydraulic switch

99

DS1.100 **On the Remote BOP panel, the by-pass valve handle (or button) directs full 3000 psi accumulator pressure to which of the following?**

Answer

- C**
- (A) Pipe rams only
 - (B) Annular preventer only
 - (C) Pipe rams and HCR valves only
 - (D) Both Annular preventer and Diverter

100

DS1.101 **From the BOP Remote panel, Upper Pipe Ram is closed. If the open light goes out, the close light does not illuminate while the accumulator and manifold pressures dropped and returned to normal?**

Answer

- D**
- (A) The close line is blocked
 - (B) The close line is leaking
 - (C) The pressure switch is faulty
 - (D) There is a fault in the close light electrical circuit

101

DS1.102 **At BOP Control Unit, annular preventer is closed. Both Accumulator and Annular gauges drop, but don't return to normal pressure while the pneumatic pump is running continuously. What problem has occurred?**

Answer

- A**
- (A) There is a leak in the annular closing line
 - (B) The 3-position valve did not operate properly
 - (C) The annular closing line is most likely blocked
 - (D) The accumulator line going to the 3-position valve is blocked

102

DS1.103 **From the BOP Remote panel, the pipe ram is closed observing 1) Open light went out, Close light is on, 2) Annular pressure didn't change, 3) Manifold pressure dropped, returns to original pressure, 4) Accumulator pressure dropped to 2300 psi and remained**

Answer

- B**
- (A) A blockage in the hydraulic line between the BOP to the BOP control unit
 - (B) The pressure switch or the pumps on the BOP control unit did not function properly
 - (C) A leakage in the hydraulic line between the BOP to the BOP control unit
 - (D) The 3-position/ 4-way valve is stuck in the open position

103

DS1.104 **From the BOP Remote panel (non-Mux), after the air lever handle has been engaged, the annular preventer is closed. The open light goes out and the close light illuminates. What problem has occurred?**

Answer

C

- (A) Confirms the annular preventer shut in the well
- (B) A signal has been sent successfully to the annular preventer regulating unit
- (C) The annular 3-position valve on the accumulator unit has functioned properly
- (D) The close line is most likely blocked

104

DS1.105 **If the air pressure reads zero on the BOP Remote panel, which of the following statement is true?**

Answer

D

- (A) All functions on the remote panel will operate properly
- (B) The annular preventer will operate properly
- (C) The pipe rams will operate properly
- (D) No BOP functions can be operated from the remote panel

105

DS1.106 **For a BOP Remote panel which functions using push-buttons, what is the main function of the "master" or "push to operate" button?**

Answer

A

- (A) To enable the functions on the driller's remote panel to become active
- (B) To adjust the pipe rams operating pressure
- (C) To adjust the annular prevent operating pressure
- (D) To close the diverter element

106

DS1.107 **According to API Standard 53, what is the minimum recommended precharge pressure for 3,000 psi accumulator bottles housed within a 3,000 psi BOP Closing Unit?**

Answer

A

- (A) 1000 psi
- (B) 1200 psi
- (C) 3000 psi
- (D) 5000 psi

107

DS1.108 **Why are fluids stored under pressure within the accumulator bottles of a BOP Closing Unit?**

Answer

- B**
- (A) To reduce the actual closing time of the remote choke
 - (B) To reduce the actual closing time of the BOP functions
 - (C) To operate the pipe ram and the annular preventer at same time
 - (D) To operate BOP functions in HPHT conditions

108

DS1.109 **What type of gas is used to establish the precharge pressures of accumulator bottles within a BOP Closing Unit?**

Answer

- C**
- (A) Hydraulic Fluid
 - (B) Carbon Dioxide
 - (C) Nitrogen
 - (D) Water

109

DS1.110 **During drilling operations, the crews noted an increase in returns. The crew spaced out, shut down and shut-in successfully. While waiting on the pressures to stabilize, what is the next step for the Driller?**

Answer

- B**
- (A) Notify the Supervisor
 - (B) Check the well is secure (no leaks)
 - (C) Verify the choke has been closed.
 - (D) Ensure pumps are completely off

110

DS1.111 **Of all the gauges on rig floor such as Driller's console or remote panel, which is used to record Slow Circulating Rate Pressures (SCRPs)?**

Answer

- A**
- (A) Drill pipe pressure gauge on the choke panel
 - (B) Casing pressure gauge on the driller's panel
 - (C) Drill pipe pressure gauge on the driller's panel
 - (D) Casing pressure gauge on the choke panel

111

DS1.112 **On a hydraulic BOP control system (non-Mux), we apply air pressure then close the annular preventer using the rig floor remote BOP panel. If the open light goes out and the close light illuminates, what does this light change indicate?**

Answer

- D**
- (A) The manifold 3-position, 4-way valve on the accumulator unit has functioned
 - (B) The accumulator by-pass 2-position, 4-way valve on the accumulator unit has functioned
 - (C) The HCR 3-position, 4-way valve on the accumulator unit has functioned
 - (D) The annular 3-position, 4-way valve on the accumulator unit has functioned

112

DS1.113 **Manifold readback pressures are affected when which equipment is operated?**

Answer

- B**
- (A) Annular preventer
 - (B) Rams and failsafe valve
 - (C) Only HCR valve
 - (D) Hydraulic choke

113

DS1.114 **Operations have spaced out, shut-down and shut-in the well. SIDPP reads 300 psi on remote choke panel. SIDPP reads 325 psi on Driller's panel. SIDPP reads 400 psi on Standpipe gauge. Which gauge is used to determine Kill Mud Weight (KMW)?**

Answer

- B**
- (A) Use the average pressure of 500 psi to calculate the Kill Mud Weight (KMW)
 - (B) With these differences in gauge pressure, inform supervisor and investigate the difference in pressures first
 - (C) Always use the gauge that shows the lowest pressure
 - (D) Always use the gauge that was calibrated most recently

114

DS1.115 **During shallow flow, after the overboard diverter valve(s) is opened and the diverter is closed, what other equipment action needs to take place?**

Answer

- B**
- (A) Direct flow through the mud-gas separator
 - (B) Increase pump rate
 - (C) Maintain constant pump rate
 - (D) Decrease pump rate

115

DS1.116 **As Kill Weight Mud (KWM) is circulated up the annulus and Final Circulation Pressure (FCP) is being held constant, what has occurred to Bottomhole Pressure (BHP)?**

Answer

- A** (A) Stay the same
- (B) Increases
- (C) Decrease
- (D) Increases rapidly followed by decreasing slowly

116

DS1.117 **For Horizontal well shut-in, the SIDPP reads the same as the SICP. During the 1st Circulation of the Driller's Method, why does the casing pressure increase so quickly when the influx moves from the horizontal section into the vertical section?**

Answer

- C** (A) This is normal for any well when the Driller's Method is used.
- (B) ECD is much greater in the horizontal section
- (C) There is no loss of hydrostatic pressure until gas arrives at the inclined or vertical section of the well.
- (D) For OBM, gas remains in solution until the gas is at the surface and breaks out of solution.

117

DS1.118 **If a leak were to occur in a rotary hose during a well control operation, what would be the next proper step to take?**

Answer

- A** (A) Pick-up off bottom, shut down the pumps and close the annular preventer.
- (B) Reduce pressure on the head and continue drilling.
- (C) Stop rotation, pick-up off bottom, close pipe rams and change out RCD insert.
- (D) Run to nearest exit

118

DS1.119 **What are the major advantages of using Managed Pressure Drilling (MPD) when drilling a well?**

Answer

- C** (A) MPD enables the well to be drilled within a tight window between pore pressure and fracture gradient.
- (B) MPD enables the well to be drilled faster as it aids in maximizing rate of penetration
- (C) All the above.
- (D) None of the above

119

DS1.120 **Why is it important to follow a pressure step down schedule when stopping the rig pumps during Well Control or connection, for a well with small gap between Pore Pressure and Fracture Gradient?**

Answer

- C** (A) To keep bottom hole pressure constant while varying the pumps.
- (B) As the pumps slow down, the wellbore friction is reduced. The surface pressure needs to be increased to replace the friction loss to prevent an influx while reducing the pumps.
- (C) Both A and B
- (D) None of the above

120

DS1.121 **Which formation characteristics affect size and speed of a kick?**

Answer

- B** (A) Influx size is not affected by permeability and differential pressure
- (B) Low permeability formation with high differential pressure
- (C) Low permeability formation with a low differential pressure.
- (D) Pour throat size and viscosity of formation fluids

121

DS1.122 **Drillstring parted about 1,500 feet above the BHA. With the well shut-in, we have noted SICP is rising at a rate of 50 psi every thirty minutes. What action can we take to restore primary control of the well?**

Answer

- C** (A) Shut well in and raise the MW.
- (B) Bullhead influx back into the formation
- (C) Stop pumping. Control the well using Volumetric method until the influx is above the washout. Then resume pumping operation.
- (D) Open the well and POH. Shut-in. Strip back to bottom with fishing assembly, engage fish and resume 1st Circulation of Driller's Method

122

DS1.123 **A well was drilled with 10.7 ppg mud. A 450-ft cement plug is set and tested. The mud above the plug is displaced by a 11.6 ppg brine. If the cement plug failed, what direction would the fluid move across the cement plug?**

Answer

- D** (A) Pressure from below would cause fluid to move up
- (B) Fluid would not move across the plug as the pressure differential is zero.
- (C) Fluid would not move across the plug until its temperature has increased and allow for expansion
- (D) Pressure from above would cause fluid to move down.

123

DS1.124 **What is the best practice for obtaining Slow Circulating Rate Pressure (SCRCP)?**

Answer

- C**
- (A) Obtain SCRCP from only mud pumps
 - (B) Obtain SCRCP from low-volume, high-pressure pump(s).
 - (C) Obtain SCRCP for a minimum of two mud pumps
 - (D) None of the above

124

DS1.125 **What drilling practice should be followed when connection gas is noticed?**

Answer

- D**
- (A) Increase the mud viscosity.
 - (B) Decrease WOB
 - (C) Decrease in rotary torque
 - (D) Control ROP to keep a minimum number of connection gas events in the hole at a time.

125

DS1.126 **Which of the gauges on the Driller's Console or Remote Choke panel are normally used to record the Slow Circulating Rate Pressure (SCRCP)?**

Answer

- B**
- (A) The Casing gauge on the Choke Control panel.
 - (B) The DP gauge on the Choke Control panel.
 - (C) The standpipe pressure.
 - (D) Gauges are not used, just sensors.

126

DS1.127 **How do we measure Slow Circulating Rate Pressure (SCRCP)?**

Answer

- C**
- (A) Record pressure at the mud pump gauge
 - (B) With the bit inside the Casing Shoe.
 - (C) With a minimum of two mud pumps
 - (D) Record pressures at the Driller's console

127

DS1.128 **What can cause inaccuracy within a Slow Pump Rate Pressure (SCRP) reading?**

Answer

- B**
- (A) Rotating the drillstring slowly when taking SCR
 - (B) An extended period of non-circulating time
 - (C) Reciprocating the drillstring slowly while taking the SCR
 - (D) Improperly calibrated gauges

128

DS1.129 **A new Slow Circulating Rate Pressure (SCRP) should be taken after which of the following operations?**

Answer

- A**
- (A) When mud weight changes
 - (B) After each connection that we drill with a top drive.
 - (C) Before killing the well.
 - (D) Anytime a swab is changed within a pump

129

DS1.130 **What statement best describes "negative" pressure test?**

Answer

- A**
- (A) A test performed on a barrier where pressure on top of a barrier is reduced to a value lower than the pressure below the barrier
 - (B) A test performed on a barrier where pressure on top of a barrier is increased to a value higher than the pressure below the barrier
 - (C) A test performed on a barrier where pressure on top of a barrier is equal to the pressure below the barrier
 - (D) A test performed on a barrier where gas is injected downhole to test the integrity of the barrier

130

DS1.131 **Which statement is true for a non-return valve or Internal Blowout Preventer (IBOP) while tripping?**

Answer

- A**
- (A) Extremely difficult to stab, if strong flow is encountered up the drillstring.
 - (B) Must never be run in the hole during stripping operations.
 - (C) Once installed, can immediately read SIDPP.
 - (D) The non-return valve has a manual closing key must be kept on the rig floor.

131

DS1.132 **While circulating out the influx, we do not see the casing pressure or pit volume going up. What could be happening?**

Answer

- A** (A) A liquid influx is in the well.
- (B) The casing pressure gauge needs to be fixed.
- (C) The PVT may need to be recalibrated.
- (D) A gaseous influx is in the well

132

DS1.133 **Why is it important to actively monitor the top of the well on the trip tank when killing a well?**

Answer

- D** (A) To verify that trip tank volumes are accurate
- (B) To check and verify of leaks from the HCR valve.
- (C) To verify the TDS system is not leaking
- (D) To check of leaks across the BOP

133

DS1.134 **Using the 1st Circulation of the Driller's method and when the gas influx reaches the surface, what happens to the mud pit level?**

Answer

- C** (A) Pit level would increase due to gas expansion then increase as gas exits the choke.
- (B) Pit level would remain the same as gas expansion does not occur in OBM.
- (C) Pit level would increase due to gas expansion then decrease as gas exits the choke.
- (D) None of the above

134

DS1.135 **Which statement best describes when the well should be monitored with flow rate and pit level equipment?**

Answer

- B** (A) During leak off tests
- (B) Any time there is a possibility or potential for a barrier to fail.
- (C) During connections
- (D) Any time there is circulation through the mud gas separator

135

DS1.136 **Operations observes flow rate increase, drillstring is spaced out, pumps stopped. The Driller notes a small amount of flow and small pit gain. Pumps are restarted with returns back to normal and no addition pit gain. What is happening?**

Answer

- C**
- (A) The pump pressure is less than mud hydrostatic pressure
 - (B) The pump pressure is greater than mud hydrostatic pressure
 - (C) The annular pressure loss is creating an overbalance against formation pressure while circulating
 - (D) The pump pressure is equal to the mud hydrostatic pressure

136

DS1.137 **A wellbore barrier cannot be verified by which method?**

Answer

- A**
- (A) Leak off test
 - (B) Using checklist for testing plug
 - (C) Monitoring fluid level
 - (D) BOP Pressure Test

137

DS1.138 **Why would the Shut-in Drillpipe Pressure (SIDPP) and the Shut-in Casing Pressure (SICP) begin to rise after initial stabilization? Why is this happening?**

Answer

- D**
- (A) The mud pump might have been left on after shutting-in the well
 - (B) The driller has been reciprocating the string causing the pressure to increase.
 - (C) The FOSV has been left closed
 - (D) The influx is gas and is migrating.

138

DS1.139 **During a flow check after a drilling break, the trip tank gains 3 barrels in 5 minutes. What should be done next?**

Answer

- B**
- (A) Trip tank may be indicating a false gain in the pit. Raise the MW up by 0.1 ppg and resume drilling operations.
 - (B) Shut well in.
 - (C) Mix 10 sacks of barite in suction pit and resume drilling operations.
 - (D) See if we can increase the rate to 7 barrels in 5 minutes and shut-in on a 10 bbl kick

139

DS1.140 **If you do not have a reduction in pit volume while drilling for the last hour while making hole what possible action should be taken?**

Answer

- B**
- (A) Continue drilling, all is good.
 - (B) Make a flow check and understand why pit volume is stable.
 - (C) Tell Derrickhand to slow water down going to active pits.
 - (D) Tell Derrickhand in increase water going into active pits

140

DS1.141 **How does using a slow pump rate aid in killing a well?**

Answer

- D**
- (A) The pump may not be rated for excessive pressure
 - (B) To prevent bursting the casing
 - (C) Allows rig crews time to mix KMW on-the-fly
 - (D) To allow choke operator time to react during a well kill.

141

DS1.142 **After kick, well was spaced out, shut down and shut-in. The mud engineer states 3700 sacks of barite are needed for KMW. Site has only 2800 sacks on site, next delivery 2 days. Which kill method should we use to kill the well?**

Answer

- D**
- (A) Wait and Weight Method
 - (B) Volumetric Method
 - (C) Lube and Bleed Method
 - (D) Driller's method

142

DS1.143 **What is the fastest action the Driller can take to shut-in the well if it is flowing while a non-shearable tool is across the BOP stack?**

Answer

- B**
- (A) Drop the string in the hole.
 - (B) Make up safety valve in the string and close the annular
 - (C) Start circulating to use ECD to the stop the kick
 - (D) Run non-shearable tool into well and shut-in on Upper Pipe Ram.

143

DS1.144 **Which of these practices could cause a shallow hazard to flow into the well?**

Answer

- A** (A) Not filling the hole properly while tripping out
- (B) Pumping a cement slurry with short transition time.
- (C) Keeping the hole full with a continuous tank
- (D) Mixing fluid density on the fly, while pumping.

144

DS1.145 **Operations spaced out, shut down and shut-in the well. After stablized SIDPP and SICP, the crew noticed the SICP has begun to increase. If the float is open, what would happen to SIDPP and SICP as the gas influx migrates?**

Answer

- D** (A) Only the annulus pressure will increase
- (B) Only drillpipe pressure will increase
- (C) Both drillpipe and annulus pressures will decrease
- (D) Both drillpipe and annulus pressures will increase

145

DS1.146 **The crew has successfully spaced out, shut down and shut-in. During the Well Control Prejob Safety Meeting (PJSM), which statement best describes Derrickhand's responsibility?**

Answer

- C** (A) Mud weight in all pits
- (B) Pit levels and measuring devices
- (C) Monitor pressures on remote choke panel
- (D) None of the above

146

DS1.147 **We have successfully spaced out, shut down and shut-in on a kick. Which of the following is an important responsibility for the Supervisor to complete?**

Answer

- D** (A) Check pit levels and line-ups
- (B) Mixing KMW
- (C) Call office and notify them of change in operations
- (D) Communicate plan with crew.

147

DS1.148 **Which is best way to mitigate top-hole section well control problems within gas-bearing formations?**

Answer

- A** (A) Control the rate of penetration
- (B) Maintain high ROP to finish the risky section as quickly as possible
- (C) Regularly pump light pump sweeps
- (D) Control rotary or top drive torque

148

DS1.149 **During well control operations, what are the most important reasons to follow good handover procedures?**

Answer

- B** (A) To allow blame to be assigned in the event of an incident during well kill operations
- (B) Ensure continuity in operations and pass on trend information to identify potential problems
- (C) It is a regulatory form which must be completed and turned into the local regulatory agency.
- (D) None of the above

149

DS1.150 **After the crew has responded to a kick, spaced out, shut down and shut-in, which statement best describes delegation of responsibilities to a crew team member?**

Answer

- C** (A) Calculate KMW needed to kill the well.
- (B) Communicating kill plan to crewmembers
- (C) Checking for leaks at the pumps, pipework or pit areas
- (D) Lowering the mud in the active pits to allow for controlled gas expansion

150

DS1.151 **When preparing for a tour change during a well kill operation, what are the best practices for conducting the handover?**

Answer

- D** (A) Each tour change individual should bring a hot cup of coffee to their relief.
- (B) Ensure the supervisor knows of the tour change and make it his/her responsibility to brief relief.
- (C) Communicate to persons on duty when its time for their relief to show up
- (D) All parties involved should stay on duty for a sufficient time to allow for thorough communication of current activities.

151

DS1.152 **We have been drilling a horizontal well and now have successfully shut-in on a kick. We do not have records of Slow Circulating Rate Pressure (SCRPP). What processes can we use to ensure we can find the correct Initial Circulating Pressure (ICP)?**

Answer

A (A) Follow the correct start-up procedures. Read the pressure on the drillpipe gauge, subtracting any overbalance/safety margin pressure

(B) Use the SIDPP as circulating pressure.

(C) Follow the ICP that was used in the previous well control operation.

(D) Since $ICP = SIDPP + SICP$, determine SICP and stage pumps up to speed holding SIDPP constant

152

DS1.153 **A written record of strokes pumped, surface drillpipe and casing pressures, choke position, active pit volume and other useful information for effective handover to a relief crew during a well kill operation is best defined as what?**

Answer

A (A) Kill log

(B) Information written in a tally book

(C) Kill sheet

(D) Circulating procedure to be followed

153

DS1.154 **If the well was full of hydrocarbons/water what term is used for the highest-pressure possible at the surface?**

Answer

C (A) Kick Volume

(B) Maximum Allowable Annular Pressure

(C) Maximum Anticipated Surface Pressure

(D) Maximum Allowable Casing Pressure

154

DS1.155 **After the first circulation of the Driller's Method, there is still influx in the annulus. What should be done to maintain constant BHP while circulating KWM to the bit?**

Answer

C (A) Maintain constant casing pressure

(B) Maintain constant drillpipe pressure

(C) Follow a Weight and Wait step down chart or graph

(D) None of the above

155

DS1.156 **The Shut-in Casing Pressure (SICP) is affected by which parameter?**

Answer

- B**
- (A) Slow circulating rate
 - (B) Annulus capacity per foot
 - (C) Internal capacity per foot
 - (D) Temperature of the well

156

DS1.157 **What action is required if the drillpipe gauge on the remote panel reads 400 psi and the drillpipe pressure gauge on the standpipe manifold reads 625 psi?**

Answer

- D**
- (A) Start the kill using 400 psi because the remote choke will be used during kill operation
 - (B) Start the kill using 625 psi due to lag time.
 - (C) Use driller's console pressure gauges since they are electronic sensor reading and much more accurate than analog pressure gauges on the remote choke panel
 - (D) Investigate the reason for the difference

157

DS1.158 **What valve would be stabbed onto the drillstring if the well takes a kick?**

Answer

- A**
- (A) Full opening safety valve
 - (B) Non-return valve
 - (C) Both non-return valve and full opening safety valve.
 - (D) Top drive non-return valve

158

DS1.159 **The crew has successfully spaced out, shut down and shut-in on a kick. After stabilization, which gauge is used to calculate the formation pressure?**

Answer

- C**
- (A) The rig air gauge on the choke panel
 - (B) The Casing pressure gauge on the choke panel
 - (C) The Drill pipe pressure gauge on the choke panel
 - (D) The choke position indicator gauge on the choke panel

159

DS1.160 **Describe the steps in order for a proper soft shut-in procedure used if there is positive flow check while tripping out of the hole at 5500 ft?**

Answer

- D** (A) Stab FOSV, Open HCR valve and choke, Close BOP, Close choke, Close safety valve.
- (B) Stab FOSV, Close BOP, Open HCR valve and choke, Close choke, Close safety valve.
- (C) Break off the Top Driver/Kelly. Install and close IBOP, tighten and torque same. Open HCR and close BOP, record pressures
- (D) Stab a full opening safety valve, close the safety valve, Open the HCR valve, close BOP and Close choke.

160

DS1.161 **What could be occurring if there is no increase in pit level or casing pressure while circulating out a kick?**

Answer

- C** (A) Pumps are failing
- (B) A gas influx is in the well
- (C) A liquid influx is in the well
- (D) BOPs are failing

161

DS1.162 **Operations has spaced out, shut down and shut-in the well. Observed 25 bbls kick with SIDPP of 300 psi and SICP of 600 psi. If faster reactions resulted in 10 bbl kick, how would this have affected shut-in pressures?**

Answer

- D** (A) SIDPP is higher
- (B) SIDPP is lower
- (C) SICP is higher
- (D) SICP is lower

162

DS1.163 **Without a float in the drillstring, what happens to shut-in pressures as gas migrates?**

Answer

- C** (A) Shut-in casing pressure may be bled off to zero.
- (B) Shut-in pressures will remain constant
- (C) Both drillpipe and annulus pressures will increase
- (D) Both drillpipe and annulus pressures will decrease

163

DS1.164 **While tripping into the hole at 6,000 feet, a flow check indicates the well is flowing. The current shut-in procedure calls for the soft shut-in procedure due to weak zones. What is the correct procedure to shut-in?**

Answer

- B**
- (A) Stab a full opening valve, Open the HCR valve, Close BOP, Close the safety valve, Close choke.
 - (B) Stab a full opening valve, close the safety valve, open the HCR valve, Close BOP, Close choke.
 - (C) Stab an inside BOP, Open the HCR valve, Close BOP, Close choke.
 - (D) Stab a full opening safety valve, open HCR, close the FOSV, Close choke and Close BOP.

164

DS1.165 **Why is it important to recognize a kick as early as possible?**

Answer

- B**
- (A) To allow the Volumetric method to be used
 - (B) To reduce the risk of formation breakdown during the kill.
 - (C) To reduce KMW required to kill the well
 - (D) So that the rig crews have more time to shut-in the well.

165

DS1.166 **While tripping into the hole at 6,000 feet, a flow check shows there's flow. The rig's procedure is to use the soft shut-in method. What steps should be followed to shut-in the well properly?**

Answer

- C**
- (A) Open HCR valve and choke, close BOP, stab full opening safety valve, close safety valve, close choke.
 - (B) Stab an inside blowout preventer (open), open HCR valve, close BOP, close choke.
 - (C) Stab a full opening safety valve (open), close the safety valve, open HCR valve, close BOP, close choke.
 - (D) None of the above

166

DS1.167 **What are the steps to shut-in when a well flows during tripping using a hard shut-in procedure?**

Answer

- C**
- (A) Open HCR valve and choke, close the BOP, stab open FOSV, Close choke and record pressures.
 - (B) Close the BOP, stable the FOSV, Close the FOSV, open choke and record pressure.
 - (C) Stab an opened FOSV, close the FOSV, Close the BOP, Open HCR valve and record pressure.
 - (D) Break off the Top Driver/Kelly. Install and close IBOP, tighten and torque same. Open HCR and close BOP, record pressures

167

DS1.168	The crew has successfully spaced out, shut down and shut-in on a kick. After stabilization, the Shut-in Drillpipe Pressure (SIDPP) reads 200 psi while the Shut-in Casing Pressure (SICP) reads 0 psi. What is the probable cause of SICP reading 0 psi?
Answer	
A	(A) Hole packed off around the BHA
	(B) Hole in the Drillpipe
	(C) Hole in the BHA
	(D) Leak in the annular preventer
168	
DS1.169	After the crew spaces out, shuts down and shut-in, Shut-in Drillpipe Pressure (SIDPP) and Shut-in Casing Pressure (SICP) stabilization time is based upon which of the following?
Answer	
B	(A) Porosity
	(B) Permeability
	(C) Gas migration
	(D) Influx density
169	
DS1.170	The crew has spaced out, shut down and successfully shut-in on a kick. Which statement best describes the derrickhand's responsibility?
Answer	
D	(A) Check current mud density
	(B) Check current mud viscosity
	(C) Check for leaks at the pumps or surface lines
	(D) Monitor and record pressures at the choke
170	
DS1.171	Which of the following statements does NOT describe a warning sign of increasing formation pressures downhole?
Answer	
A	(A) Increase in the shale density
	(B) Gradual increase in ROP
	(C) Decrease in the shale density
	(D) Increase in the connection gas
171	

DS1.172 **What is the most important reason for detecting a kick as soon as possible?**

Answer

- A** (A) To reduce risk of formation breakdown during the kill operation
- (B) To prevent gas expansion as it is circulated up the wellbore
- (C) To prevent gas migration
- (D) To prevent oil expansion as it is circulated up the wellbore

172

DS1.173 **What statement best describes "abnormal pressure"?**

Answer

- B** (A) Formation pressures that differ from the hydrostatic pressure created by a full column of fresh water
- (B) Formation pressures that differ from the hydrostatic pressure created by a full column of formation water
- (C) Formation pressures that equal normal formation pore pressure
- (D) Pressure that differs from the hydrostatic pressure caused by a full column of formation gas

173

DS1.174 **What drilling data is affected, as we drill a transition zone into an abnormally high-pressure formation?**

Answer

- D** (A) Increase in WOB to maintain constant ROP
- (B) Increase in Torque
- (C) Gradual increase in RPM
- (D) Gradual increase in ROP

174

DS1.175 **After circulating bottoms-up, which of the following is NOT a warning sign of increased formation pressure?**

Answer

- B** (A) Decreasing Shale Density
- (B) Increasing Shale Density
- (C) Increasing Cuttings Volume
- (D) Increasing Mud Temperature

175

DS1.176 **How does sudden abnormal pressure impact primary well control?**

Answer

- B** (A) Reduces the ability to use Annular Preventer as primary BOP to close in on well.
- (B) Decreases the overbalance
- (C) Allows Driller to increase tripping speed per stand.
- (D) Increases the overbalance

176

DS1.177 **How can connection gas be limited while drilling?**

Answer

- D** (A) Decrease the mud viscosity.
- (B) Increase the mud viscosity.
- (C) Increase mud pH
- (D) Control ROP to keep a minimum number of connection gas events in the hole at a time.

177

DS1.178 **If a sudden increase in gas levels are noted, what is the Driller's safest next step?**

Answer

- D** (A) Stop drilling, circulate bottoms up, monitor & record gas levels, inform your supervisor
- (B) Begin to circulate weighted sweeps periodically
- (C) Continue to monitor and record the gas level
- (D) Check for flow and call the supervisor

178

DS1.179 **During drilling of a wild cat exploratory well, what is the best method for determining formation strength for these wells with little or no downhole pressure information?**

Answer

- C** (A) Equivalent Mud Weight Test (EMW)
- (B) Formation Integrity Test (FIT)
- (C) Leak Off Test (LOT)
- (D) Casing Test

179

DS1.180 **Due to ballooning and after shutting-in, Operations decides to bleed 5 bbls of mud into the trip tank, then circulate bottoms up. What is the safest next step?**

Answer

- A** (A) Circulate through a fully open choke with the BOP closed and monitor active pit level closely
- (B) Circulate bottoms up until you see gas at surface then shut in the well and circulate through the fully open choke
- (C) Circulate bottoms up at the maximum pump speed
- (D) Circulate bottoms up through the kill line

180

DS1.181 **How will oil-based mud density be affected by increasing downhole pressure?**

Answer

- B** (A) Decreases mud density
- (B) Increases mud density
- (C) Mud density remains unchanged
- (D) Oil based muds have no density

181

DS1.182 **After noting an increase in flow, the crew spaced out, shut down and successfully shut-in the well. After pressure stabilization, what is the Driller's safest next step?**

Answer

- C** (A) Calculate the correct kill mud density
- (B) Monitor and record surface pressures
- (C) Check and verify the well is secure (no leaks)
- (D) Monitor and record the pit gain

182

DS1.183 **During a well control circulation through a choke, what occurs as a gas influx is circulated out while keeping bottom hole pressure (BHP) constant?**

Answer

- B** (A) The volume of gas will decrease
- (B) The volume of gas will increase
- (C) The volume of gas will stay the same
- (D) The gas will disperse into the drilling mud

183

DS1.184 **Excessive flowback time during connection. Shut-in. SIDPP is 200 psi, SICP is 200 psi. Operations suspects ballooning is occurring and pressures have been bled off in increments of 50 psi until they read 0 psi. What is the next safest step?**

Answer

- D**
- (A) Open the BOP and drill ahead
 - (B) Open the choke and perform a flow-check. If negative, circulate bottoms-up through the open riser
 - (C) Circulate at the maximum pump rate through the mud gas separator
 - (D) Open the choke and perform a flow-check. If negative, circulate bottoms-up through the choke

184

DS1.185 **Which statement best describes the main indicator that the influx is below the bit and potentially requires stripping back to bottom?**

Answer

- B**
- (A) The SIDPP is greater than SICP
 - (B) The SIDPP is equal to SICP
 - (C) The SIDPP is lower than SICP
 - (D) None of the above

185

DS1.186 **Operations has spaced out, shut down and shut-in on a kick. The SIDPP is 200 psi, SICP is 300 psi. Both pressures start rising due to gas migration. If the SIDPP is kept constant at 200 psi, what happens to BHP?**

Answer

- C**
- (A) Increase
 - (B) Decrease
 - (C) Stay the Same
 - (D) None of the above

186

DS1.187 **During W&W Method, KMW has been circulated to the bit and the pumps were staged down. SIDPP is 200 psi and SICP is 600 psi. Which of the following may result in an additional influx entering the wellbore?**

Answer

- B**
- (A) Bleed off approximately 50 psi to check for trapped pressure
 - (B) Bleed the SIDPP to 0 psi
 - (C) Bleed off approximately 100 psi to check for trapped pressure
 - (D) Re-check the calculations of surface to bit strokes and kill mud density

187

DS1.188 **During tripping operations, well swabbed. Spaced out, installed and closed FOSV, closed Annular Preventer using Hard Shut-in. SIDPP and SICP is 175 psi. We are currently 1,412 feet off bottom. What is the safest next step?**

Answer

- B**
- (A) Circulate the kick out using the Driller's method while 30 stands off bottom
 - (B) Strip to bottom and start circulation using first circulation of Driller's method
 - (C) Use volumetric procedure until the entire influx has migrated to the bit and start circulation using the Driller's Method
 - (D) Use Lube and Bleed Method to remove gas from the well

188

DS1.189 **When stripping back to bottom with a migrating kick, what is the proper procedure to be followed in order to maintain constant Bottomhole Pressure (BHP)?**

Answer

- C**
- (A) Bleed off the drill pipe steel displacement while tripping each stand
 - (B) Bleed off the volume of mud equal to the open-end displacement of the drill pipe tripped into the well
 - (C) Bleed off the volume of mud equal to the closed end displacement of the drill pipe stripped into the well
 - (D) Do not bleed mud while stripping in with drill pipe

189

DS1.190 **Which statement best describes a kill plan?**

Answer

- A**
- (A) A document stating duties, procedures, and specific lines of communication and safety requirements for the entire kill operation
 - (B) A kill sheet completed by the supervisor
 - (C) A document that all rig personnel must sign
 - (D) A document stating the correct kill mud density used to kill the well

190

DS1.191 **If the u-tube (well) has lost integrity due to losses, how can we identify this condition?**

Answer

- A**
- (A) Manipulations of the choke result in unpredictable fluctuations of the casing and drillpipe pressures.
 - (B) Decrease in drillpipe pressure followed by a increase in casing pressure.
 - (C) Manipulation of the choke results in a change in casing pressure and a predictable pressure response on the drillpipe pressure gauge.
 - (D) None of the above

191

Drilling Supervisor Practice Test Day 1

DST1.1 **From the BOP Closing Unit, the gauge on the manifold pressure reads above 1500 psi and all the other gauges are normal, what would be a possible problem?**

Answer

- D**
- (A) The pressure switch controlling the pump is faulty
 - (B) The electrically driven triplex pump is not set for automatic.
 - (C) The pneumatic pump does not have sufficient pressure
 - (D) A malfunction in the manifold regulator

1

DST1.2 **On a hydraulic BOP control system (non-Mux), we initiate air pressure then close the annular preventer using the rig floor remote BOP panel. If the open light goes out and the close light illuminates, what is the problem indicated by the light change?**

Answer

- B**
- (A) The drillpipe 3-position, 4-way valve on the accumulator unit has functioned
 - (B) The annular 3-position, 4-way valve on the accumulator unit has functioned
 - (C) The HCR 3-position, 4-way valve on the accumulator unit has functioned
 - (D) The emergency 2-position, 4-way valve on the accumulator unit has functioned

2

DST1.3 **From the BOP Remote panel after the air lever has been engaged and the Upper Pipe Ram is closed, what has happened if the open light stays on and the close light does not illuminate. Manifold pressures remains static.**

Answer

- A**
- (A) The 3-position/4-way valve failed to operate
 - (B) The hydraulic closing line to the BOP is most likely plugged
 - (C) The hydraulic closing line to the BOP is most likely leaking
 - (D) The manifold regulator is malfunctioning

3

DST1.4 **If we suspect ballooning, what is the safest next step the Driller can take?**

Answer

- C** (A) Record the time it takes for 5 bbls of mud to flow into the trip tank, then shut in well
- (B) Continue drilling ahead
- (C) Shut the well in, call the supervisor and record pressures and pit gain
- (D) Check and verify that the cuttings returned over the shakers is shale

4

DST1.5 **While drilling, we are losing mud at a rate of 10 bbls per hour. During a connection, the well started slowly flowing. When the Driller restarts the pumps, losses begin again. What could be occurring downhole?**

Answer

- B** (A) The mud is u-tubing due to different weights in the drill string and annulus
- (B) The well is indicating that the formation may be “ballooning”
- (C) The well is losing mud to the formation, causing a reduction in hydrostatic pressure
- (D) An influx in the wellbore is “ballooning”

5

DST1.6 **Which one of the following is NOT considered a barrier?**

Answer

- B** (A) Mud hydrostatic pressure.
- (B) Drill string float valve
- (C) Shut-in choke and valve combination.
- (D) Annular preventer along with mud hydrostatic pressure

6

DST1.7 **What is the minimum number of barriers that are usually needed for any part of the operation?**

Answer

- A** (A) Two
- (B) One
- (C) Three
- (D) Four

7

DST1.8 **If a kick is swabbed in when tripping because pipe was pulled too fast, which type of barrier failed?**

Answer

- B** (A) Mechanical barrier
- (B) Operational barrier
- (C) Tertiary barrier
- (D) Physical barrier

8

DST1.9 **After setting and cementing the liner, we are to perform a (API: Inflow Test) across the liner-lap. The liner is displaced with a lightweight fluid while well is shut-in and circulating on the choke with predetermined surface backpressure held constant.**

Answer

- B** (A) Not enough information to make an informed decision.
- (B) Failed test. Shut the well in and notify Supervisor.
- (C) Successful test. Continue with planned operations.
- (D) Repeat test to ensure that it was performed correctly

9

DST1.10 **Which two barriers are NOT independent of each other?**

Answer

- C** (A) Fluid and cement plug
- (B) Cement plug and cast-iron bridge plug
- (C) Annular and Ram BOP
- (D) None of the above

10

DST1.11 **While drilling a slug of 20 bbls of light mud is displaced into the drillstring followed by the Current Mud Weight (CMW) of 12 ppg. The Driller stages the pump down holding casing pressure constant and shut-in the well with light mud still inside the dri**

Answer

- D** (A) BHP will increase
- (B) BHP will decrease
- (C) BHP will increase, then Decrease
- (D) BHP will remain the same

11

DST1.12 **A well is shut in because of a kick, and the shut-in pressures on the gauges have stabilized. The Shut-In Drillpipe Pressure (SIDPP) is 400 psi, and the Shut-In Casing Pressure (SICP) is 800 psi. Is the well balanced?**

Answer **C**

- (A) The well is not balanced, bottomhole pressure (BHP) on drillpipe side is not correct due to poor drillpipe float operation.
- (B) The well is not balanced, bottomhole pressure (BHP) on annular side is higher than on the drillpipe side of the U-tube.
- (C) The well is balanced; pressure at the bottom balanced on both sides of the well.
- (D) Ballooning

12

DST1.13 **The well is shut in on a kick. The well is shut in. The shut-in drillpipe pressure (SIDPP) reads 300 psi, shut-in casing pressure (SICP) reads 400 psi. Both pressures are rising due to gas migration. If SICP is held constant, what happens to bottom hole p**

Answer **C**

- (A) Stay the same
- (B) Increase
- (C) Decrease
- (D) None of the above

13

DST1.14 **A well has been shut-in after a kick. The Shut-In Drillpipe Pressure (SIDPP) is 300 psi, and the Shut-In Casing Pressure (SICP) is 500 psi. If CP start to incrementally rise because of gas migration and DP stays at 300 psi, what will happen to the Bottom**

Answer **B**

- (A) Increase.
- (B) Stay the same
- (C) Decrease
- (D) None of the above

14

DST1.15 **During a well kill, the choke operator sees that both drillpipe and casing pressures are slowly going down. He adjusts the choke to keep the original pump pressure, while keeping the pump speed steady. What effect will this adjustment have on the Bottom H**

Answer **C**

- (A) BHP is not affected by pump rate.
- (B) BHP is not affected by choke adjustment.
- (C) BHP is returned to correct value.
- (D) BHP is not affected by mud density

15

DST1.16 **Why is it important to keep Bottom Hole Pressure (BHP) constant during a well kill?**

Answer

- C** (A) To prevent the influx from expanding as it is circulated up hole.
- (B) To prevent the drillpipe pressure from fluctuating while circulating kick out of the hole.
- (C) To prevent a further influx or formation breakdown during the kill process
- (D) None of the above

16

DST1.17 **If a well is shut-in on an annular preventer with larger diameter casing, what may result?**

Answer

- D** (A) Annular closing pressure is too low to seal around the larger diameter casing
- (B) Annular closing pressure should be increased for proper seal around the casing
- (C) Annular closing pressure should not be adjusted for casing
- (D) Annular closing pressure is too high and damages the larger diameter casing

17

DST1.18 **During drilling operations, the well is losing mud at a rate of 10 barrels per hour. During connections, the well has a trickle flow. When we restart the pumps, we start losing mud. The Driller performs a flow check and the well starts flowing again. W**

Answer

- A** (A) Stop drilling, shut in the well and call the Supervisor.
- (B) Stop drilling and flow check for 5 minutes to see if the flow rate stops
- (C) The well is ballooning and there is no need to shut-in the well.
- (D) If the flow was less than a finger width, return to drilling. Most likely, higher temperatures are causing fluids to expand within the hole

18

DST1.19 **During the 1st Circulation of the Driller's Method, the Initial Circulating Pressure (ICP) has been stable at 600 psi and the Initial Circulating Choke Pressure (CP) reads 500 psi during the past one hour. Suddenly, the ICP drops to 525 psi and the CP re**

Answer

- A** (A) The bit nozzle has washed out.
- (B) You have a washout in the drillpipe
- (C) The choke is plugging up
- (D) The choke is washing out

19

DST1.20 **Once kill weight mud is pumped to the bit, if the pumps were shut-down and there was still pressure on the drillpipe, what condition would exist in the well?**

Answer

- B**
- (A) The annulus is underbalanced.
 - (B) The drillpipe is underbalanced
 - (C) The DP pressure sensor on the standpipe has failed.
 - (D) The drillpipe has trapped air

20

DST1.21 **What action would provide the most effective handover during a well kill operation?**

Answer

- C**
- (A) Instruct them to write down the key points of their behavior and hand to new crew.
 - (B) Think about what to tell the replacement at the end of the shift.
 - (C) Instruct crew to write down the key points of their handover and discuss each point with their replacement.
 - (D) Stagger rig crew handovers so that everyone works at every station

21

DST1.22 **What should the Driller do if there is a sudden jump in mud gas levels?**

Answer

- A**
- (A) Check for flow and call Supervisor.
 - (B) Request Mud Engineer to increase mud weight by 0.3 ppg
 - (C) Stop drilling, circulate bottoms-up, record gas levels and inform we supervisor.
 - (D) Ignore gas level increases as bubbles continually migrate to surface.

22

DST1.23 **We are drilling ahead at a steady rate of penetration (ROP), but the flow cleaners can't manage the amount of cuttings coming from the well. What is the safest action to take?**

Answer

- B**
- (A) Slow down the mud pump until the shakers can handle the amount of cuttings in the returns and continue drilling
 - (B) Check for flow – if there is none, circulate bottoms up at a reduced rate so that the shakers can handle the cuttings volume.
 - (C) Stop circulating and pull the bit up into the shoe.
 - (D) Bring online the desander

23

DST1.24 **What is the proper definition of Maximum Anticipated Annular Surface Pressure (MAASP)?**

Answer

- C** (A) The total pressure which will cause losses at the casing shoe.
- (B) The pressure at which the bottom of the hole fractures
- (C) The pressure which will cause losses at the casing shoe minus the mud hydrostatic at the casing shoe.
- (D) The pressure at which the Annular Preventer will fail

24

DST1.25 **Which one of the following conditions can occur if additional pressure is put on a non-Aqueous (oil-based) mud?**

Answer

- A** (A) Increases density
- (B) Decreases density
- (C) Density remains the same
- (D) All the above

25

DST1.26 **What happens to fluid density when you increase the pressure?**

Answer

- C** (A) It decreases fluid density.
- (B) Fluid density remains the same.
- (C) It increases the fluid density
- (D) All the above

26

DST1.27 **Which statement best describes a choke drill?**

Answer

- B** (A) To test that the choke is working properly before drilling out the last casing shoe
- (B) To help the rig personnel understand how the choke and well pressures react during a kill operation
- (C) To ensure all alarms are functioning properly
- (D) To ensure the choke panel is functioning properly

27

DST1.28 **Which statement best describes the purpose of a pit drill?**

Answer

- C** (A) To ensure all alarms are functioning properly
- (B) To ensure the rig personnel is trained to kill the well
- (C) To ensure the rig personnel is able to effectively recognize and react to a kick
- (D) To ensure the choke panel is functioning properly

28

DST1.29 **Why are choke drills performed?**

Answer

- B** (A) To test to ensure the choke is operating correctly before drilling out the casing shoe.
- (B) To help the crew understand how the choke and well pressures react during a kill operation.
- (C) To ensure the crew is trained to line up correctly for a reverse circulation well kill.
- (D) To meet IADC emergency response times.

29

DST1.30 **The 1st Circulation of the Driller's Method was going as planned. Then, the drillpipe pressure began to drop, followed by a decrease in casing pressure (CP). What could be the possible problems?**

Answer

- A** (A) Pump is washing out
- (B) Choke is washing out
- (C) Mud weight is non-homogeneous
- (D) Choke is plugging up

30

DST1.31 **As pumps are shut down and we go from a dynamic state to static state, what happens to the downhole Equivalent Circulating Density (ECD)?**

Answer

- C** (A) Remains the same.
- (B) Increases
- (C) Decreases
- (D) None of the above

31

DST1.32 **Why is Measured Depth (MD) used to calculate Annular Pressure Loss (APL) instead of True Vertical Depth (TVD) in the well?**

Answer

- B**
- (A) Because of the friction of mud solids and cuttings impact the entire drillstring.
 - (B) Because the friction of the mud solids and cuttings impact the entire annulus.
 - (C) Because the friction of the mud solids and cutting impact only the drillstring
 - (D) Because APL cannot be calculated at all. We can estimate it to be 200 psi, regardless of mud density

32

DST1.33 **After adding air and functioning the close control to close the annular preventer, the "open" light turns off, but the "close" light does not come on. The pressure gauge for the annular drops and then goes back to the right pressure. What has occurred?**

Answer

- B**
- (A) The 3-position/4-way valve did not operate correctly.
 - (B) The close light circuit or bulb has failed.
 - (C) A plug exists between 3-position / 4-way valve and the BOP
 - (D) Isolation valve is inoperable

33

DST1.34 **On the Driller's air-operated panel for a surface BOP, the ram is closed and the following happened: 1) the open light turned off, 2) the close light turned on, 3) annular pressure did not change, 4) manifold pressure dropped and then came back, and 5) ac**

Answer

- A**
- (A) The pressure switch or the pumps on the BOP control unit did not function.
 - (B) The 3-position / 4-way valve did not operate.
 - (C) A plug exists between 3-position / 4-way valve and the surface BOP.
 - (D) Rig power outage

34

DST1.35 **On the surface stack remote panel, the High-Pressure Bypass handle or button lets we apply full accumulator pressure to which of the following?**

Answer

- B**
- (A) Annular only
 - (B) Rams and HCR valves only
 - (C) Annular, Rams and HCR valves
 - (D) Rams only

35

DST1.36 **Which of the following are not integral components of a Rotating Control Device (RCD)?**

Answer

- C** (A) Bearing
- (B) Housing
- (C) Flowline and the orbit valve in the flowline
- (D) Lock down devices

36

DST1.37 **What information is needed to calculate the correct formation strength at the shoe?**

Answer

- A** (A) A calibrated pressure gauge.
- (B) An accurate stroke counter
- (C) An accurate hole capacity
- (D) The formation porosity and permeability

37

DST1.38 **Which statement does NOT require a flow check?**

Answer

- D** (A) A 10% increase in ROP during a 10 foot interval
- (B) A 20% increase in ROP during a 10 foot interval
- (C) A sudden decrease in pump pressure
- (D) After the Driller has increased the WOB

38

DST1.39 **The crew notes an increase in flow rate, so the drillstring is spaced out and the flow check is performed. The crew cannot determine if the well is flowing. What is the safest next step?**

Answer

- A** (A) Line up on the trip tank and monitor for a gain in the pit level
- (B) Shut in the well and line up through the mud-gas separator and monitor for flow at the shakers
- (C) Shut in the well and line up through the mud-gas separator and monitor for flow at the mud gas separator
- (D) Line up on the active pit and monitor for gain in the pit level

39

DST1.40 **Which of the following is NOT a warning sign of an increasing formation pressure (FormP)?**

Answer

- B**
- (A) Increasing background gas
 - (B) Increasing shale density
 - (C) Decreasing shale density
 - (D) Increasing ROP at same WOB

40

DST1.41 **What is the best lagging indicator to describe an increase in formation pressure?**

Answer

- D**
- (A) Decrease in volume of cuttings
 - (B) Increase in ROP.
 - (C) Decrease in WOB
 - (D) Change in background gas

41

DST1.42 **What is the pressure inside the pore spaces where pressurized fluid is found called?**

Answer

- A**
- (A) Formation pressure
 - (B) Porosity
 - (C) Permeability
 - (D) Pore throat pressure

42

DST1.43 **We shut-in a well on a kick with the bit on bottom. Which gauge reading do we use to calculate the formation pressure?**

Answer

- C**
- (A) Casing pressure on the driller's console
 - (B) The casing pressure gauge on the choke panel
 - (C) The drillpipe pressure gauge on the choke panel
 - (D) The drillpipe pressure gauge on the Driller's console

43

DST1.44

What occurs to the gas influx expansion rate as it is pumped from the horizontal section to the vertical section of the well?

Answer

B

(A) Same

(B) More

(C) Less

(D) None of the above

44

DST1.45

How does gas migration affect wellbore pressures?

Answer

C

(A) All the pressures decrease.

(B) Does not affect wellbore pressures

(C) All the pressures increase.

(D) None of the above

45

DST1.46

During drilling operations, no Slow Circulating Rate Pressures (SCRPs) have been recorded. If a kick ensues, what is the proper procedure for determining correct Initial Circulating Pressure (ICP)?

Answer

D

(A) Turn pumps on quickly and read ICP

(B) Allow casing pressure to increase to 500 psi regardless of pump velocity and record ICP

(C) ICP = 500 psi over SIDPP

(D) Follow correct start-up procedure. Read the pressure on the drill pipe gauge, subtract any overbalance safety margin pressure. This is the ICP

46

DST1.47

Which statement best describes a Tally Book containing, as a minimum, surface pressures, choke position and other useful information for future handovers?

Answer

A

(A) Kill Log

(B) Kill Plan

(C) Kill Sheet

(D) Step-down Chart

47

DST1.48	What is the best way to describe a written record of strokes pumped, surface pressures, choke position, and other information which helps the relief crew during a well kill operation?
Answer	
D	(A) Drillpipe pressure schedule.
	(B) Pre-job safety meeting with step-by-step directions for 1st and 2nd circulation of the Driller's Method.
	(C) Killsheet
	(D) Kill Log
48	
DST1.49	As Current Mud Weight (CMW) density increases, how does this effect Maximum Allowable Annular Surface Pressure (MAASP)?
Answer	
B	(A) MAASP will increase
	(B) MAASP will decrease.
	(C) MAASP remains the same.
	(D) None of the above
49	
DST1.50	What is the first thing we should do if we are circulating out a kick in a deep well and the casing pressure gets close to the Maximum Allowable Annular Surface Pressure (MAASP) while the gas is still in the open hole?
Answer	
A	(A) Minimize any extra pressure in the annulus without allowing the BHP to fall below the pore pressure.
	(B) Continue pumping at the same SPM. The MAASP is only a problem when the influx is inside casing.
	(C) Keep the casing pressure at the MAASP by opening the choke.
	(D) Switch to a Water Based Mud as WBM are not compressible
50	
DST1.51	What should we do if we are circulating gas inside the casing and the casing pressure gets close to the Maximum Allowable Annular Surface Pressure (MAASP)?
Answer	
B	(A) Continue with the kill procedure, but open the choke as needed to maintain casing pressure equal to MAASP thereby risking additional influx.
	(B) Continue with the correct kill procedure, the increasing MAASP will not cause formation breakdown.
	(C) Shut operation down and bullhead influx back into the formation.
	(D) Switch to a Water Based Mud as WBM are not compressible
51	

DST1.52 **If a heavy pill is pumped into the well and the pipe is tripped out of the well, what will happen to the bottom hole pressure?**

Answer

- A** (A) Decrease
- (B) Increase then decrease rapidly
- (C) Increase
- (D) Stay the same

52

DST1.53 **How are weighted pills used during drilling and tripping operations?**

Answer

- C** (A) Used to clear the mud within the drillpipe before tripping out of the hole.
- (B) To compensate for hydrostatic underbalance when tripping out of the hole.
- (C) All the above.
- (D) None of the above

53

DST1.54 **Which statement describes a "Kill Plan"**

Answer

- A** (A) Document stating duties, procedures, lines of communication and safety requirements
- (B) Document stating kill method to use and barite stock levels
- (C) A completed Kill sheet
- (D) A tally book with Kill Log

54

DST1.55 **While completing a kill sheet, why do we take the time to calculate the volume and strokes of open hole from bit to casing shoe?**

Answer

- A** (A) To determine when the influx has entered the casing.
- (B) It is useful for calculating how much more gas expansion will occur.
- (C) This is used to estimate the maximum casing pressure.
- (D) This is used to estimate Maximum Allowable Annular Surface Pressure (MAASP)

55

DST1.56 **For worldwide operations, what is the most common cause of abnormal pressure?**

Answer

- C** (A) Artesian flow
- (B) Carbonate layers of formation
- (C) Trapped fluid in shale formations
- (D) Faults

56

DST1.57 **If the surface pumps fail during well control circulation, what is the next best step?**

Answer

- D** (A) Switch immediately to another available pump
- (B) Increase the pump rate
- (C) Decrease the pump rate
- (D) Shut in the well

57

DST1.58 **Which of the following statements best explains the conditions which makes kick detection more difficult for the Driller?**

Answer

- D** (A) High permeability formations with oil-based mud
- (B) High permeability formations with water-based mud
- (C) Low permeability formations with water-based mud
- (D) Low permeability formations with oil-based mud

58

DST1.59 **What occurs to surface pressure as gas breaks out of an oil-based mud, while influx is being circulated to surface?**

Answer

- C** (A) Surface pressure will remain the same
- (B) Surface pressure will decrease rapidly
- (C) Surface pressure will increase rapidly
- (D) None of the above

59

DST1.60 **An inaccurate Slow Circulating Rate Pressure (SCRCP) may be recorded by which of the following?**

Answer

- D**
- (A) Recording the drill pipe pressure at the choke control panel
 - (B) Maintaining a consistent mud density throughout the well
 - (C) Accurate pump stroke counters
 - (D) Gelled mud in the well immediately after a round trip

60

DST1.61 **Slow Circulating Rate Pressures (SCRCP) are needed to perform which of the following?**

Answer

- A**
- (A) To accurately calculate the Initial Circulating Pressure (ICP)
 - (B) To accurately calculate the kill mud density
 - (C) To accurately calculate the pump strokes from surface to bit
 - (D) To accurately calculate the correct volume of barrels pumped

61

DST1.62 **When would we consider taking a new Slow Circulation Rate Pressure (SCRCP)?**

Answer

- A**
- (A) When we have drilled long hole sections rapidly.
 - (B) When MW has changed by 0.2 ppg.
 - (C) During each tour change.
 - (D) Slow Circulating Rate Pressures are never to be taken

62

DST1.63 **What can lead to a bigger kick size and a higher Shut-In Casing Pressure (SICP)?**

Answer

- D**
- (A) Formation integrity
 - (B) Formation porosity
 - (C) Formation pressure
 - (D) None of the above

63

DST1.64 **If a kick is shut-in and we do not know the Slow Circulating Rate Pressure (SCRП), what steps should we follow to find the correct Initial Circulating Pressure (ICP)?**

Answer

- B**
- (A) Use the SICP as the circulating pressure.
 - (B) Follow the correct start-up procedure. Read the pressure on the drillpipe gauge subtracting any overbalance / safety margin pressure.
 - (C) Use the SIDPP as the circulating pressure.
 - (D) Read the pressure on the casing gauge subtracting any overbalance / safety margin pressure.

64

DST1.65 **What will happen to the casing pressure when gas is being circulated from the horizontal section to the vertical section of the well?**

Answer

- (A) Casing pressure will stay the same.
- (B) Casing pressure will decrease.
- (C) Nothing will be affected
- (D) Casing pressure will increase.

65

DST1.66 **After the crew has spaced out, shut down and shut-in, they recorded a stabilized Shut-in Casing Pressure (SICP) of 600 psi. The Driller pumped at 5 SPM until the float was bumped and then turned off the pump. Now, the casing pressure gauge reads 700 psi,**

Answer

- D**
- (A) 500 psi
 - (B) 700 psi
 - (C) 250 psi
 - (D) 100 psi

66

DST1.67 **Which statement describes Shut-in Drillpipe Pressure (SIDPP)?**

Answer

- D**
- (A) The formation fluid pressure
 - (B) The mud hydrostatic pressure in the drillstring
 - (C) Formation pressure minus shut-in casing pressure
 - (D) The difference between the hydrostatic pressure in the drillstring and the formation pressure.

67

DST1.68 **What causes SICP to normally be higher than SIDPP?**

Answer

- B**
- (A) Cuttings in the annulus help reduce the hydrostatic pressure
 - (B) The hydrostatic in the drillpipe is greater than the hydrostatic in the annulus
 - (C) There is usually more volume in the annulus
 - (D) SICP gauges are more accurate than SIDPP gauges

68

DST1.69 **What changes may we expect at the flowlines or shakers as we enter a transition zone to abnormally high-pressured formation?**

Answer

- B**
- (A) Reduction in size of cuttings returned at the shakers
 - (B) Decreasing density of shale cuttings
 - (C) Increasing density of shale cuttings
 - (D) All the above

69

DST1.70 **For a well where the Drillstring Volume is less than the Bit to Shoe Volume, which kill method would minimize pressure at the casing shoe and lower the risks of lost circulation?**

Answer

- C**
- (A) Volumetric Method
 - (B) Driller's Method
 - (C) Wait and Weight Method.
 - (D) Bullheading Method

70

Drilling Supervisor Study Guide Day 2

DS2.1 **Porosity can be defined by which of the following statements?**

Answer

- B**
- (A) When KMW is at the shoe
 - (B) The amount of void space in the rock, expressed as a percent.
 - (C) The measure of ease or ability of a rock to transmit a one-phase fluid under conditions of laminar flow. The unit of porosity is the Darcy.
 - (D) Pore throat pressure

1

DS2.2 **The drilling crew has successfully spaced out, shut down and shut-in the well. Which pressure do we want to keep the Bottom Hole Pressure (BHP) at least equal to?**

Answer

- B**
- (A) At least equal to the drillpipe pressure
 - (B) At least equal to the formation fluid pressure
 - (C) Initial circulating pressure
 - (D) At least equal to the casing pressure

2

DS2.3 **Which of the following factors can change the value of Shut-in Casing Pressure (SICP) after we shut-in a well during a kick?**

Answer

- C**
- (A) Choke line length
 - (B) SCRCP pressure
 - (C) The formation fluid pressure (pore pressure)
 - (D) Low surface temperatures

3

DS2.4 **In order to kill a well, which statement is correct?**

Answer

- D**
- (A) Keep BHP 100 psi over Formation Pressure
 - (B) Keep BHP 100 psi under formation pressure
 - (C) Keep BHP equal to Casing Shoe Pressure
 - (D) Keep BHP at least equal to formation pressure

4

DS2.5 **Which of the following equipment is used to secure a well while the drilling line is cut and the drillstring is in the well?**

Answer

- B**
- (A) Inside Blowout Preventer
 - (B) Full Opening Safety Valve or Non-Return Valve
 - (C) Check valve
 - (D) Secure hanging wire rope in derrick

5

DS2.6 **The operator's drilling engineer has calculated a kick tolerance of 35 bbls based on a 1 ppg kick intensity. If we recalculate for a kick intensity of 1.5 ppg, what will happen to the maximum allowable kick volume?**

Answer

- B**
- (A) Maximum allowable kick volume will increase
 - (B) Maximum allowable kick volume will decrease
 - (C) Maximum allowable kick volume would not be affected
 - (D) Both A and C

6

DS2.7 **What can raise the risk of going over the Maximum Allowable Annulus Surface Pressure (MAASP) during a kill operation?**

Answer

- A**
- (A) Long open hole section
 - (B) Short open hole section
 - (C) Weak casing shoe
 - (D) No open hole section

7

DS2.8 **While circulating the gas influx inside the casing, if the casing pressure gets close to the Maximum Allowable Annulus Surface Pressure (MAASP), what should we do?**

Answer

- D** (A) Continue with the kill procedure, but open the choke as needed to maintain casing pressure equal to MAASP, thereby risking additional influx.
- (B) Shut-in well, use Volumetric Well Control Method after mobilizing a Well Control provider.
- (C) Shut-in well, use the Lube and Bleed Method
- (D) Continue with the correct kill procedure, the increasing MAASP will not cause formation breakdown.

8

DS2.9 **During a well kill operation, which of the following is LEAST likely to cause us to exceed the Maximum Anticipated Annular Surface Pressure (MAASP)?**

Answer

- C** (A) Increase mud weight
- (B) Small difference between formation fracture pressure and drilling fluid hydrostatic pressure.
- (C) Short open hole section
- (D) None of the above

9

DS2.10 **At what point during a kill operation could the choke pressure exceed the Maximum Anticipated Annular Surface Pressure (MAASP) without causing the formation at the shoe to break down?**

Answer

- C** (A) When performing the pump start-up procedure
- (B) When the influx is in the open hole section
- (C) When the influx is above the casing shoe.
- (D) When the influx is at the choke

10

DS2.11 **The greatest pressure at the casing shoe occurs at what point in a well kill circulation?**

Answer

- B** (A) During the pump start-up
- (B) Either at initial shut-in or when influx reaches the shoe.
- (C) Only when the top of the influx reaches the shoe.
- (D) When gas reaches the choke

11

DS2.12 **Which operation affects the Maximum Allowable Annular Surface Pressure (MAASP) and requires that it be recalculated?**

Answer

- A** (A) After mud weight change
- (B) Every 300 to 500 feet drilled
- (C) After changing out a swab on the mud pump.
- (D) When alternate choke is used

12

DS2.13 **Most problems in well control can be recognized early. For a Mud Gas Separator, what indicators would suggest that blow through was about to occur?**

Answer

- D** (A) Increasing trend in hydrostatic seal pressure
- (B) Loss of mud leg or dip tube
- (C) Reduction in flow coming from vent line
- (D) Increasing trend in vent line pressure

13

DS2.14 **After noting an increase in flow, the crew spaced out, shut down and successfully shut-in the well. Which of the following statement best describes a crewmember responsibility?**

Answer

- D** (A) Communicate kill plans and procedures to crewmembers
- (B) Calculate the correct kill mud density
- (C) Choose which kill method to use
- (D) Check for leaks at the mud pumps, pipework and pit areas

14

DS2.15 **Which statement best describes when a flow check is needed?**

Answer

- C** (A) Immediately after a 5% decrease in the weight on bit
- (B) Immediately after a 5% increase in the rate of penetration
- (C) Before the BHA is pulled up through the BOP stack
- (D) After the BHA is pulled up to the surface

15

DS2.16 **Which of the following well is NOT a wellbore barrier test?**

Answer

- A** (A) Leak-off Test
- (B) Positive Test
- (C) Negative Test
- (D) Cumulative Pressure Test

16

DS2.17 **What is a definition of a negative barrier test?**

Answer

- A** (A) A test on a barrier where pressure on the formation side of the barrier is greater than the pressure on the surface side of the barrier.
- (B) A test on a barrier where pressure on the surface side of the barrier is greater than the formation side of the barrier..
- (C) A test to see if the safety valve is holding
- (D) Opposite of a positive barrier test

17

DS2.18 **What is the proper definition of a "negative" pressure test?**

Answer

- B** (A) A test where the barrier is made to fail.
- (B) A test on a barrier where pressure is reduced to a value lower than the other side of the barrier
- (C) A test on a barrier where pressure is increased to a value higher than the other side of the barrier.
- (D) The opposite of a positive test

18

DS2.19 **What is a proper definition of a "physical barrier" according to the following statements?**

Answer

- A** (A) A cement plug
- (B) Proper BOP shut-in procedure
- (C) Monitoring the trip tank
- (D) Monitoring DP and CP at the remote choke

19

DS2.20 **During tripping operations, if an influx is swabbed into the well, what primary type of barrier failed?**

Answer

- B** (A) Permanent mechanical barrier
- (B) Operational barrier
- (C) Temporary mechanical barrier
- (D) Retrievable mechanical barrier

20

DS2.21 **"Procedural barriers" can be defined by which of the following statements**

Answer

- A** (A) Monitoring well for gains and losses
- (B) Calculating KMW
- (C) Mixing mud
- (D) Procedural barriers are those barriers with written operating procedures

21

DS2.22 **Procedural barrier can be defined as?**

Answer

- B** (A) Mud pumped into well
- (B) Monitoring the well for gains or losses.
- (C) Running casing in well
- (D) Tripping pipe from well

22

DS2.23 **Concerning most rig operations, the API recommends at least two independent barriers to be in place throughout operations. Which pair of barriers are NOT independent of each other?**

Answer

- C** (A) Drilling fluid and an annular preventer
- (B) A tubing plug set downhole and a backpressure valve installed in the hanger
- (C) An annular and a pipe ram on a BOP stack
- (D) Drilling fluid and a pipe ram

23

DS2.24 **For most operations, B.S.S.E. always recommends installation of two independent barriers. Which pair of barriers are NOT independent of each other?**

Answer

- C**
- (A) Viscosity and mud weight
 - (B) Kill mud in the well and the BOP
 - (C) Annular and a Ram on the BOP stack
 - (D) Drillpipe Safety Valve and Kill Mud

24

DS2.25 **A 9.7 ppg brine has been circulated above a plug set from 7500 ft to 7850 ft across a casing shoe. The highest-pressure zone below the plug was drilled with a 10.5 ppg mud. What would happen if there is communication across the plug?**

Answer

- C**
- (A) Well will remain static.
 - (B) Well will lose circulation
 - (C) Well will flow
 - (D) The well will remain static until the brine begins to increase in temperature

25

DS2.26 **If a safety margin is added to the mud weight what is the result?**

Answer

- A**
- (A) Cannot be removed easily
 - (B) Increases MAASP at the shoe
 - (C) Decreases casing shoe pressure
 - (D) Decreases MAASP at the shoe

26

DS2.27 **What is the definition of a positive test?**

Answer

- C**
- (A) A test on a barrier where the pressure on the formation side of the barrier is equal to the pressure on the surface side of the barrier.
 - (B) A test on a barrier where the pressure applied to the formation side of the barrier is greater than the pressure on the surface side of the barrier.
 - (C) A test on a barrier where the pressure on the formation side of the barrier is less than the pressure on the surface side of the barrier.
 - (D) A test performed on a barrier where gas is injected downhole to test the integrity of the barrier

27

DS2.28 **Which of the following statements best describes a "procedural barrier"?**

Answer

- C** (A) Writing handover notes for your relief
- (B) Setting a packer assembly
- (C) Actively monitoring the well for gains or losses
- (D) Decreasing the rate of penetration

28

DS2.29 **Which type of barrier has failed when a kick is swabbed into the wellbore during tripping?**

Answer

- B** (A) Secondary Hydrostatic barrier
- (B) Primary Hydrostatic barrier
- (C) Tertiary Hydrostatic barrier
- (D) Primary Hydrostatic Pressure

29

DS2.30 **Which type of barrier has failed when a kick is swabbed into the wellbore during tripping?**

Answer

- A** (A) Primary hydrostatic barrier
- (B) Temporary mechanical barrier
- (C) Permanent mechanical barrier
- (D) Secondary barrier

30

DS2.31 **Which statement is NOT a wellbore barrier verification method?**

Answer

- A** (A) Leak off test
- (B) Weight set confirmation (Tag top of cement)
- (C) Inflow test (Negative test)
- (D) Measuring the mud density

31

DS2.32 Which statement best describes a positive test?

Answer

- C**
- (A) A test performed on a barrier where the pressure applied to the formation side of the barrier is equal to the pressure on the surface side of the barrier
 - (B) A test performed on a barrier where the pressure applied to the formation side of the barrier is greater than the pressure on the surface side of the barrier
 - (C) A test performed on a barrier where the pressure on the formation side of the barrier is less than the pressure on the surface side of the barrier
 - (D) A good pressure test on surface equipment

32

DS2.33 Which statement best describes a "Barrier System" (Envelope)?

Answer

- D**
- (A) A set of processes and procedures that prevent the well from kicking
 - (B) The BOP only
 - (C) Drilling fluid and barite
 - (D) More than one barrier that act together to prevent flow from the formation

33

DS2.34 Which statement best describes an item that is NOT classified as a barrier?

Answer

- B**
- (A) The mud hydrostatic pressure
 - (B) A drill string float valve
 - (C) The pipe rams
 - (D) A cement plug

34

DS2.35 We are drilling a development well within an established producing field. Which statement best describes the pressure test normally used to determine formation strength at the casing shoe?

Answer

- B**
- (A) Leak Off Test
 - (B) Formation Integrity Test
 - (C) Drill Stem Test
 - (D) BOP Test

35

DS2.36 **Please select the best statement which describes what is needed to determine the casing shoe formation strength?**

Answer

- A** (A) The exact TVD of the casing shoe
- (B) Accurate hole capacity
- (C) Accurate pump stroke counters
- (D) Kill weight mud

36

DS2.37 **Which of the following statements may NOT warn of an increase in formation pressure?**

Answer

- C** (A) WOB display or gauge
- (B) ROP display or gauge
- (C) Accumulator Pressure display or gauge
- (D) Torque display or gauge

37

DS2.38 **Which statement describes the best definition of Maximum Allowable Annular Surface Pressure (MAASP)?**

Answer

- C** (A) The pressure which will cause losses at the bottom of the hole
- (B) The pressure which will cause flow at the casing
- (C) The pressure which will cause losses at the casing shoe minus the mud hydrostatic at the casing shoe
- (D) The pressure which will cause pit gains when bit is on bottom

38

DS2.39 **Which statement defines the percentage of void space in a formation?**

Answer

- A** (A) Porosity
- (B) Permeability
- (C) Viscosity
- (D) Mud density

39

DS2.40 **When ballooning occurs, Bottomhole Pressure (BHP) is slightly increased more than the formation fracture pressure. Which statement best describes what causes an increase?**

Answer

- C**
- (A) Mud hydrostatic pressure in the well
 - (B) Abnormal formation pore pressure
 - (C) Annular friction pressure while circulating
 - (D) Subnormal formation pore pressure

40

DS2.41 **Operations shut-in on kick. The stabilized SIDPP is 300 psi and SICP is 450 psi. While preparing to kill the well, the crew maintains a SIDPP of 300 psi. What's happening to BHP?**

Answer

- C**
- (A) Increase
 - (B) Decrease
 - (C) Stay the same
 - (D) None of the above

41

DS2.42 **Which statement best describes the amount of pressure a formation can withstand before fracturing?**

Answer

- B**
- (A) Mud Hydrostatic pressure
 - (B) Formation strength
 - (C) Mud type
 - (D) Annular pressure loss while circulating

42

DS2.43 **What circumstances require a flow check?**

Answer

- D**
- (A) Before the drill bit is run-in past the BOP
 - (B) Before the drill bit is run-in past the casing shoe
 - (C) After pulling the entire drill string out of hole
 - (D) Before beginning to pull out of the hole

43

DS2.44 **Which of the following statements describes the LEAST likely event which will cause pressures to exceed Maximum Allowable Annular Surface Pressures (MAASP)?**

Answer

- D**
- (A) A small pit gain
 - (B) A large pit gain
 - (C) A long open hole section
 - (D) A short open hole section

44

DS2.45 **Which statement best describes Maximum Allowable Annular Surface Pressure (MAASP)?**

Answer

- B**
- (A) Total pressure applied at the shoe that will cause fluid losses
 - (B) Total pressure that will cause fluid losses at the casing shoe minus the mud hydrostatic at the casing shoe
 - (C) The difference in pressure between the annulus hydrostatic pressure and formation pressure
 - (D) Minimum pump pressure required to start a well kill operation

45

DS2.46 **As a gas influx is circulated to surface within an oil-based mud, which statement best describes the cause of gas to break out of solution?**

Answer

- C**
- (A) The large inside diameter of the riser causes gas to break out of solution
 - (B) The temperature change of the mud as it gets cooler near surface
 - (C) The decrease in mud hydrostatic pressure above the gas when it gets close to surface
 - (D) The increasing mud hydrostatic pressure in the annulus

46

DS2.47 **While drilling ahead at a constant rate of penetration, returns on the flowline become excessive. What is the safest next step?**

Answer

- A**
- (A) Check for flow – if there is none, circulate bottoms up at a reduced pump rate so that the shale shakers can handle the cuttings volume
 - (B) Reduce the mud pump rate until the shakers can handle the amount of cuttings in the returns and continue drilling
 - (C) By pass the shale shakers until the problem is fixed
 - (D) Increase pump rate to the maximum pump speed

47

DS2.48 **The crew has spaced out, shut down and successfully shut-in on a kick. The stabilized Shut-in Drillpipe Pressure (SIDPP) reads 300 psi and the Shut-in Casing Pressure (SICP) reads 450 psi. What is occurring in the shut-in well?**

Answer

B

- (A) The well is not balanced, bottom hole pressure on the annulus side is higher than on the drill string side of the U-tube
- (B) The well is currently balanced, pressures at the bottom are balanced on both sides of the well
- (C) The well is not balanced, bottom hole pressure on the annulus side is lower due to the lighter density of formation fluids
- (D) The well is overbalanced causing loss of returns

48

DS2.49 **Which statement best describes Shut-in Casing Pressure (SICP)?**

Answer

- D**
- (A) Total pressure applied at the casing shoe likely to cause losses
 - (B) Total pressure inside the drill string minus the bottom hole pressure
 - (C) Total pressure applied below the casing shoe likely to cause losses
 - (D) The difference between the hydrostatic pressure in the annulus and the formation pressure

49

DS2.50 **Which of the following statements best describes factors needed to determine bottomhole pressure?**

Answer

- D**
- (A) Influx gradient psi/ft
 - (B) Type of influx in the well
 - (C) Mud density in the annulus
 - (D) Mud density inside the drill string

50

DS2.51 **During well control, Kill Weight Mud (KMW) has been pumped to the bit. If the pumps were correctly staged down and the well shut-in, what would the Shut-in Drillpipe Pressure (SIDPP) reading be? JUST THEORY, DO NOT ATTEMPT IN REAL WORLD**

Answer

- D**
- (A) The drill pipe pressure gauge will read the same pressure as the original SIDPP
 - (B) The drill pipe pressure gauge will read the same pressure as the original SICP
 - (C) The drill pipe pressure gauge will read greater than the new SICP value
 - (D) The drill pipe pressure gauge should read 0 psi due to the string being full of kill mud

51

DS2.52 **Why is Staging Up to Speed (SPUTS) so important in well control?**

Answer

- D** (A) To maintain a pressure safety margin of 100 psi while bringing the pump up to the kill rate
- (B) To allow the choke operator to reach the calculated initial circulating pressure
- (C) To reduce the time it takes for the hydraulic delay
- (D) To ensure the correct bottom hole pressure is set for the kill operation

52

DS2.53 **What is the lagging indicator of an increase in formation pressure?**

Answer

- D** (A) Increasing ROP
- (B) Decreasing WOB
- (C) Decrease in background gas
- (D) Change in background gas

53

DS2.54 **During a well kill, the pump rate is increased while keeping the drill pipe pressure constant. If we ignore any effects from the Equivalent Circulating Density (ECD), how will the increase in pump rate affect Bottom Hole Pressure (BHP)?**

Answer

- D** (A) BHP will increase risking formation damage.
- (B) BHP will remain constant, risking no further influx or damage to the formation.
- (C) BHP will decrease slowly and increase dramatically, risking further influx.
- (D) None of the above

54

DS2.55 **Operations shut-in on kick. Stabilized SIDPP is 300 psi and SICP is 500 psi. If the drillpipe pressure rises to 450 psi, what happens to BHP?**

Answer

- B** (A) Stay the same
- (B) Increases
- (C) Decreases
- (D) None of the above

55

DS2.56 **What is the purpose of a Staging Pumps Up to Speed (SPUTS) procedure on a surface stack rig?**

Answer

- B** (A) To ensure correct drillpipe pressure is maintained.
- (B) To ensure correct bottomhole pressure is maintained.
- (C) To ensure correct casing pressure is maintained.
- (D) To ensure the driller can respond to orders

56

DS2.57 **What causes the increased Bottom Hole Pressure when ballooning formations occur and the BHP is slightly higher than the formation fracture pressure?**

Answer

- D** (A) Failure to fill up when tripping out.
- (B) Abnormal formation pressure
- (C) Drillpipe friction causes bottomhole pressure to decrease
- (D) Annular friction causes bottomhole pressure to exceed formation fracture pressure.

57

DS2.58 **How will reducing the pump rate while keeping drillpipe pressure constant during a well kill on a surface BOP affect Bottom Hole Pressure (BHP)?**

Answer

- C** (A) BHP will drop dramatically
- (B) There is no way to know what will happen to BHP
- (C) BHP will increase
- (D) Both B and C

58

DS2.59 **A well is shut-in after taking a 20 bbl kick with a Shut-in Drillpipe Pressure (SIDPP) reads 400 psi and a Shut-in Casing Pressure (SICP) reads 500 psi. If the well had been shut-in on a 10 bbl kick, how would the shut-in pressure change?**

Answer

- A** (A) SICP would be lower
- (B) SIDPP would be lower
- (C) SICP would be higher
- (D) Shut-in pressures are not affected by influx size

59

DS2.60 **Both drillpipe and casing pressures are rising because of gas migration. If we keep the casing pressure constant at 500 psi, what will happen to the Bottom Hole Pressure (BHP)?**

Answer

- A** (A) Decrease
- (B) Increase
- (C) BHP will not be affected
- (D) Slowly increase

60

DS2.61 **During the drilling of a high temperature well, our MWD shows the downhole temperature is significantly higher than the surface mud temperature. What effect will this temperature increase have on downhole hydrostatic pressure?**

Answer

- A** (A) It will decrease the effective downhole mud weight and cause a lower hydrostatic pressure.
- (B) It will increase the effective downhole mud weight and cause a higher hydrostatic pressure.
- (C) Increased temperature has no effect on downhole hydrostatic pressure.
- (D) All the above

61

DS2.62 **During the 1st Circulation of the Driller's Method, the influx is currently being circulated out the hole. If the gas influx is not allowed to expand, what will happen to the Bottom Hole Pressure (BHP)?**

Answer

- D** (A) Stay the same
- (B) It will decrease
- (C) Both A and B
- (D) It will increase

62

DS2.63 **When we properly circulate out a kick, what is the main goal of well control?**

Answer

- B** (A) To have an overbalance pressure as close to the fracture pressure as possible.
- (B) To maintain a BHP at least equal to formation pressure
- (C) To use the highest pump rate possible.
- (D) To ensure the kick does not lead to a blowout during the 2nd Circulation of the Driller's Method

63

DS2.64 **When soft shales break down in the mud and increase mud viscosity, what happens to Bottom Hole Pressure (BHP)?**

Answer

- C** (A) BHP will decrease.
- (B) BHP will stay constant
- (C) BHP will increase.
- (D) All the above

64

DS2.65 **If a gas influx is not allowed to expand while being displaced up the annulus, what will happen to Bottom Hole Pressure (BHP)?**

Answer

- D** (A) Decrease
- (B) Stay the same
- (C) Both B and C
- (D) Increase

65

DS2.66 **Operations has shut-in. The stabilized SIDPP is 250 psi and SICP is 350 psi. The crew notes that SIDPP remains at 250 psi. What's happened to BHP?**

Answer

- A** (A) It will stay the same
- (B) It will decrease
- (C) It will increase
- (D) Both A and B

66

DS2.67 **What does the Bottom Hole Pressure do if the drillpipe pressure is kept constant as kill weight mud is pumped down the drillstring?**

Answer

- A** (A) Increase
- (B) Stay the same
- (C) Decrease
- (D) Circulating CP will be variable

67

DS2.68 **What will happen to Bottom Hole Pressure (BHP) if the Final Circulating Pressure (FCP) is held constant as Kill Weight Mud (KMW) is circulated up the annulus?**

Answer

- A** (A) Stay the same
- (B) Decrease
- (C) Increase
- (D) Both A and B

68

DS2.69 **Operations shut-in. Stabilized SIDPP is 310 psi and SICP is 540 psi. If the casing pressure is held constant at 540 psi while both pressures rise due to gas migration, what happens to BHP?**

Answer

- B** (A) Increase
- (B) Decrease
- (C) Stay the same
- (D) All the above

69

DS2.70 **IF the choke is closed to raise drillpipe pressure while circulating at constant SPM, what happens to bottom-hole pressure?**

Answer

- C** (A) It will stay the same
- (B) It will decrease
- (C) It will increase
- (D) All the above

70

DS2.71 **What is an uncontrolled flow of formation fluids to the surface called?**

Answer

- B** (A) Kick
- (B) Blowout
- (C) Porosity
- (D) Underground flow

71

DS2.72 **The Equivalent Circulating Density (ECD) determines the actual Bottom Hole Pressure (BHP) while circulating down the drillstring and up the annulus. Which part of the system's pressure losses is used to calculate the ECD?**

Answer

- D**
- (A) The pressure loss across the nozzles
 - (B) The pressure loss in the drillstring
 - (C) The pressure loss in surface lines
 - (D) The pressure loss in the annulus

72

DS2.73 **A loss of Equivalent Circulating Density (ECD) may show that the well has become underbalanced. For our operations, what downhole tool would be the best option to accurately measure ECD while drilling?**

Answer

- A**
- (A) A pressure while drilling (PWD) tool
 - (B) A rotary steerable tool
 - (C) An adjustable gauge stabilizer (AGS) tool
 - (D) A cement bond log tool

73

DS2.74 **The Pressure While Drilling (PWD) tool in the bottomhole assembly (BHA) gives us data which shows a drop in Equivalent Circulating Density (ECD) while drilling. What could lead to this drop in ECD during drilling?**

Answer

- A**
- (A) A loss of overbalance with formation fluids contaminating the mud in the annulus.
 - (B) A change in the ROP
 - (C) Increasing the RPR's on the drillstring.
 - (D) A washout in the drillstring above the BHA

74

DS2.75 **Shut-in Casing Pressure (SICP) value is affected by which condition?**

Answer

- A**
- (A) Annulus capacity per foot
 - (B) Slow circulating rate pressure
 - (C) Length of choke line
 - (D) Pump rate

75

DS2.76 **Concerning shallow hazards, which statement best describes a procedure which could result in an influx into the wellbore?**

Answer

- B**
- (A) Pumping out of the hole
 - (B) Not filling the hole properly while tripping out of the hole
 - (C) Reducing the pump rate
 - (D) Performing a flow check

76

DS2.77 **Which statement can increase risk for exceeded the Maximum Allowable Annular Surface Pressure (MAASP) during a kill operation?**

Answer

- B**
- (A) Short open hole section
 - (B) Small difference between formation fracture pressure and mud hydrostatic pressure
 - (C) Large difference between formation fracture pressure and mud hydrostatic pressure
 - (D) Long casing section

77

DS2.78 **What happens to bottom-hole pressure if the pumps are speeded up while holding drillpipe pressure constant?**

Answer

- A**
- (A) It will decrease
 - (B) It will increase
 - (C) It will stay the same
 - (D) All the above

78

DS2.79 **During the second circulation of the Driller's Method, the final circulating pressure is 350 psi at 150 gallons per minute. What would happen to the casing pressure if circulation was stopped and the choke closed once kill weight mud enters the annulus?**

Answer

- B**
- (A) Decreases
 - (B) Increases
 - (C) Stays the same.
 - (D) All the above

79

DS2.80 **What are the changes in Drillpipe Pressure which can be expected if the mud density is decreased?**

Answer

- D** (A) Pump pressure will increase
- (B) Pump pressure will stay the same
- (C) Pump pressure will slowly decline to zero
- (D) Pump pressure will decrease

80

DS2.81 **Which of the following statements best describes needed action for updating and recording a new Slow Circulating Rate Pressure (SCR)?**

Answer

- C** (A) Every time 50 ft of new formation is drilled
- (B) Only after recharging the mud pump pulsation dampener
- (C) When you drill long sections of hole rapidly
- (D) Only at the beginning of a new hole section

81

DS2.82 **2000 feet were drilled since SCR was recorded. After at kick and shut-in, the Driller's Method will be used. After Staging Pumps up to speed (SPUTS), what will happen to ICP?**

Answer

- C** (A) ICP actual = ICP calculated
- (B) ICP actual < ICP calculated
- (C) ICP actual > ICP calculated
- (D) ICP actual = FCP

82

DS2.83 **When circulating out an influx from a non-aqueous (oil-based) mud, what happens to the choke pressure as gas is liberated from the non-aqueous (oil-based) mud?**

Answer

- C** (A) Stay the same.
- (B) Decrease rapidly
- (C) Increase rapidly
- (D) None of the above

83

DS2.84 **Which of the following properties will affect the value of the initial Shut-In Casing Pressure (SICP)?**

Answer

- A** (A) The formation fluid pressure (pore pressure)
- (B) Permeability of formation
- (C) Porosity of formation.
- (D) Type of formation fluids

84

DS2.85 **If a kick is shut-in and the Slow Circulating Rate Pressure (SCRП) is not known, what procedure should be used to get the correct Initial Circulating Pressure (ICP)?**

Answer

- A** (A) Follow the correct start-up procedure. When the kill rate has been reached, the drillpipe pressure minus any overbalance or safety factor will be the ICP
- (B) Contact mud logger and request the best possible calculation of ICP
- (C) Check the records and chose the kill rate circulating pressure taken with the last BHA in the hole nearest to the depth where the kick took place. Add 100 psi as a safety margin.
- (D) Stage pumps up to speed holding drillpipe pressure constant. When kill rate has been reached, subtract off any overbalance or safety factor will be the ICP

85

DS2.86 **SICP value is affected by which factor?**

Answer

- C** (A) The ID of the BHA and drillpipe.
- (B) Drillstring capacity per foot
- (C) Annulus capacity per foot
- (D) Hole diameter

86

DS2.87 **If a drillstring 750 ft off bottom is stripped into an influx which is not migrating and is at the bottom of the hole, what would be expected to happen to the SICP?**

Answer

- C** (A) Stay the same
- (B) Decrease
- (C) Increase
- (D) All the above

87

DS2.88 **A salt-water kick is circulated out using the Driller's Method. The shut-in pressures and influx gradient calculation determine the influx has a greater density than current mud in the well. When will the surface casing pressure be at its minimum valve?**

Answer **C**

- (A) When the influx is at the choke.
- (B) When the annular geometry causes the influx to have minimum height.
- (C) When the annular geometry causes the influx to have the maximum height
- (D) At shut-in conditions only

88

DS2.89 **After the crew has successfully spaced out, shut down and shut-in. The pressures are allowed to stabilize. The remote choke panel Shut-in Casing Pressure (SICP) reads 300 psi. The Driller's console SICP reads 750 psi. What is the Driller's safest next**

Answer **A**

- (A) Investigate the reason for the difference
- (B) Record the pressure from the pressure gauge on the choke panel.
- (C) Record the pressure from the pressure gauges on the choke manifold
- (D) Assume the remote panel gauges are inaccurate and need to be replaced

89

DS2.90 **Which of these methods could be used to find the Shut-In Drillpipe Pressure (SIDPP) if a kick occurs with a non-ported float valve in the drillstring?**

Answer **D**

- (A) Pump slowly down the drillpipe until the float is observed opening, then shut down the pumps. Record the pressure when the float opened as the SIDPP.
- (B) Pump slowly down the casing, until the SIDPP begins to rise, then stop pumping. The SIDPP is the SICP minus any trapped pressure.
- (C) Pump quickly down the drillpipe until the SICP starts to increase, then increase pumping until 500 psi worth of trapped pressure is built. The SIDPP is the drillpipe pressure minus 500 psi.
- (D) Pump slowly down the drillpipe until the SICP starts to increase, then stop pumping. The SIDPP is the drillpipe pressure minus current trapped pressure.

90

DS2.91 **A well has been shut-in after a kick, and the drillpipe pressure is zero because there is a Non-Return Float Valve in the string. How should the operator obtain the Shut-In Drill Pipe Pressure (SIDPP)?**

Answer **C**

- (A) Pump very slowly into the drillstring with the well shut-in and continue pumping until the casing pressure reaches 100 psi over SICP. Stop the pumps. Record current SICP and subtract 100 psi to achieve SIDPP.
- (B) Bring the pump up to kill rate holding the casing pressure constant by opening the choke. The pressure shown when the pump is at kill rate is the Shut-In Drill Pipe Pressure
- (C) Pump very slowly into the drill string with the well shut-in and as casing pressure starts to rise, stop the pump, read the pressure, and subtract any overbalance.
- (D) None of the above

91

DS2.92 **After observing a well flowing, space out, shut down and observe flow, then shut-in the BOP, what key data set must the Driller record?**

Answer

- C** (A) Rotary Torque
- (B) WOB
- (C) SIDPP
- (D) SICP and ROP

92

DS2.93 **How is the density of oil-based mud affected due to increasing wellbore temperature?**

Answer

- C** (A) Increases mud density
- (B) Mud Density remains the same
- (C) Decreases mud density
- (D) None of the above

93

DS2.94 **What effect does Annular Pressure Loss (APL) have on Bottom Hole Pressure (BHP)?**

Answer

- A** (A) It causes BHP to be greater than the hydrostatic pressure
- (B) It causes BHP to be less than the hydrostatic pressure
- (C) It causes BHP to be equal to the hydrostatic pressure
- (D) None of the above

94

DS2.95 **When would a flowcheck not be necessary?**

Answer

- B** (A) After tripping back to bottom.
- (B) After driller has increased the WOB
- (C) After seeing a pit level increase
- (D) When you are below the BOP stack during a trip out of the hole

95

DS2.96 **Why are well control drills are conducted?**

Answer

- D** (A) To meet governmental regulations and policies surrounding operations of well control
- (B) To train crew members to be able to have the fastest documented times to close in the well
- (C) Required by BSEE as listed in the Drilling Permit
- (D) To familiarize the rig crews with techniques that will be implemented in the event of a kick

96

DS2.97 **What is the function of pit drills?**

Answer

- C** (A) To ensure that the crew can fill out a kill sheet
- (B) To ensure that the BOPs are functioning properly
- (C) To ensure that the crew can effectively recognize and react to a kick.
- (D) To ensure rig crews can effectively clean pits

97

DS2.98 **On the BOP Closing Unit, the By-Pass Lever is used to perform which operation?**

Answer

- D** (A) Increase the accumulator pressure to 3000 psi
- (B) Decrease the accumulator pressure to 1500 psi
- (C) Increase air pressure for the Remote BOP panel
- (D) Increase the manifold pressure to the current accumulator pressure

98

DS2.99 **From the BOP Remote panel after the air lever handle has been engaged and the Upper Pipe Ram is closed, what has happened if the open light goes out, the close light does illuminates and the pressure gauges do not decrease?**

Answer

- B** (A) Faulty pressure switch
- (B) Close line is most likely blocked
- (C) Close line is most likely leaking
- (D) Faulty hydraulic switch

99

DS2.100 **On the Remote BOP panel, the by-pass valve handle (or button) directs full 3000 psi accumulator pressure to which of the following?**

Answer

- C**
- (A) Pipe rams only
 - (B) Annular preventer only
 - (C) Pipe rams and HCR valves only
 - (D) Both Annular preventer and Diverter

100

DS2.101 **From BOP remote panel, the Upper Pipe Ram is closed. The open light goes out, the close light does not illuminate, accumulator and manifold pressures dropped and returned to normal working pressure?**

Answer

- D**
- (A) The close line is blocked
 - (B) The close line is leaking
 - (C) The pressure switch is faulty
 - (D) There is a fault in the close light electrical circuit

101

DS2.102 **On BOP Control Unit, Annular Preventer is closed. Both accumulator and annular pressure gauges rapidly drop, but do not return back to normal pressure while the pneumatic pump is running continuously. What has occurred?**

Answer

- A**
- (A) There is a leak in the annular closing line
 - (B) The 3-position valve did not operate properly
 - (C) The annular closing line is most likely blocked
 - (D) The accumulator line going to the 3-position valve is blocked

102

DS2.103 **From the BOP Remote panel, Pipe rams closed. 1) open light went out and close light illuminates, 2) The annular pressure did not change, 4) The manifold pressure dropped and did not return back to normal pressure. What has occurred?**

Answer

- B**
- (A) A blockage in the hydraulic line between the BOP to the BOP control unit
 - (B) The pressure switch or the pumps on the BOP control unit did not function properly
 - (C) A leakage in the hydraulic line between the BOP to the BOP control unit
 - (D) The 3-position/ 4-way valve is stuck in the open position

103

DS2.104 **From the BOP Remote panel (non-Mux), after the air lever handle has been engaged, the annular preventer is closed. The open light goes out and the close light illuminates. What problem has occurred?**

Answer

C

- (A) Confirms the annular preventer shut in the well
- (B) A signal has been sent successfully to the annular preventer regulating unit
- (C) The annular 3-position valve on the accumulator unit has functioned properly
- (D) The close line is most likely blocked

104

DS2.105 **If the air pressure reads zero on the BOP Remote panel, which of the following statement is true?**

Answer

D

- (A) All functions on the remote panel will operate properly
- (B) The annular preventer will operate properly
- (C) The pipe rams will operate properly
- (D) No BOP functions can be operated from the remote panel

105

DS2.106 **For a BOP Remote panel which functions using push-buttons, what is the main function of the "master" or "push to operate" button?**

Answer

A

- (A) To enable the functions on the driller's remote panel to become active
- (B) To adjust the pipe rams operating pressure
- (C) To adjust the annular prevent operating pressure
- (D) To close the diverter element

106

DS2.107 **According to API Standard 53, what is the minimum recommended precharge pressure for 3,000 psi accumulator bottles housed within a 3,000 psi BOP Closing Unit?**

Answer

A

- (A) 1000 psi
- (B) 1200 psi
- (C) 3000 psi
- (D) 5000 psi

107

DS2.108 **Why are fluids stored under pressure within the accumulator bottles of a BOP Closing Unit?**

Answer

- B**
- (A) To reduce the actual closing time of the remote choke
 - (B) To reduce the actual closing time of the BOP functions
 - (C) To operate the pipe ram and the annular preventer at same time
 - (D) To operate BOP functions in HPHT conditions

108

DS2.109 **What type of gas is used to establish the precharge pressures of accumulator bottles within a BOP Closing Unit?**

Answer

- C**
- (A) Hydraulic Fluid
 - (B) Carbon Dioxide
 - (C) Nitrogen
 - (D) Water

109

DS2.110 **During drilling operations, the crews noted an increase in returns. The crew spaced out, shut down and shut-in successfully. While waiting on the pressures to stabilize, what is the next step for the Driller?**

Answer

- B**
- (A) Notify the Supervisor
 - (B) Check the well is secure (no leaks)
 - (C) Verify the choke has been closed.
 - (D) Ensure pumps are completely off

110

DS2.111 **Of all the gauges on rig floor such as Driller's console or remote panel, is used to record Slow Circulating Rate Pressures (SCRPs)?**

Answer

- A**
- (A) Drill pipe pressure gauge on the choke panel
 - (B) Casing pressure gauge on the driller's panel
 - (C) Drill pipe pressure gauge on the driller's panel
 - (D) Casing pressure gauge on the choke panel

111

DS2.112 **On a hydraulic BOP control system (non-Mux), we apply air pressure then close the annular preventer using the rig floor remote BOP panel. If the open light goes out and the close light illuminates, what does this light change indicate?**

Answer

- D**
- (A) The manifold 3-position, 4-way valve on the accumulator unit has functioned
 - (B) The accumulator by-pass 2-position, 4-way valve on the accumulator unit has functioned
 - (C) The HCR 3-position, 4-way valve on the accumulator unit has functioned
 - (D) The annular 3-position, 4-way valve on the accumulator unit has functioned

112

DS2.113 **Manifold readback pressure are affected when which equipment is operated?**

Answer

- B**
- (A) Annular preventer
 - (B) Rams and failsafe valve
 - (C) Only HCR valve
 - (D) Hydraulic choke

113

DS2.114 **Operations shut-in well. Remote Panel gauge shows stablized SIDPP is 200 psi, Driller's panel gauge shows SIDPP is 325 psi and Standpipe gauge shows 400 psi. What is the next step?**

Answer

- B**
- (A) Use the average pressure of 500 psi to calculate the Kill Mud Weight (KMW)
 - (B) With these differences in gauge pressure, inform supervisor and investigate the difference in pressures first
 - (C) Always use the gauge that shows the lowest pressure
 - (D) Always use the gauge that was calibrated most recently

114

DS2.115 **During shallow flow, what action is simultaneously performed as the diverter is closed?**

Answer

- B**
- (A) Direct flow through the mud-gas separator
 - (B) Increase pump rate
 - (C) Maintain constant pump rate
 - (D) Decrease pump rate

115

DS2.116 **As Kill Weight Mud (KWM) is circulated up the annulus and Final Circulation Pressure (FCP) is being held constant, what is occurred to Bottomhole Pressure (BHP)?**

Answer

- A** (A) Stay the same
- (B) Increases
- (C) Decrease
- (D) Increases rapidly followed by decreasing slowly

116

DS2.117 **A Kick in horizontal well has been shut-in. The stabilized SIDPP = SICP. Operations begins circulating out the kick using the 1st Circulation of the Driller's Method. What can we expect?**

Answer

- C** (A) This is normal for any well when the Driller's Method is used.
- (B) ECD is much greater in the horizontal section
- (C) There is no loss of hydrostatic pressure until gas arrives at the inclined or vertical section of the well at which point the vertical height of the gas increase causing a reduction of hydrostatic pressure.
- (D) For OBM, gas remains in solution until the gas is at the surface and breaks out of solution. Therefore, surface pressure gauges are in error and should be replaced.

117

DS2.118 **If a leak were to occur in a rotary hose during a well control operation, what would be the next proper step to take?**

Answer

- A** (A) Pick-up off bottom, shut down the pumps, close choke and close standpipe valve.
- (B) Reduce pressure on the head and continue drilling.
- (C) Stop rotation, pick-up off bottom, close pipe rams and change out RCD insert.
- (D) Run to nearest exit

118

DS2.119 **What are the major advantages of using Managed Pressure Drilling (MPD) when drilling a well?**

Answer

- C** (A) MPD enables the well to be drilled within a tight window between pore pressure and fracture gradient.
- (B) MPD enables the well to be drilled faster as it aids in maximizing rate of penetration
- (C) All the above.
- (D) None of the above

119

DS2.120	Why is it important to follow a pressure ramp schedule when stopping the rig pumps for a connection on a well with a small gap between Pore Pressure and Fracture Gradient?
Answer	
C	(A) To keep bottom hole pressure constant while varying the pumps.
	(B) As the pumps slow down, the wellbore friction is reduced. The surface pressure needs to be increased to replace the friction loss to prevent an influx while reducing the pumps.
	(C) Both A and B
	(D) None of the above
120	
DS2.121	Which formation characteristics affect size and speed of a kick?
Answer	
B	(A) Influx size is not affected by permeability and differential pressure
	(B) High permeability formation with a high differential pressure.
	(C) Low permeability formation with a low differential pressure.
	(D) Pour throat size and viscosity of formation fluids
121	
DS2.122	Drillstring parted about 1,500 feet above BHA. With the well shut-in, Shut-in Casing Pressure (SICP) is rising at a rate of 50 psi every thirty minutes. What action can we take?
Answer	
C	(A) Shut well in and raise the MW.
	(B) Bullhead influx back into the formation
	(C) Stop pumping. Control the well using Volumetric method until the influx is above the washout due to parting. Then resume pumping operation.
	(D) Open the well and POH. Shut-in. Strip back to bottom with fishing assembly, engage fish and resume 1st Circulation of Driller's Method
122	
DS2.123	A well was drilled with 10.7 ppg mud. A 450-ft cement plug is set and tested. The mud above the plug is displaced by a 11.6 ppg brine. If the cement plug failed, what direction would the fluid move across the cement plug?
Answer	
D	(A) Pressure from below would cause fluid to move up
	(B) Fluid would not move across the plug as the pressure differential is zero.
	(C) Fluid would not move across the plug until its temperature has increased and allow for expansion
	(D) Pressure from above would cause fluid to move down.
123	

DS2.124 **What is the best practice for obtaining Slow Circulating Rate Pressure (SCRCP)?**

Answer

- C** (A) Obtain SCRCP from only mud pumps
- (B) Obtain SCRCP from low-volume, high-pressure pump(s).
- (C) Obtain SCRCP for a minimum of two mud pumps
- (D) None of the above

124

DS2.125 **What drilling practice should be followed when connection gas is noticed?**

Answer

- D** (A) Increase the mud viscosity.
- (B) Decrease WOB
- (C) Decrease in rotary torque
- (D) Control ROP to keep a minimum number of connection gas events in the hole at a time.

125

DS2.126 **Which of the gauges on the Driller's Console or Remote Choke panel are normally used to record the Slow Circulating Rate Pressure (SCRCP)?**

Answer

- B** (A) The Casing gauge on the Choke Control panel.
- (B) The DP gauge on the Choke Control panel.
- (C) The standpipe pressure.
- (D) Gauges are not used, just sensors.

126

DS2.127 **How do we measure Slow Circulating Rate Pressure (SCRCP)?**

Answer

- C** (A) Record pressure at the mud pump gauge
- (B) With the bit inside the Casing Shoe.
- (C) With a minimum of two mud pumps
- (D) Record pressures at the remote choke panel

127

DS2.128 **What can cause inaccuracy within a Slow Pump Rate Pressure (SCRP) reading?**

Answer

- B**
- (A) Rotating the drillstring slowly when taking SCR
 - (B) An extended period of non-circulating time
 - (C) Reciprocating the drillstring slowly while taking the SCR
 - (D) Improperly calibrated gauges

128

DS2.129 **A new Slow Circulating Rate Pressure (SCRP) should be taken after which of the following operations?**

Answer

- A**
- (A) When mud weight changes
 - (B) After each connection that we drill with a top drive.
 - (C) Before killing the well.
 - (D) Anytime a swab is changed within a pump

129

DS2.130 **What statement best describes "negative" pressure test?**

Answer

- A**
- (A) A test performed on a barrier where pressure on top of a barrier is reduced to a value lower than the pressure below the barrier
 - (B) A test performed on a barrier where pressure on top of a barrier is increased to a value higher than the pressure below the barrier
 - (C) A test performed on a barrier where pressure on top of a barrier is equal to the pressure below the barrier
 - (D) A test performed on a barrier where gas is injected downhole to test the integrity of the barrier

130

DS2.131 **Which statement is false for a non-return valve while tripping?**

Answer

- A**
- (A) Easier to stab if strong flow is encountered up the drillstring.
 - (B) While stripping to bottom, must not be run into the hole in closed position.
 - (C) Must be pumped open to read SIDPP.
 - (D) The non-return valve has a manual closing key which must be kept on the rig floor.

131

DS2.132 **While circulating out the influx, we do not see the casing pressure or pit volume going up. What could be happening?**

Answer

- A** (A) A liquid influx is in the well.
- (B) The casing pressure gauge needs to be fixed.
- (C) The PVT may need to be recalibrated.
- (D) A gaseous influx is in the well

132

DS2.133 **Why is it important to actively monitor the top of the well on the trip tank when killing a well?**

Answer

- D** (A) To verify that trip tank volumes are accurate
- (B) To check and verify of leaks from the HCR valve.
- (C) To verify the TDS system is not leaking
- (D) To check of leaks across the BOP

133

DS2.134 **Using the 1st Circulation of the Driller's method and when the gas influx reaches the surface, what happens to the mud pit level?**

Answer

- C** (A) Pit level would increase due to gas expansion then increase as gas exits the choke.
- (B) Pit level would remain the same as gas expansion does not occur in OBM.
- (C) Pit level would increase due to gas expansion then decrease as gas exits the choke.
- (D) None of the above

134

DS2.135 **Which statement best describes when the well should be monitored with flow rate and pit level equipment?**

Answer

- B** (A) During leak off tests
- (B) Any time there is a possibility or potential for a barrier to fail.
- (C) During connections
- (D) Any time there is circulation through the mud gas separator

135

DS2.136 **During flow check, the Driller notes a small amount of flow and small pit gain. Pumps are restarted with returns back to normal and no addition pit gain. What is occurring?**

Answer

- C**
- (A) The pump pressure is less than mud hydrostatic pressure
 - (B) The pump pressure is greater than mud hydrostatic pressure
 - (C) The annular pressure loss is creating an overbalance against formation pressure while circulating
 - (D) The pump pressure is equal to the mud hydrostatic pressure

136

DS2.137 **A wellbore barrier cannot be verified by which method?**

Answer

- A**
- (A) Leak off test
 - (B) Using checklist for testing plug
 - (C) Monitoring fluid level
 - (D) BOP Pressure Test

137

DS2.138 **Why would the Shut-in Drillpipe Pressure (SIDPP) and the Shut-in Casing Pressure (SICP) begin to rise after initial stabilization? Why is this happening?**

Answer

- D**
- (A) The mud pump might have been left on after shutting-in the well
 - (B) The driller has been reciprocating the string causing the pressure to increase.
 - (C) The FOSV has been left closed
 - (D) The influx is gas and is migrating.

138

DS2.139 **During a flow check after a drilling break, the trip tank gains 3 barrels in 5 minutes. What should be done next?**

Answer

- B**
- (A) Trip tank may be indicating a false gain in the pit. Raise the MW up by 0.1 ppg and resume drilling operations.
 - (B) Shut well in.
 - (C) Mix 10 sacks of barite in suction pit and resume drilling operations.
 - (D) See if we can increase the rate to 7 barrels in 5 minutes and shut-in on a 10 bbl kick

139

DS2.140 **If you do not have a reduction in pit volume while drilling for the last hour while making hole what possible action should be taken?**

Answer

- B** (A) Continue drilling, all is good.
- (B) Make a flow check and understand why pit volume is stable.
- (C) Tell Derrickhand to slow water down going to active pits.
- (D) Tell Derrickhand in increase water going into active pits

140

DS2.141 **How does using a slow pump rate aid in killing a well?**

Answer

- D** (A) The pump may not be rated for excessive pressure
- (B) To prevent bursting the casing
- (C) Allows rig crews time to mix KMW on-the-fly
- (D) To allow choke operator time to react during a well kill.

141

DS2.142 **Operations shut-in on kick. The mud engineer says we need 3100 sacks of barite to make Kill Weight Mud. Only 1000 sacks of barite are available. What is the best well control method to be used?**

Answer

- D** (A) Wait and Weight Method
- (B) Volumetric Method
- (C) Lube and Bleed Method
- (D) Driller's method

142

DS2.143 **What is the fastest action the Driller can take to shut-in the well if it is flowing while a non-shearable tool is across the BOP stack?**

Answer

- B** (A) Drop the string in the hole.
- (B) Make up safety valve in the string and close the annular
- (C) Start circulating to use ECD to the stop the kick
- (D) Run non-shearable tool into well and shut-in on Upper Pipe Ram.

143

DS2.144 **Which of these practices could cause a shallow hazard to flow into the well?**

Answer

- A** (A) Not filling the hole properly while tripping out
- (B) Pumping a cement slurry with short transition time.
- (C) Keeping the hole full with a continuous tank
- (D) Mixing fluid density on the fly, while pumping.

144

DS2.145 **After shut-in, stablized SIDPP = 100 psi and SICP = 250 psi. During preparations, he crew noticed the SICP has begun to increase. If the float is open, what would happen to the shut-in pressures?**

Answer

- D** (A) Only the annulus pressure will increase
- (B) Only drillpipe pressure will increase
- (C) Both drillpipe and annulus pressures will decrease
- (D) Both drillpipe and annulus pressures will increase

145

DS2.146 **The crew has successfully spaced out, shut down and shut-in. During the Well Control Prejob Safety Meeting (PJSM), which of the following responsibilities of the Derrickhand will be discussed?**

Answer

- A** (A) Monitor pressures on choke
- (B) Mud weight in all pits
- (C) Pit levels and measuring devices
- (D) Ensure flair remains ignited throughout well control circulation

146

DS2.147 **We have successfully spaced out, shut down and shut-in on a kick. Which of the following is an important responsibility for the Supervisor to complete?**

Answer

- D** (A) Check pit levels and line-ups
- (B) Mixing KMW
- (C) Call office and notify them of change in operations
- (D) Communicate plan with crew.

147

DS2.148 **Which is best way to mitigate top-hole section well control problems within gas-bearing formations?**

Answer

- A** (A) Control the rate of penetration
- (B) Maintain high ROP to finish the risky section as quickly as possible
- (C) Regularly pump light pump sweeps
- (D) Control rotary or top drive torque

148

DS2.149 **During well control operations, what are the most important reasons to follow good handover procedures?**

Answer

- B** (A) To allow blame to be assigned in the event of an incident during well kill operations
- (B) Ensure continuity in operations and pass on trend information to identify potential problems
- (C) It is a regulatory form which must be completed and turn into the local regulatory agency.
- (D) None of the above

149

DS2.150 **After the crew has responded to a kick, spaced out, shut down and shut-in, which statement best describes delegation of responsibilities to a crew team member?**

Answer

- C** (A) Calculate KMW needed to kill the well.
- (B) Communicating kill plan to crewmembers
- (C) Checking for leaks at the pumps, pipework or pit areas
- (D) Lowering the mud in the active pits to allow for controlled gas expansion

150

DS2.151 **When preparing for a tour change during a well kill operation, what are the best practices for conducting the handover?**

Answer

- D** (A) Each tour change individual should bring a hot cup of coffee to their relief.
- (B) Ensure the supervisor knows of the tour change and make it his/her responsibility to brief relief.
- (C) Communicate to persons on duty when its time for their relief to show up
- (D) All parties involved should stay on duty for a sufficient time to allow for thorough communication of current activities.

151

DS2.152 **We have been drilling a horizontal well and now have successfully shut-in on a kick. We do not have records of Slow Circulating Rate Pressure (SCRPP). What processes can we use to ensure we can find the correct Initial Circulating Pressure (ICP)?**

Answer

- A**
- (A) Follow the correct start-up procedures. Read the pressure on the drillpipe gauge, subtracting any overbalance/safety margin pressure
 - (B) Use the SIDPP as circulating pressure.
 - (C) Follow the ICP that was used in the previous well control operation.
 - (D) Since $ICP = SIDPP + SICP$, determine SICP and stage pumps up to speed holding SIDPP constant

152

DS2.153 **A written record of strokes pumped, surface drillpipe and casing pressures, choke position, active pit volume and other useful information for effective handover to a relief crew during a well kill operation is best defined as what?**

Answer

- A**
- (A) Kill log
 - (B) Information written in a tally book
 - (C) Kill sheet
 - (D) Circulating procedure to be followed

153

DS2.154 **If the well was full of hydrocarbons/water what term is used for the highest-pressure possible at the surface?**

Answer

- C**
- (A) Kick Volume
 - (B) Maximum Allowable Annular Pressure
 - (C) Maximum Anticipated Surface Pressure
 - (D) Maximum Allowable Casing Pressure

154

DS2.155 **After the first circulation of the Driller's Method, there is still influx in the annulus. What should be done to maintain constant BHP while circulating KWM to the bit?**

Answer

- C**
- (A) Maintain constant casing pressure
 - (B) Both A and B
 - (C) Follow a Weight and Wait step down chart or graph
 - (D) None of the above

155

DS2.156 **The Shut-in Casing Pressure (SICP) is affected by which parameter?**

Answer

- B**
- (A) Slow circulating rate
 - (B) Annulus capacity per foot
 - (C) Internal capacity per foot
 - (D) Temperature of the well

156

DS2.157 **What action is required if the drillpipe gauge on the remote panel reads 400 psi and the drillpipe pressure gauge on the standpipe manifold reads 625 psi?**

Answer

- D**
- (A) Start the kill using 400 psi because the remote choke will be used during kill operation
 - (B) Start the kill using 625 psi due to lag time.
 - (C) Use driller's console pressure gauges since they are electronic sensor reading and much more accurate than analog pressure gauges on the remote choke panel
 - (D) Investigate the reason for the difference

157

DS2.158 **What valve would be stabbed onto the drillstring if the well takes a kick?**

Answer

- A**
- (A) Full opening safety valve
 - (B) Non-return valve
 - (C) Both non-return valve and full opening safety valve.
 - (D) Top drive non-return valve

158

DS2.159 **The crew has successfully spaced out, shut down and shut-in on a kick. After stabilization, which gauge is used to calculate the formation pressure?**

Answer

- C**
- (A) The rig air gauge on the choke panel
 - (B) The Casing pressure gauge on the choke panel
 - (C) The Drill pipe pressure gauge on the choke panel
 - (D) The choke position indicator gauge on the choke panel

159

DS2.160 **Describe the steps in order for a proper soft shut-in procedure used if there is positive flow check while tripping out of the hole at 5500 ft?**

Answer

- D** (A) Stab FOSV, Open HCR valve and choke, Close BOP, Close choke, Close safety valve.
- (B) Stab FOSV, Close BOP, Open HCR valve and choke, Close choke, Close safety valve.
- (C) Break off the Top Driver/Kelly. Install and close IBOP, tighten and torque same. Open HCR and close BOP, record pressures
- (D) Stab a full opening safety valve, close the safety valve, Open the HCR valve, close BOP and Close choke.

160

DS2.161 **What could be occurring if there is no increase in pit level or casing pressure while circulating out a kick?**

Answer

- C** (A) Pumps are failing
- (B) A gas influx is in the well
- (C) A liquid influx is in the well
- (D) BOPs are failing

161

DS2.162 **The crew has spaced out, shut down and shut-in the well with a 25 bbls kick. Stabilized SIDPP is 300 psi and SICP is 600 psi. If the crew had reached faster and shut-in on a kick of 10 bbl, how would shut-in pressures be affected?**

Answer

- D** (A) SIDPP is higher
- (B) SIDPP is lower
- (C) SICP is higher
- (D) SICP is lower

162

DS2.163 **While MPD drilling at 12,050' MD/7,800' TVD, the well is full of 11.0 ppg mud, formation pressure is 4,725 psi, surface back pressure is 200 psi and APL is 275 psi. The well is?**

Answer

- C** (A) Overbalanced
- (B) Hydrostatically Underbalanced
- (C) Both A and B
- (D) None of the above

163

DS2.164 **Without a float in the drillstring, what happens to shut-in pressures as gas migrates?**

Answer

- C** (A) Shut-in casing pressure may be bled off to zero.
- (B) Shut-in pressures will remain constant
- (C) Both drillpipe and annulus pressures will increase
- (D) Both drillpipe and annulus pressures will decrease

164

DS2.165 **While tripping into the hole at 6,000 feet, a flow check indicates the well is flowing. The current shut-in procedure calls for the soft shut-in procedure due to weak zones. What is the correct procedure to shut-in?**

Answer

- B** (A) Stab a full opening valve, Open the HCR valve, Close BOP, Close the safety valve, Close choke.
- (B) Stab a full opening valve, close the safety valve, open the HCR valve, Close BOP, Close choke.
- (C) Stab an inside BOP, Open the HCR valve, Close BOP, Close choke.
- (D) Stab a full opening safety valve, open HCR, close the FOSV, Close choke and Close BOP.

165

DS2.166 **Why is it important to recognize a kick as early as possible?**

Answer

- B** (A) To allow the Volumetric method to be used
- (B) To reduce the risk of formation breakdown during the kill.
- (C) To reduce KMW required to kill the well
- (D) So that the rig crews have more time to shut-in the well.

166

DS2.167 **While tripping into the hole at 6,000 feet, a flow check shows there's flow. The rig's procedure is to use the soft shut-in method. What steps should be followed to shut-in the well properly?**

Answer

- C** (A) Open HCR valve and choke, close BOP, stab full opening safety valve, close safety valve, close choke.
- (B) Stab an inside blowout preventer (open), open HCR valve, close BOP, close choke.
- (C) Stab a full opening safety valve (open), close the safety valve, open HCR valve, close BOP, close choke.
- (D) None of the above

167

DS2.168 **What are the steps to shut-in when a well flows during tripping using a hard shut-in procedure?**

Answer

- C** (A) Open HCR valve and choke, close the BOP, stab open FOSV, Close choke and record pressures.
- (B) Close the BOP, stable the FOSV, Close the FOSV, open choke and record pressure.
- (C) Stab an opened FOSV, close the FOSV, Close the BOP, Open HCR valve and record pressure.
- (D) Break off the Top Driver/Kelly. Install and close IBOP, tighten and torque same. Open HCR and close BOP, record pressures

168

DS2.169 **The crew has successfully spaced out, shut down and shut-in on a kick. After stabilization, the Shut-in Drillpipe Pressure (SIDPP) reads 200 psi while the Shut-in Casing Pressure (SICP) reads 0 psi. What is the probable cause of SICP reading 0 psi?**

Answer

- A** (A) Hole packed off around the BHA
- (B) Hole in the Drillpipe
- (C) Hole in the BHA
- (D) Leak in the annular preventer

169

DS2.170 **After the crew spaces out, shuts down and shut-in, Shut-in Drillpipe Pressure (SIDPP) and Shut-in Casing Pressure (SICP) stabilization time is based upon which of the following?**

Answer

- B** (A) Porosity
- (B) Permeability
- (C) Gas migration
- (D) Influx density

170

DS2.171 **The crew has spaced out, shut down and successfully shut-in on a kick. Which statement best describes the derrickhand's responsibility?**

Answer

- D** (A) Check current mud density
- (B) Check current mud viscosity
- (C) Check for leaks at the pumps or surface lines
- (D) Monitor and record pressures at the choke

171

DS2.172 **Which of the following statements does NOT describe a warning sign of increasing formation pressures downhole?**

Answer

- A** (A) Increase in the shale density
- (B) Gradual increase in ROP
- (C) Decrease in the shale density
- (D) Increase in the connection gas

172

DS2.173 **What is the most important reason for detecting a kick as soon as possible?**

Answer

- A** (A) To reduce risk of formation breakdown during the kill operation
- (B) To prevent gas expansion as it is circulated up the wellbore
- (C) To prevent gas migration
- (D) To prevent oil expansion as it is circulated up the wellbore

173

DS2.174 **What statement best describes "abnormal pressure"?**

Answer

- B** (A) Formation pressures that differ from the hydrostatic pressure created by a full column of fresh water
- (B) Formation pressures that differ from the hydrostatic pressure created by a full column of formation water
- (C) Formation pressures that equal normal formation pore pressure
- (D) Pressure that differs from the hydrostatic pressure caused by a full column of formation gas

174

DS2.175 **What drilling data is affected, as we drill a transition zone into an abnormally high formation pressure?**

Answer

- D** (A) Increase in WOB to maintain constant ROP
- (B) Increase in Torque
- (C) Gradual increase in RPM
- (D) Gradual increase in ROP

175

DS2.176 **After circulating bottoms-up, which of the following is NOT a warning sign of increased formation pressure?**

Answer

- B**
- (A) Decreasing Shale Density
 - (B) Increasing Shale Density
 - (C) Increasing Cuttings Volume
 - (D) Increasing Mud Temperature

176

DS2.177 **How does sudden abnormal pressure impact primary well control?**

Answer

- B**
- (A) Reduces the ability to use Annular Preventer as primary BOP to close in on well.
 - (B) Decreases the overbalance
 - (C) Allows Driller to increase tripping speed per stand.
 - (D) Increases the overbalance

177

DS2.178 **How can connection gas be limited while drilling?**

Answer

- D**
- (A) Decrease the mud viscosity.
 - (B) Increase the mud viscosity.
 - (C) Increase mud pH
 - (D) Control ROP to keep a minimum number of connection gas events in the hole at a time.

178

DS2.179 **If a sudden increase in gas levels are noted, what is the Driller's safest next step?**

Answer

- D**
- (A) Stop drilling, circulate bottoms up, monitor & record gas levels, inform your supervisor
 - (B) Begin to circulate weighted sweeps periodically
 - (C) Continue to monitor and record the gas level
 - (D) Check for flow and call the supervisor

179

DS2.180 **During drilling of a wild cat exploratory well, what is the best method for determining formation strength for these wells with little or no downhole pressure information?**

Answer

- C**
- (A) Equivalent Mud Weight Test (EMW)
 - (B) Formation Integrity Test (FIT)
 - (C) Leak Off Test (LOT)
 - (D) Casing Test

180

DS2.181 **Flow check is small trickle. After starting pumps, the Driller notes small losses and believes ballooning is occurring. After shutting-in, the decision was made to bleed 5 barrels of of mud. What is the next proper step?**

Answer

- A**
- (A) Circulate through a fully open choke with the BOP closed and monitor active pit level closely
 - (B) Circulate bottoms up until you see gas at surface then shut in the well and circulate through the fully open choke choke
 - (C) Circulate bottoms up at the maximum pump speed
 - (D) Circulate bottoms up through the kill line

181

DS2.182 **How will oil-based mud density be affected by increasing downhole pressure?**

Answer

- B**
- (A) Decreases mud density
 - (B) Increases mud density
 - (C) Mud density remains unchanged
 - (D) Oil based muds have no density

182

DS2.183 **After noting an increase in flow, the crew spaced out, shut down and successfully shut-in the well. After pressure stabilization, what is the Driller's safest next step?**

Answer

- C**
- (A) Calculate the correct kill mud density
 - (B) Monitor and record surface pressures
 - (C) Check and verify the well is secure (no leaks)
 - (D) Monitor and record the pit gain

183

DS2.184 **During a well control circulation through a choke, what occurs as a gas influx is circulated out while keeping bottom hole pressure (BHP) constant?**

Answer

- B**
- (A) The volume of gas will decrease
 - (B) The volume of gas will increase
 - (C) The volume of gas will stay the same
 - (D) The gas will disperse into the drilling mud

184

DS2.185 **During a connection, flowback time has been excessive. After bumping the float, the stabilized SIDPP and SICP is 70 psi. We suspect ballooning is occurring. What is the next proper step?**

Answer

- D**
- (A) Open the BOP and drill ahead
 - (B) Open the choke and perform a flow-check. If negative, circulate bottoms-up.
 - (C) Circulate at the maximum pump rate through the mud gas separator
 - (D) Open the choke and perform a flow-check. If negative, circulate bottoms-up through the choke

185

DS2.186 **Which statement best describes the main indicator that the influx is below the bit and potentially requires stripping back to bottom?**

Answer

- B**
- (A) The SIDPP is greater than SICP
 - (B) The SIDPP is equal to SICP
 - (C) The SIDPP is lower than SICP
 - (D) None of the above

186

DS2.187 **Shut-in on kick. Stabilized SIDPP is 200 psi and SICP is 300 psi. Crews suspect pressures will start rising due to gas migration. If the SIDPP remains at 200 psi, what happens to BHP?**

Answer

- C**
- (A) Increase
 - (B) Decrease
 - (C) Stay the Same
 - (D) None of the above

187

DS2.188 **During W&W method, KMW has been circulated to the bit and the pumps were staged down. Stabilized SIDPP is 150 psi and SICP is 600 psi. If the KMW is correct density, what is the next step?**

Answer

- B**
- (A) Bleed off approximately 50 psi to check for trapped pressure
 - (B) Bleed the SIDPP to 0 psi
 - (C) Bleed off approximately 100 psi to check for trapped pressure
 - (D) Re-check the calculations of surface to bit strokes and kill mud density

188

DS2.189 **Swabbed well. After spacing out, installing and closing Full Opening Safety Valve (FOSV), the Annular Preventer is closed using the Hard Shut-in Procedure. What is the safest next step?**

Answer

- B**
- (A) Circulate the kick out using the Driller's method while 30 stands off bottom
 - (B) Strip to bottom and start circulation using first circulation of Driller's method
 - (C) Use volumetric procedure until the entire influx has migrated to the bit and start circulation using the Driller's Method
 - (D) Use Lube and Bleed Method to remove gas from the well

189

DS2.190 **When stripping back to bottom with a migrating kick, what is the proper procedure to be followed in order to maintain constant Bottomhole Pressure (BHP)?**

Answer

- C**
- (A) Bleed off the drill pipe steel displacement while tripping each stand
 - (B) Bleed off the volume of mud equal to the open-end displacement of the drill pipe tripped into the well
 - (C) Bleed off the volume of mud equal to the closed end displacement of the drill pipe stripped into the well
 - (D) Do not bleed mud while stripping in with drill pipe

190

DS2.191 **Which statement best describes a kill plan?**

Answer

- A**
- (A) A document stating duties, procedures, and specific lines of communication and safety requirements for the entire kill operation
 - (B) A kill sheet completed by the supervisor
 - (C) A document that all rig personnel must sign
 - (D) A document stating the correct kill mud density used to kill the well

191

DS2.192 **If the u-tube (well) has lost integrity due to losses, how can we identify this condition?**

Answer

- A** (A) Manipulations of the choke result in unpredictable fluctuations of the casing and drillpipe pressures.
- (B) Decrease in drillpipe pressure followed by a increase in casing pressure.
- (C) Manipulation of the choke results in a change in casing pressure and a predictable pressure response on the drillpipe pressure gauge.
- (D) None of the above

192

Drilling Supervisor Practice Test Day 2

DST2.1 **During the 1st Circulation of the Driller's Method, ICP is at 500 psi. Choke pressure showing 400 psi for the past 30 minutes. ICP dramatically increases to 800 psi , Choke Pressure remains at 400 psi. What is the problem?**

- B**
- (A) Washout in the drill pipe
 - (B) One bit nozzle has plugged
 - (C) Washout in the choke
 - (D) The choke has partially plugged

1

DST2.2 **If a gas influx is taken while drilling the horizontal section of a well, what conditions exist when comparing Shut-in Drillpipe Pressure (SIDPP) to Shut-in Casing Pressure (SICP)?**

Answer

- C**
- (A) $SIDPP < SICP$
 - (B) $SIDPP > SICP$
 - (C) $SIDPP = SICP$
 - (D) $SIDPP = SICP = 0 \text{ psi}$

2

DST2.3 **SCR are recorded every 500 feet. Due to a crew change, 1000 feet were drilled without recording a SCRP. Kick occurred and shut-in. During the 1st Circulation of Driller's Method what would the actual ICP be when compared to the calculated ICP?**

Answer

- D**
- (A) Lower
 - (B) The same
 - (C) $ICP = 500 \text{ psi over SIDPP}$
 - (D) Higher

3

DST2.4 **How will Bottomhole Pressure (BHP) be affected, if during the well kill operation, the pump rate is reduced while holding drillpipe pressure constant?**

Answer

- B** (A) BHP will decrease
- (B) BHP will increase
- (C) BHP will remain constant
- (D) BHP will decrease rapidly then increase slowly

4

DST2.5 **In a horizontal well, as a gas kick is being circulated within the horizontal section of well. What happens to the active mud pit level at this time, provided a correct kill procedure is being followed?**

Answer

- C** (A) Increase as the kill mud is displaced into the horizontal section of the hole
- (B) Decrease as the kill mud is displaced into the horizontal section of the hole
- (C) Remain approximately constant
- (D) None of the above

5

DST2.6 **During a horizontal well kill, what happens to the surface casing pressure as the gas influx moves from the horizontal to the vertical section of the well?**

Answer

- B** (A) The casing pressure will decrease
- (B) The casing pressure will increase
- (C) The casing pressure will stay the same
- (D) CP will decrease, then increase

6

DST2.7 **We are killing a well using the Driller's method. The pressure in the MGS is increasing. The Drillpipe Pressure (DP) is 680 psi at 40 spm. If we slow down pump but keep the DP at 680 psi, what will happen to the BHP?**

Answer

- A** (A) Bottomhole pressure will increase.
- (B) Bottomhole pressure will decrease.
- (C) Bottomhole pressure remains the same.
- (D) All the above

7

DST2.8 **We are drilling a vertical well when kick is taken and shut-in. The stabilized SIDPP = 350 psi and the SICP = 450 psi. Why is there a difference in the two readings?**

Answer

- B**
- (A) Because the gauges are out of calibration and should be replaced.
 - (B) Because the influx is in the annulus and has a lower density than the mud.
 - (C) The DP has a smaller ID than the hole ID, therefore the DP should have smaller readings.
 - (D) Because only digital sensors can read stabilized pressures

8

DST2.9 **Tapered string of 3,500' of 6-5/8" DP and 3500' of 5" DP. Strokes to bit is 1600 strokes. Pressure schedule ICP = 500 psi, FCP = 225 psi. Step-down is 18 psi per 100 strokes. How will this impact BHP after the KMW reaches the top of the 5" drillpipe?**

Answer

- D**
- (A) Bottomhole pressure will be too high
 - (B) Bottomhole pressure will be too low
 - (C) Bottomhole pressure will not be affected
 - (D) Bottomhole pressure will be correct

9

DST2.10 **MW = 12.2 ppg, bit size = 9-7/8", OH Cap = 0.0947 bbls/ft. 5" OD DP cap = 0.01776 bbls/ft, Metal Disp = 0.0075 bbls/ft, 10-3/4" csg at 6,000 ft. What data sheet changes are needed after drilling from 6,500 ft to 6,605 ft in last hour?**

Answer

- B**
- (A) Change SCR
 - (B) Change in surface-to-bit strokes
 - (C) Change in pump rate
 - (D) Change in MAASP

10

DST2.11 **We have completed the 1st Circulation of the Driller's Method, but annulus is not clear of the influx. We have decided to start the 2nd Circulation. How do we maintain the correct BHP when circulating the Kill Weight to the bit?**

Answer

- C**
- (A) All casing pressure to increase until KMW reaches the bit
 - (B) Hold drillpipe pressure constant until KMW reaches the bit.
 - (C) Follow a wait and weight step-down chart or graph
 - (D) Vary pump rates without holding casing pressure constant

11

DST2.12 **What is the correct I.A.D.C. definition of the "closing ratio" of a ram preventer?**

Answer

- B** (A) The amount of gallons that are needed to close the ram preventer.
- (B) Ratio of the packer area against the piston rod area
- (C) The amount of volume needed in reservoir.
- (D) Ratio of the Weep Hole to the Rod Size

12

DST2.13 **Why is a Vacuum Degasser not used in place of a Mud Gas Separator (MGS)?**

Answer

- A** (A) Because vacuum degasser has design limitations.
- (B) Because MGS has design limitations.
- (C) Because MGS is always poor at separating gas from liquid.
- (D) Because vacuum degasser is farther from the flowline

13

DST2.14 **A Pressure While Drilling (PWD) tool within the Bottom Hole Assembly (BHA) can show signs of an influx while drilling. What information from the PWD tool would show that there is an influx in the well?**

Answer

- D** (A) Pit level decreases
- (B) An increase in Equivalent Circulating Density (ECD)
- (C) Mud salinity
- (D) A reduction in Equivalent Circulating Density (ECD)

14

DST2.15 **The crew closed a ram preventer using the remote BOP panel. The open light turns off and the close light turns on, but accumulator gauge does not move. What went wrong?**

Answer

- C** (A) The charging pumps did not operate correctly.
- (B) Three-position / 4-way valve did not operate correctly.
- (C) Possible blockage in the line between the accumulator unit and the BOP stack.
- (D) Air pressure does not read between 100 and 120 psi

15

DST2.16	Why is it important to control the flow rate when circulating out a kick through the Mud Gas Separator (MGS)?
Answer	
D	(A) The only dimension to consider when circulating out an influx is the height of the separator.
	(B) A separator's flow rate is unlimited and therefore, "blow-through" is not a problem under any condition
	(C) Because pressure within the choke manifold, will not allow personnel to use the panic line if blow-through occurs.
	(D) None of the above
16	
DST2.17	A Pressure While Drilling (PWD) Tool in the Bottom Hole Assembly (BHA) can show that the Equivalent Circulating Density (ECD) is reducing during drilling. What could cause the ECD to decrease while drilling?
Answer	
B	(A) A change in Rate of Penetration (ROP)
	(B) A reduction of hydrostatic pressure with formation fluids contaminating the mud in the annulus.
	(C) A change in Weight on Bit (WOB)
	(D) A change in Rotary Torque
17	
DST2.18	Which two gauges on the remote BOP panel would we expect to show a pressure drop when the annular preventer is closing?
Answer	
C	(A) Accumulator pressure and manifold pressure
	(B) Annular pressure and manifold pressure.
	(C) Accumulator pressure and annular pressure
	(D) Annular pressure
18	
DST2.19	What are some possible problems that can happen with a Mud Gas Separator (MGS)?
Answer	
C	(A) Vent line restrictions (small diameter/excessive curves) causes too much back pressure on the MGS and risks a blow-through.
	(B) The mud leg (dip tube) may become plugged with solids during well control operations.
	(C) Both A and B
	(D) None of the above
19	

DST2.20 **How does the Annular Pressure Loss (APL) impact Bottom Hole Pressure (BHP)?**

Answer

- C** (A) BHP would equalize to the hydrostatic pressure.
- (B) It will cause the BHP to be less than the hydrostatic pressure
- (C) It will cause the BHP to be greater than the hydrostatic pressure.
- (D) Annular pressure loss equals drillpipe pressure loss and therefore does not impact BHP

20

DST2.21 **When testing a Surface BOP stack with a test plug, why should the side outlet valves below the plug be left open?**

Answer

- C** (A) Otherwise, reverse circulation will be needed to release the test plug.
- (B) To prevent pressure lock
- (C) Because of potential damage to the wellhead/casing/open hole.
- (D) To meet regulations as specified in the Drilling permit

21

DST2.22 **What usually happens to the pit level when the Mud Engineer starts the vacuum degasser and desilter?**

Answer

- C** (A) Pit level will stay the same
- (B) Pit level will increase
- (C) Pit level will decrease
- (D) None of the above

22

DST2.23 **Which of the following statements about the Annular Preventer is correct?**

Answer

- C** (A) Will not allow tool joints to pass through
- (B) It is not designed to seal on a square or hexagonal kelly.
- (C) It can provide secondary well control
- (D) The annular preventer holds pressure due to metal inserts within the element

23

DST2.24 **Which parts of the BOP stack get their operating fluid from the Ram manifold?**

Answer

- A** (A) Rams and HCRs
- (B) Pipe Rams only
- (C) Annular Preventer
- (D) Both A and C

24

DST2.25 **What are the blind rams designed to do?**

Answer

- A** (A) To seal off open hole
- (B) As a back-up to the Annular
- (C) Cut the pipe
- (D) Hold pressure from both ways

25

DST2.26 **We operated a function from the remote panel. The open light turns off, and the close light comes on, but the pressure gauges did not drop. What could be the likely cause of this problem?**

Answer

- D** (A) Accumulator is not plugged it
- (B) There is a leak in the close line
- (C) There are no lights on a remote BOP panel
- (D) Close line is blocked.

26

DST2.27 **Which of the statements about fixed bore ram type BOP's is correct?**

Answer

- D** (A) Drawdown tests should be performed daily
- (B) They are not recommended on BLM locations.
- (C) Fixed bore rams can close on any type of wireline
- (D) Fixed bore ram type BOPs should be equipped with a mechanical locking system.

27

DST2.28 **We are displacing the drillstring with Kill Weight Mud (KWM). Which of the following sections of the drillstring will have the fastest displacement per foot?**

Answer

- B**
- (A) 4" with an ID of 3.340"
 - (B) 3-1/2" with an ID of 2.764"
 - (C) 5" with a capacity factor of 0.01776 bbl/ft
 - (D) 6-5/8" with an ID of 6"

28

DST2.29 **What can be a problem when we shut-in on larger diameter casing without adjusting annular preventer?**

Answer

- C**
- (A) The fast-closing speed of the annular preventer compared to the casing ram causes the formation to fracture.
 - (B) It may take longer than expected for the annular preventer to close
 - (C) Annular closing pressure is too high and damages the casing
 - (D) The annular preventer will not be able to affect a seal on larger diameter pipe

29

DST2.30 **When a gas influx begins to be circulated through the choke manifold, what major problem may arise?**

Answer

- C**
- (A) The increased volume of gas at the MGS will increase the BHP
 - (B) The gas will change to a liquid and increase pit level.
 - (C) The increased volume can overload the MGS
 - (D) Choke can easily plug up

30

DST2.31 **During wireline operations, which of the following statements is true about detecting a kick?**

Answer

- C**
- (A) There is no need to monitor the well during wireline operations because the well is static.
 - (B) The well should be lined up to active pits due to the large displacement volumes of the wireline tools.
 - (C) The well should be put on the trip tank to monitor displacement as wireline is run in/out of the hole.
 - (D) The casing annulus should be put on the trip tank to monitor for leakage while running wireline.

31

DST2.32 **During drilling operations, what condition could affect our flow gauge?**

Answer

- C** (A) Faulty pit level indicator
- (B) Depth of the water
- (C) Cuttings build-up in the flowline
- (D) Near freezing temperatures

32

DST2.33 **During well control operations, an Annular Preventer element begins to leak fluid routed to the trip tank. What is the proper corrective action to mitigate this risk?**

Answer

- A** (A) Increase closing pressure to maximum value
- (B) Monitor mud level above the BOPs on trip tank.
- (C) Monitor active pipe levels for mud level increases
- (D) Add motor oil to the top of the Annular preventer. Strip pipe into the well until a tooljoint enters the Annular preventer and plugs the leak

33

DST2.34 **During well control operations while circulating the Wait and Weight method, one or more drilling nozzles become plugged. What is the proper correction action to mitigate this risk?**

Answer

- B** (A) Add the increase in DP to each valve on the drillpipe pressure schedule and continue circulating.
- (B) Note the increase in drillpipe pressure. Recalculate SCR pressure, FCP and the drillpipe pressure schedule.
- (C) Open the choke to hold drillpipe pressure constant.
- (D) Open the choke and hold casing pressure constant

34

DST2.35 **What equipment is specifically designed to aid in proper shut-in during casing installation operations?**

Answer

- D** (A) Full Open Safety Valve (FOSV)
- (B) Inside Blowout Preventer (IBOP)
- (C) Cementing head with FOSV
- (D) A suitable crossover (Swage)

35

DST2.36 **Concerning kick indicators for most drilling rigs, which indicator normally detects problems first?**

Answer

- B** (A) Gain in pit volume.
- (B) Flow rate increase.
- (C) Increase in background gas.
- (D) Mud Gas Separator shakes

36

DST2.37 **When the annular preventer is being closed, which gauge(s) on the remote panel would show a drop in pressure?**

Answer

- C** (A) Air Pressure
- (B) Manifold pressure.
- (C) Accumulator gauge and Annular pressure gauge.
- (D) Choke position indicator

37

DST2.38 **What is the purpose of a choke for well control operations?**

Answer

- A** (A) Creates a back-pressure when killing the well.
- (B) Keep fluids from flowing out of the drillstring.
- (C) Holds premixed fluids
- (D) Route fluids between MGS and Choke Manifold

38

DST2.39 **After tripping out of the hole, a visual inspection of the BOPs demonstrated the weep-hole is leaking wellbore fluid. What is the safest next-step?**

Answer

- A** (A) Piston rod mud seal is leaking and should be replaced then retested
- (B) Place Teflon tape over piston rod mud seal and retest the seal
- (C) Replace the ram blocks.
- (D) Weld up the weep-hole and continue tripping out of the hole

39

DST2.40 **What is the precharge pressure on a 3000 psi BOP accumulator, if the minimum operating pressure is designed to be 1200 psi?**

Answer

- A** (A) 1000 psi
- (B) 1200 psi
- (C) 3000 psi
- (D) 850 psi

40

DST2.41 **When does "blow through" occur while circulating out a kick and using the mud gas separator?**

Answer

- A** (A) When the vent line back pressure exceeds the hydrostatic pressure of the mud leg.
- (B) When the mud leg contains water, the hydrostatic pressure has been reduced allowing gas to enter the vent line without restrictions.
- (C) Both A and B
- (D) None of the above

41

DST2.42 **The 1st Circ of the Driller's Method is complete, well shut-in. The initial stabilized SIDPP = 300 psi and initial SICP = 500 psi. After the 1st Circ, current stabilized SIDPP = 300 psi and current SICP = 400 psi. What's happening?**

Answer

- B** (A) Nothin, Proceed to 2nd Circulation and kill the well.
- (B) There is still a kick in the well.
- (C) There is still a kick in the well, providing greater hydrostatic pressure on the annulus side of the well when compared to the HP of the drillpipe side
- (D) Losing mud to permeable formation

42

DST2.43 **After the 1st Circulation of the Driller's Method, if the well is properly staged down and shut-in with no trapped pressure, overbalance or safety factor, what should the Shut-in Casing Pressure (SICP) be?**

Answer

- A** (A) Equal to the shut in drill pipe pressure (SIDPP)
- (B) Higher than the shut in drill pipe pressure (SIDPP)
- (C) Lower than the shut in drill pipe pressure (SIDPP)
- (D) Zero

43

DST2.44 **Well shut-in on kick. Mud Engineer shows 2000 sacks of barite needed for KMW. Only 1300 sacks on site and requires 24 hours for resupply. Which well control method should be used?**

Answer

- B**
- (A) Volumetric Method
 - (B) Driller's Method
 - (C) Begin to Lubricate and Bleed
 - (D) Bullheading Method

44

DST2.45 **Which statement demonstrated the correct procedure when using the Driller's Method for well control?**

Answer

- A**
- (A) While circulating at the kill rate, hold the surface pressure constant on the side of the U-tube that has the constant fluid density
 - (B) Anytime you start, stop, or change pump rate, hold DP pressure constant
 - (C) Only requires one circulation
 - (D) Barite is required to begin the procedure

45

DST2.46 **Initial SIDPP = 200 psi and SICP = 400 psi. After completing 1st Circulation of Driller's Method, well is shut-in. Current stabilized SIDPP = 200 psi and SICP = 350 psi. What is happening downhole?**

Answer

- B**
- (A) Nothing, the kill mud has not been pumped yet, so the well is not killed
 - (B) There is still a kick in the well
 - (C) There is trapped pressure that can be either kept as a safety margin or bled off upon startup
 - (D) BHP is increasing due to gas migration

46

DST2.47 **During the 2nd Circulation of the Driller's Method, KMW has been pumped down the drillpipe and up the annulus. The DP pressure increases dramatically and does not respond to opening the choke. In turn, the CP decreases dramatically. What is occurring?**

Answer

- D**
- (A) One bit nozzle has plugged
 - (B) One bit nozzle has washed out
 - (C) The wellbore is ballooning
 - (D) The annulus has packed off with cuttings

47

DST2.48 **After drilling ahead at 30 ft/hr a trip is made due to a Pressure While Drilling (PWD) failure. The bit is green and is re-run. After returning to bottom and running the same WOB, the ROP is 55 ft/hr. What if any actions should you take?**

Answer

- B**
- (A) Shut-in the well and circulate bottoms up.
 - (B) Carry out a flow check and consider circulating bottoms up through choke line.
 - (C) Both A and B
 - (D) All the above

48

DST2.49 **The drill string is being displaced with Kill Weight Mud (KMW) while holding the casing pressure constant during the 2nd Circulation of the Driller's Method, why is it important that the annulus is clear of gas?**

Answer

- D**
- (A) Gas in the annulus will remain the same size, annulus hydrostatic pressure will be reduced causing the well to go underbalanced.
 - (B) Gas in the annulus will decrease in size, therefore casing pressure will cause well to become overbalanced.
 - (C) Gas will remain in solution and does not affect CP
 - (D) Gas in the annulus will expand, therefore casing pressure will cause well to go underbalanced.

49

DST2.50 **When using the Driller's Method to control a saltwater influx, how does the Casing Pressure (CP) change as the influx rises in the annulus? Assume the mud in the well is heavier than the incoming saltwater?**

Answer

- A**
- (A) Casing pressure will only change due to changes in annular size.
 - (B) As influx expands, the casing pressure will expand.
 - (C) Casing pressure will remain the same.
 - (D) As the influx contracts, the casing pressure will decrease.

50

DST2.51 **During the 2nd Circulation of the Driller's Method, Kill Weight Mud (KWM) is pumped into the drillpipe while keeping the Casing Pressure (CP) constant. Why is it crucial that all gas be removed annulus before the 2nd Circulation begins?**

Answer

- C**
- (A) Gas in the annulus will expand, therefore a constant casing pressure will cause the well to go overbalanced.
 - (B) Gas in the annulus will increase the bottom hole pressure during the 2nd circulation.
 - (C) Gas in the annulus will expand, therefore a constant casing pressure will cause the well to go underbalanced.
 - (D) Gas in the annulus will remain constant, therefore casing pressure will decrease

51

DST2.52 **What is the correct procedure used during the 2nd circulation of the Driller's Method, when there is no influx remaining in the well, to maintain a constant bottom hole pressure (BHP)?**

Answer

- C** (A) Hold CP constant until KMW reaches bit hold ICP constant until KMW reaches surface.
- (B) Follow the step-down chart, once KMW is at the bit, hold ICP constant until KMW reaches surface.
- (C) Maintain the drillpipe pressure constant once kill fluid is at the bit.
- (D) Maintain drillpipe pressure during entire circulation

52

DST2.53 **We have finished the first part of the Driller's method and think the annulus still has some kick. We are going to do a second circulation. How can we keep the bottom hole pressure the same while pumping Kill Mud Weight (KMW) down to the bit?**

Answer

- D** (A) Hold DP pressure constant by manipulating the choke until the KMW reaches the bit.
- (B) Vary pumps holding DP pressure constant to compensate for heavier density KMW being pumped to the bit.
- (C) Follow the Driller's Method step-down chart
- (D) Follow the Wait and Weight step down chart.

53

DST2.54 **After 1st Circulation of Driller's method is completed, well is shut-in. Initial SIDPP = 250 psi, Initial SICP = 320 psi. After circulation, current SIDPP = 250 psi and current SICP = 300 psi. What is the current condition of the well?**

Answer

- C** (A) The well is balanced.
- (B) Well is experiencing parital losses while circulating.
- (C) There is still a kick in the well, providing less hydrostatic on the annulus side of the well.
- (D) The well is overbalanced

54

DST2.55 **After the 1st Circulation of the Driller's Method, the well is shut-in. Initial SIDPP =250 psi and Initial SICP = 320 psi. After the 1st Circulation, current SIDPP = 250 psi and SICP = 300 psi. What is the current status of the well?**

Answer

- B** (A) There is trapped pressure on the annulus side which can be bled off once pumps are started.
- (B) There is still a kick in the well, providing less hydrostatic on the annulus side of the well.
- (C) Well is experiencing parital losses while circulating.
- (D) Well is experiencing total losses while circulating

55

- DST2.56 **After the 1st Circulation of the Driller's Method, the well is shut-in. Initial SIDPP = 250 psi and Initial SICP = 350 psi. After the 1st Circulation, current SIDPP = 300 psi and SICP = 400 psi. What is the current status of the well?**
- Answer
- A**
- (A) There is trapped pressure which can be kept as a safety margin or bled off upon pump start-up.
 - (B) There is another kick in the hole.
 - (C) The well is dead
 - (D) The annular preventer is leaking. Close the Upper Pipe Ram
- 56
- DST2.57 **During the 1st Circulation of the Driller's Method, a gas influx is being circulated up the annulus and is above the casing shoe. We are keeping the drillpipe pressure constant. What happens to the pressure at the casing shoe?**
- Answer
- B**
- (A) Casing shoe pressure increases
 - (B) Casing shoe pressure stays the same
 - (C) Casing shoe pressure decreases
 - (D) The casing shoe pressure cannot be measured or estimated. Therefore, casing shoe pressure is unknown and can be ignored.
- 57
- DST2.58 **After 1st Circulation of the Driller's Method, annulus is not clear of the influx. We're instructed to start the 2nd Circulation. How do we maintain the correct BHP when circulating the KMW to the bit?**
- Answer
- B**
- (A) Go ahead and start the second circulation of the Driller's Method
 - (B) Follow a Wait and Weight step-down chart or graph
 - (C) Maintain drillpipe pressure constant as kill mud circulates down the drillpipe
 - (D) After staging up to speed, decrease the ICP by 50 psi for every 200 strokes until KMW reaches the bit. Read, record, and maintain FCP.
- 58
- DST2.59 **During the 1st Circulation of the Driller's Method, ICP is at 660 psi, while Choke Pressure is at 500 psi for an hour. Suddenly, the ICP spikes to 780 psi while the choke pressure remains at 500 psi. What type of problem has most likely occurred?**
- Answer
- C**
- (A) The choke is plugging up
 - (B) You have a washout in the drillpipe
 - (C) A bit nozzle has plugged
 - (D) The choke is washing out
- 59

DST2.60 **What is the goal of the 1st Circulation of the Driller's Method?**

Answer

- C** (A) Circulate out the influx while displacing the drillstring and annulus with kill mud weight
- (B) Displace the drillstring and annulus with KMW
- (C) Circulate out the influx using CMW
- (D) None of the above

60

DST2.61 **After completing the 1st Circulation of the Driller's Method, well is shut-in. Stabilized SICP reads 400 psi higher than original SIDPP. What is the safest next-step?**

Answer

- C** (A) Bleed casing pressure down until it is equal to SIDPP
- (B) A SICP is always greater than the SIDPP, the second circulation can be started.
- (C) Resume circulation and continue until all the influx is out of the hole and SICP=SIDPP.
- (D) All the above

61

DST2.62 **What is the best indication that gas has been removed from the well on a surface stack at the end of the first circulation using the Driller's Method prior to shutting down the pumps and closing in the well?**

Answer

- C** (A) The weight of the string has increased.
- (B) The circulating casing pressure has decreased to zero.
- (C) Gas has been circulated from well and have completed circulating at least one bottoms-up strokes with complete returns
- (D) Never shut down pumps as APL is lost and the well will kick again

62

DST2.63 **What should be the shut-in pressures after the first circulation to the Driller's Method has been completed?**

Answer

- C** (A) SIDPP will be lower than the SICP
- (B) SIDPP will be the same as original SIDPP
- (C) SICP and SIDPP will be equal to the original SIDPP
- (D) ICP will be lower in the 2nd Circulation

63

DST2.64 **For a kick in the horizontal hole section, what will the SIDPP and SICP be if the gas comes in while we're in the horizontal section? (Assuming there is no float valve in the drill string.)**

Answer

- C**
- (A) SIDPP will be much higher than the SICP.
 - (B) The SICP will be much higher than the SIDPP.
 - (C) Both are about the same
 - (D) None of the above

64

DST2.65 **If we take a kick in the horizontal hole section, what will the SIDPP and SICP be if the gas comes in while we're in the horizontal section? (Assuming there is no float valve in the drill string.)**

Answer

- D**
- (A) The SIDPP will be much higher than the SICP
 - (B) The SIDPP will be zero
 - (C) The SICP will be much higher than the SIDPP
 - (D) Both will be about the same.

65

DST2.66 **A gas influx is being circulated through the horizontal section. What should happen to the mud pit volume if we are following the correct kill procedure?**

Answer

- B**
- (A) Increase
 - (B) Remain approximately constant
 - (C) Decrease
 - (D) Slowly decrease

66

DST2.67 **If the choke line fails during well control, what is the next safest step?**

Answer

- C**
- (A) Immediately stop the pumps and close the choke
 - (B) Continue circulating at the maximum pump rate
 - (C) Immediately stop the pumps and close a hydraulic valve upstream of the failure
 - (D) Immediately stop the pumps and check for flow

67

DST2.68 **During the 1st Circulation of the Driller's Method, the drillpipe pressure drops followed by a drop in casing pressure. What is the probable cause of this?**

Answer

- C**
- (A) The choke is plugging
 - (B) The choke is washing-out
 - (C) The mud pump is washing out
 - (D) The drill bit is plugging

68

DST2.69 **If the Slow Circulating Rate Pressure (SCRPP) is unknown at a time of a kick, what would be the correct procedure for obtaining the Initial Circulating Pressure (ICP)?**

Answer

- D**
- (A) Check the previous recorded pressures and choose the kill rate circulating pressure taken with the last BHA in the hole closest to the depth where the kick occurred. Add 100 psi as a safety margin
 - (B) Begin pumping slowly down the drill pipe, when the pressure fluctuates, record the drill pipe pressure, this is the initial circulating pressure
 - (C) Use the SIDPP as the new initial circulating pressure
 - (D) Follow correct start up procedure. When kill circulation rate has been reached, the drillpipe pressure will be the initial circulating pressure

69

DST2.70 **During the 1st Circulation of the Driller's Method, the influx is currently being circulated out the hole. What should happen to the active pits as the gas is circulated to the surface?**

Answer

- C**
- (A) The active pit level should decrease due to gas expansion then increase as gas is circulated out of the well
 - (B) The active pit level should increase due to gas expansion then stay constant as gas is circulated out of the well
 - (C) The active pit level should increase due to gas expansion then decrease as gas is circulated out of the well
 - (D) The active pit level will stay constant

70

Drilling Supervisor Study Guide Day 3

DS3.1 **The best way abnormal formation pressure can be described is as follows?**

Answer

- C** (A) Pressure which is less than the hydrostatic pressure caused by a full column of regional formation water.
- (B) Pressure which is equal to the hydrostatic pressure caused by a full column of regional formation water.
- (C) Pressure that is greater than the hydrostatic pressure caused by a full column of formation water.
- (D) All the above

1

DS3.2 **What does the IADC definition of "uncontrollable flow of formation fluids at surface or mudline" mean?**

Answer

- A** (A) Blowout
- (B) Kick
- (C) Partial Loss Circulation
- (D) Total Lost Circulation

2

DS3.3 **What describes the basic function of a diverter?**

Answer

- A** (A) To direct formation fluids a safe distance away from the rig floor.
- (B) To act as a backup system if the annular preventer fails
- (C) To shut-in a shallow kick
- (D) Allow formation fluids to escape to the rig floor in a controlled manner.

3

DS3.4 **What do we call the highest-pressure that could be measured at the surface if the well were filled with formation fluids, which are usually thought to be gas?**

Answer

- A** (A) Maximum Anticipated Surface Pressure.
- (B) Worst Assumed Surface pressure
- (C) Maximum Allowable Casing Pressure
- (D) Maximum Allowable Drillpipe Pressure

4

DS3.5 **What should be done when drilling a section of the well with low kick tolerance?**

Answer

- D** (A) Set cement plug and sidetrack
- (B) Pull bit into casing shoe as quickly as possible
- (C) Drill section as quickly as possible to decrease chances of taking a kick.
- (D) Consider setting casing/liner

5

DS3.6 **Before drilling out the casing shoe and using information provided by the onsite Geologist, a kick tolerance of 30 bbls is determined for the next hole section. What does the 30 bbl kick tolerance mean?**

Answer

- C** (A) A gas kick of 30 bbls is the maximum that can be circulated out without bursting the casing at the surface.
- (B) A saltwater kick of 30 bbls is the maximum that can be shut-in and bullheaded without possible lost circulation.
- (C) A gas kick of 30 bbls is the maximum that can be shut-in and circulated out without possible lost circulation.
- (D) Kick tolerance can be ignored in an OBM, as gas stays in solution regardless of size

6

DS3.7 **What will happen to the Maximum Allowable Annular Surface Pressure (MAASP) if the mud weight (MW) is lowered?**

Answer

- C** (A) MAASP will decrease
- (B) MAASP will stay the same.
- (C) MAASP will increase
- (D) None of the above

7

DS3.8 **What is the term for the maximum surface pressure for a the well in which the casing shoe may fail?**

Answer

- C** (A) MACP (Maximum Allowable Casing Pressure)
- (B) MAASP (Maximum Allowable Annular Pressure)
- (C) MASP (Maximum Anticipated Surface Pressure)
- (D) SIDPP (Shut-in Drillpipe Pressure)

8

DS3.9 **In a horizontal well, where should a heavy pill be spotted on a trip out of the hole?**

Answer

- B** (A) At the beginning of the horizontal section.
- (B) In the vertical part of the well.
- (C) At the end of the horizontal section.
- (D) At the Kick-Off Point (KOP)

9

DS3.10 **While running in the hole with a slick, non-shearable tool through the surface BOP, the well begins flowing. What is the safest next step?**

Answer

- A** (A) Make up safety valve in drill string and close the annular BOP
- (B) Pick up a joint or stand of shearable drill pipe, run into position, and install a safety valve
- (C) Open downwind diverter line and close diverter packing element
- (D) Space out and close shear rams

10

DS3.11 **Ballooning occurs when the Bottom Hole Pressure (BHP) gets close to the Formation Fracture Pressure (FracP) while circulating. What causes this increase in Bottom Hole Pressure (BHP)?**

Answer

- B** (A) Friction pressure from mud pumps
- (B) Annular friction while circulating
- (C) Friction pressure from drillpipe
- (D) Friction pressure from surface lines

11

DS3.12 **During a flow check, the well flowed with a small trickle and ballooning is suspected. What would be the next proper step to take?**

Answer

- B**
- (A) Perform an extended flow check to ensure the size and rate of the gain.
 - (B) Follow shut-in procedures, record pressures, pit gain and then notify the supervisor.
 - (C) Increase the pump rate and note if increase in losses is observed.
 - (D) Investigate if water hose has been left on

12

DS3.13 **While making a connection, flowback time was excessive and well shut-in. The stabilized SIDPP and SICP is 150 psi.. Driller thinks the problem might be a ballooning formation. How can we be sure it is ballooning?**

Answer

- B**
- (A) Partially open the choke and measure the timing and volume of two or more consecutive bleeds.
 - (B) Circulate bottoms-up at a reduced rate through the remote choke.
 - (C) Switch to manual choke. Use manual choke to bleed back 100 psi into trip tank. Time to bleed off 1 bbl and shut-in. Perform second and third bleed off of 1 bbl and shut-in. Compare time to bleed. If time decreases
 - (D) Bleed off 150 psi and wait to see if the pressure will continue to decrease

13

DS3.14 **While drilling, the well is losing mud at 15 barrels per hour (bbls/hr). During a connection, the crew notices that the well is flowing. When the pumps start again, the mud loss happens again. What might be happening downhole?**

Answer

- B**
- (A) The well is underbalanced while drilling and overbalanced at the connection.
 - (B) The well is overbalanced while drilling and underbalanced at the connection.
 - (C) The well is underbalanced while drilling and underbalanced at the connection.
 - (D) None of the above

14

DS3.15 **A well has been shut-in after a kick, and the pressures on the gauges have stabilized. The Shut-In Drillpipe Pressure (SIDPP) reads 700 psi, and the Shut-In Casing Pressure (SICP) reads 900 psi. What is going on in the well?**

Answer

- A**
- (A) The well is balanced; pressure at the bottom are balanced on both sides of the well.
 - (B) The well is not balanced; bottom hole pressure (BHP) on the annular side is higher than on the drillpipe side of the U-tube.
 - (C) The well is not balanced; drillpipe pressure is missing the trapped pressure on casing due to a partially plugged drillpipe float.
 - (D) The well is not balanced; bottom hole pressure (BHP) on the drillpipe side is higher than on the annulus side of the U-tube.

15

DS3.16 **At end of 1st Circulation of Driller's Method, KWMin equals KMWout After staging pumps down and shutting-in, SIDPP = SICP of 200 psi. How do we determine if extra pressure is trapped pressure?**

Answer

- D**
- (A) Open a 2" fill line valve on the standpipe and bleed 50 psi. Then monitor pressure to see if the well is static or builds back pressure up to 200 psi.
 - (B) Bump the float
 - (C) Line up and pump down the annulus
 - (D) Bleed off 50 psi at the choke then monitor pressure to see if the well remains static or builds pressure.

16

DS3.17 **During drilling operations, we have started losing mud during the last stand. During the subsequent connection, the well flowed after the pumps were shut-off. The Driller suspects the formation is "ballooning." What is the safest next-step?**

Answer

- B**
- (A) Bleed 5 bbls back to the trip tank and check if flowback speed reduces
 - (B) Shut-in the well
 - (C) Immediately start pumping to restore BHP
 - (D) If the flow was less than a finger width, return to drilling. Most likely, higher temperatures are causing fluids to expand within the hole

17

DS3.18 **During drilling operations and within the last five connections, the Driller observed a delayed flowback. Well shut-in with SIDPP and SICP = 250 psi. After bleeding off 50 psi, SIDPP = 250 psi, and SICP = 275 psi. What's your assessment?**

Answer

- D**
- (A) This is ballooning. Bleed off and drill ahead.
 - (B) This is mud compressibility. Bleed off and drill ahead
 - (C) If the flow was less than a finger width, return to drilling. Most likely, higher temperatures are causing fluids to expand within the hole
 - (D) This is a kick. Record pressures and prepare for the kill

18

DS3.19 **Which of these statements best describe a working procedure which could lead to a swabbed-in kick while running casing?**

Answer

- B**
- (A) Running in the hole too fast.
 - (B) Pulling the casing back out after getting it stuck.
 - (C) Rotating the casing.
 - (D) Not filling the casing incrementally while running in the hole

19

DS3.20 **Cement has been mixed and displaced down the casing. According to the IADC, after dropping upper plug, cement is displaced with water. During this displacement, what should happen to the active pit level?**

Answer

A (A) Pit level will increase

(B) Pit level will decrease

(C) Pit level will stay the same.

(D) Since water is highly compressible, pit level will slightly decrease

20

DS3.21 **What potential well control problem do casing centralizers pose while running casing?**

Answer

B (A) Could get damaged as they are being run through the BOP stack and Wellhead

(B) Increase chances of surging the well while running the casing.

(C) The time spent on running the casing will increase.

(D) No problem, simply toss all the centralizers into the reserve pit as they are not needed

21

DS3.22 **What indicates that a self-fill float valve is plugged?**

Answer

D (A) Trip monitoring returns will decrease

(B) Trip monitoring returns will remain the same.

(C) Trip monitoring returns will show total losses (no returns)

(D) Trip monitoring returns will increase to closed-end displacement.

22

DS3.23 **During the 1st Circulation of the Driller's Method, the gas influx is circulated up the annulus towards the casing shoe. If Bottomhole Pressure (BHP) is maintained constant during this annulus segment, what happens to the casing shoe pressure?**

Answer

D (A) Decreases

(B) Remains constant

(C) Decreases rapidly then increases

(D) Increases

23

DS3.24 **During 1st Circulation Driller's Method and with kick in open hole, the crew had to shut down and shut-in. The initial SIDPP = 200 psi and initial SICP = 400 psi. Current SICP = 500 psi. What would cause the casing pressure to be higher?**

Answer

- A**
- (A) Pressure was trapped when the well was shut in
 - (B) The kill mud density is too low causing increased pressure
 - (C) The influx is saltwater only
 - (D) The hole is packing off around the BHA

24

DS3.25 **Cement is being displaced into the casing. After all the cement has been pumped from cement pumps, the cement is displaced by current mud weight. What should happen to the active pit level during this stage of operation?**

Answer

- B**
- (A) Pit level will increase.
 - (B) Pit level will stay constant
 - (C) Pit level will decrease
 - (D) Pit level will decrease rapidly and increase slowly

25

DS3.26 **After drilling out and testing casing shoe, our Geologist states we will enter a section with low kick tolerance. If this formation is penetrated and our kick tolerance begins to decrease, what is our next safest step?**

Answer

- C**
- (A) Keep drilling as there is no problem
 - (B) Consider Plug and Abandoning well and no way possible to continue to drill
 - (C) Consider setting casing or liner
 - (D) Consider raising mud weight

26

DS3.27 **Operational plans call for running in the hole to float shoe and circulate the casing clean before drilling out shoe. Why during displacement, should we monitor pit and flow levels?**

Answer

- A**
- (A) To check, that the float shoe is not leaking.
 - (B) To verify that casing is leaking
 - (C) To check the cement head is leaking
 - (D) To verify the high pressure, low volume pump has sufficient pressure to displace cement

27

DS3.28 **While running casing, bowed centralizer are installed every other joint. Bowed centralizers are used to ensure the casing is not touching the formation to affect a good cement job. Which statement outlines a problem when running centralizers?**

Answer

- C**
- (A) Decrease chances of swabbing the well while pulling out of the hole with casing
 - (B) Increase chances of swabbing the well while pulling out of the hole with casing
 - (C) Increase chances of surging the well while running the casing
 - (D) Decrease chances of surging the well while running the casing

28

DS3.29 **Which of the statements best describes conditions resulting from plugging of a self-fill (auto-fill) float?**

Answer

- A**
- (A) Trip monitoring returns will increase with closed-end displacement
 - (B) Trip monitoring returns will decrease with closed-end displacement
 - (C) Trip monitoring returns will decrease with open-ended displacement
 - (D) Trip monitoring returns will increase with open-ended displacement

29

DS3.30 **In a deviated well, why is it important to circulate a full bottoms-up before initiating a cement job?**

Answer

- C**
- (A) To ensure a good clean column of spacer in the casing.
 - (B) To ensure a good clean column of mud in the casing.
 - (C) To ensure a good clean column of mud in the annulus.
 - (D) To ensure a good clean column of spacer in the annulus.

30

DS3.31 **What can occur as cement sets downhole?**

Answer

- B**
- (A) The cement could plug up the choke line
 - (B) Hydrostatic pressure can decrease and potentially cause a kick
 - (C) The cement can swell and lead to casing collapse
 - (D) Hydrostatic pressure can increase and potentially cause a kick

31

DS3.32 **Why is important to circulate at least one bottoms-up prior to initiating a cement job?**

Answer

- D** (A) To remove the drilled cuttings that could potentially plug the casing float collar
- (B) To ensure the cement will not enter the drill string
- (C) To ensure the cement does not get mixed with the drilling fluid
- (D) To ensure the annulus contains a good clean column of drilling fluid

32

DS3.33 **Fixed bladed centralizers are placed on casing as it is run in the hole. Centralizers are run to improved overall cementing quality. What could be a potential problem in running centralizers on casing?**

Answer

- C** (A) It will increase the probability of swabbing the well while running in the hole
- (B) It will keep the casing string fixed in the axial center of the well
- (C) It will increase the probability of surging the well while running in the hole
- (D) It will create multiple voids in the cement between the casing and the well bore

33

DS3.34 **When casing is run in the hole with a non-return float valve, what could happen if the casing is not filled?**

Answer

- D** (A) The hook load will suddenly drop
- (B) The casing joint directly above the float assembly will burst
- (C) The entire casing string will burst
- (D) The float could fail and cause drilling fluid to u-tube up inside the casing

34

DS3.35 **When casing is run in the hole with a non-return float valve, and the casing is not filled at regular intervals, if the float assembly suddenly fails, the mud will u-tube into the casing. What will happen to Bottomhole Pressure (BHP)?**

Answer

- A** (A) Decrease bottom hole pressure
- (B) Increase bottom hole pressure
- (C) Bottom hole pressure will stay constant
- (D) Bottom hole pressure will drop quickly and increase slowly

35

DS3.36 **What is to be located on the rig floor while running casing?**

Answer

- A** (A) Suitable crossover (Swage)
- (B) Inside blow out preventer (IBOP)
- (C) Casing hanger
- (D) Test plug

36

DS3.37 **Which statement best describes the result if a self-fill (Auto-tube type) failed to convert into a check valve?**

Answer

- A** (A) Pressure would have to be added to the cement head to prevent U- tubing.
- (B) Allow U- tubing to occur as cement inside casing is no problem
- (C) U-tubing with cement cannot occur
- (D) As we displace cement with 18 ppg mud, the U-tube would place cement up the annulus

37

DS3.38 **After running and setting casing, what would happen if a self-fill (autofill tube-type) float shoe failed to convert back into a check valve?**

Answer

- A** (A) The fluids in the annulus or the formation could enter the casing
- (B) The cement would have to be reverse circulated out of the well
- (C) The pressure would have to be held on the annulus to prevent the u-tubing effect
- (D) Nothing would change

38

DS3.39 **When running casing in the well, what increases the risk of surging?**

Answer

- B** (A) Slower casing running speed
- (B) Faster casing running speed
- (C) Cleaning and removing cuttings in the hole before running the casing
- (D) Pumping a slug prior to running the casing

39

DS3.40	During drilling operations and during connection flowback, the crews suspect ballooning. The Driller bled 10 bbls of mud into a calibrated trip tank. Afterword, the Driller initiates circulating bottoms-up. What is the safest next step?
Answer	D
	(A) Circulate through a flowline with BOP open monitor pit level closely
	(B) Circulate through a fully closed choke with the BOP closed and monitor pit level closely
	(C) Circulate through casing valves monitor pit level closely
	(D) Circulate through a fully open choke with the BOP closed and monitor pit level closely
40	
DS3.41	A 300-foot-long balanced cement plug has been set at 8000 feet in 13-3/8" casing with a MW of 13.0 ppg. The mud above the plug is displaced with a 15 ppg mud. If the plug fails, what will happen to the BHP?
Answer	A
	(A) Increase BHP
	(B) Decrease BHP
	(C) BHP would remain the same
	(D) BHP would increase rapidly, then decrease slowly
41	
DS3.42	What is the major precaution for drilling fluid prior to initiating wireline operations?
Answer	D
	(A) Trip out of the hole without trip sheet.
	(B) Strip to bottom and run wireline inside DP
	(C) Under no circumstances should wireline operations be conducted in open hole.
	(D) Drilling fluid should be circulated and conditioned because the well will be static for an extended period of time
42	
DS3.43	Operations observes 20 bbl kick and shut-in well. After stabilization, SIDPP = 200 psi, SICP = 500 psi. If the crew had reacted faster and limited the kick size to 10 bbls, how would the shut-in pressures be affected?
Answer	B
	(A) SICP and SIDPP will be higher
	(B) SICP will be lower but no change in SIDPP
	(C) SIDPP will be lower but no change in SICP
	(D) SIDPP will remain the same, SICP will decrease
43	

DS3.44 **Which statement best describes a "negative test" procedure?**

Answer

- C**
- (A) A test on a barrier where pressure on the surface side of the barrier is greater than the pressure on the formation side of the barrier
 - (B) A weight test on a barrier to see if barrier moves downward
 - (C) A test on a barrier where pressure on the formation side of the barrier is greater than the pressure on the surface side of the barrier
 - (D) A weight test on a barrier to see if barrier moves upward

44

DS3.45 **Which of the following statements best describes an appropriate action needed when drilling through a transition zone?**

Answer

- B**
- (A) Circulate bottoms up at each connection
 - (B) Finger print connections for flow back trends.
 - (C) Increase mud weight by 0.2 ppg and keep drilling
 - (D) Close annular preventer and drill ahead taking returns through choke line

45

DS3.46 **We are drilling a HTHP well. Based on readings from our BHA, our downhole temperature is dramatically higher than surface mud temperatures. What effect will these higher temperatures have on our subsequent downhole Hydrostatic Pressure?**

Answer

- B**
- (A) Temperature will increase the effective downhole mud weight and cause a higher hydrostatic pressure.
 - (B) Temperature will decrease the effective downhole mud weight and cause a lower hydrostatic pressure.
 - (C) Temperature has no effect on mud weight or hydrostatic pressure.
 - (D) Increasing temperatures will always cause plugging hydrates to form

46

DS3.47 **When ballooning is suspected, what is the first course of action a Driller should take?**

Answer

- B**
- (A) Bleed 5 bbls back to trip tank and check if flow back rate decreases
 - (B) Shut in the well
 - (C) Bleed 10 bbls back to trip tank and check if flow back rate decreases
 - (D) Continue to monitor the open well

47

DS3.48 **Which statement best describes porosity?**

Answer

- A** (A) The amount of void or empty space in the rock, expressed as a percent
- (B) The ability of formation fluids to move through the formation
- (C) The amount of overburden pressure applied to the formation
- (D) The total pressure applied to the formation

48

DS3.49 **After shutting-in on a kick, stabilized SIDPP = 400 psi, SICP = 0 psi. After a walkabout, no surface line, pit and BOP leaks are detected. What is the probable cause of the zero SICP discrepancy?**

Answer

- C** (A) The choke pressure gauge failed
- (B) A plugged choke line
- (C) The hole is packed off with cuttings around BHA
- (D) The formation fractured at the casing shoe

49

DS3.50 **During stripping operations, if the Casing Pressure (CP) is held constant while stripping the drillstring through the bubble, what happens to Bottom Hole Pressure (BHP)?**

Answer

- C** (A) BHP will stay the same.
- (B) BHP will increase
- (C) BHP will decrease
- (D) BHP will increase, rapidly decrease

50

DS3.51 **Which statement best describes effect of swabbing?**

Answer

- A** (A) A reduction in bottom hole pressure
- (B) A kick
- (C) An increase in bottom hole pressure
- (D) Bottom hole pressure remains constant

51

DS3.52 **Which statement illustrates the increase in risk due to swabbing?**

Answer

- C** (A) Pulling out of hole at a slow speed
- (B) Low formation permeability
- (C) Increase in mud density
- (D) Balled up stabilizers on the BHA

52

DS3.53 **The crew has spaced out, shut down and successfully shut-in during a tripping operation. The bit and BHA are 250 feet off bottom with the influx below the bit. What are the conditions of Shut-in Drillpipe (SIDPP) and Shut-in Casing Pressure (SICP)?**

Answer

- B** (A) The SIDPP is higher than SICP
- (B) The SIDPP is equal to SICP
- (C) The SIDPP is lower than SICP
- (D) No readings can be taken

53

DS3.54 **We are circulating down the drillstring and up the annulus. Which of the following affects bottom hole pressure (BHP) of the well?**

Answer

- D** (A) Drillstring pressure loss
- (B) Pressure differential across the bit nozzles.
- (C) Surface line pressure loss
- (D) Annular pressure loss

54

DS3.55 **Kick was observed and well shut-in. Stabilized SIDPP = 400 psi and SICP = 600 psi. Both pressures begin to rise because of gas migration. If the casing pressure is kept steady at 600 psi, what will happen to the Bottom Hole Pressure (BHP)?**

Answer

- C** (A) BHP will increase
- (B) BHP will remain constant.
- (C) BHP will decrease
- (D) None of the above

55

DS3.56 **How does having cuttings in the annulus change the Bottom Hole Pressure (BHP)?**

Answer

- C** (A) BHP will stay the same.
- (B) BHP will decrease
- (C) BHP will increase.
- (D) None of the above

56

DS3.57 **On a drilling rig, which pressure is kept steady as the pump rate goes up to the kill rate?**

Answer

- A** (A) Casing pressure
- (B) Fracture Pressure
- (C) Drillpipe pressure
- (D) Formation pressure

57

DS3.58 **Well has been shut-in. SCR is 400 psi at 30 SPM. SIDPP = 400 and SICP = 550 psi. Before starting the kill operation, there is a total power failure at the pumps. Which pressure should we keep steady to ensure the constant BHP?**

Answer

- B** (A) 550 psi casing pressure
- (B) 400 psi drillpipe pressure
- (C) 725 psi casing pressure
- (D) 950 psi casing pressure

58

DS3.59 **During a well kill, the pump rate is increased while holding drillpipe pressure constant. If we ignore Equivalent Circulating Density (ECD) effects, how will the pump rate increase affect Bottom Hole Pressure (BHP)?**

Answer

- A** (A) BHP will decrease, risking further influx
- (B) BHP will stay constant
- (C) BHP will increase risking formation damage
- (D) None of the above

59

DS3.60 **The well is shut-in on a kick. Stabilized SIDPP = 250 psi, SICP = 600 psi. Both pressures begin to rise because of gas migration. If we maintain the drillpipe pressure at 250 psi, what will happen to the BHP??**

Answer

- A**
- (A) It will stay the same
 - (B) BHP will be too low
 - (C) BHP will increase
 - (D) All the above

60

DS3.61 **While preparing to circulate Kill Weight Mud (KMW), gas begins to migrate. If we do not take any action, what will happen to the Bottom Hole Pressure (BHP)?**

Answer

- D**
- (A) BHP will decrease
 - (B) BHP will stay the same.
 - (C) Both B and C
 - (D) BHP will increase

61

DS3.62 **The drilling crew has successfully spaced out, shut down and shut-in the well. The Shut-in Casing Pressure (SICP) reads 0 psi and the Shut-in Drillpipe Pressure (SIDPP) reads 300 psi, what could be the reason for having SICP which reads 0 psi?**

Answer

- A**
- (A) The hole is packed off around the BHA
 - (B) Superintendent stated it would be ok not to install casing gauges
 - (C) There is not float in the string.
 - (D) The gauge is faulty. Change out same.

62

DS3.63 **Well has been shut-in on a kick. The stabilized SIDPP = 200 psi, SICP = 300 psi. Both pressures start to rise because of gas migration. If we keep the drillpipe pressure constant at 200 psi, what will happen to the BHP?**

Answer

- D**
- (A) Increase
 - (B) Decrease
 - (C) Both A and C
 - (D) Stay the same.

63

DS3.64 **The crew has successfully spaced out, shut down and shut-in. During the 2nd Circulation of the Driller's Method, Kill Weight Mud (KWM) is pumped to the bit holding drillpipe pressure constant. How does this affect Bottom Hole Pressure (BHP)?**

Answer

- D**
- (A) Decrease
 - (B) Stay the same.
 - (C) Both A and C
 - (D) Increase

64

DS3.65 **If a safety margin is applied to the mud weight, what could occur during well control?**

Answer

- B**
- (A) Cannot be easily added to the mud weight
 - (B) Cannot be removed easily.
 - (C) There is no danger in adding safety margins to mud weight.
 - (D) All the above

65

DS3.66 **What is going on downhole if the well is flowing with the pumps off, but there is no increase in the mud pit level when the pumps are running?**

Answer

- C**
- (A) Cannot determine exactly what is happening downhole without fingerprinting data used for comparison.
 - (B) The mud hydrostatic pressure is greater than the formation pressure when the pumps are on.
 - (C) The annular pressure loss is creating an overbalance against formation pressure when the pumps are on.
 - (D) Minor losses are occurring and masking a potential kick

66

DS3.67 **What establishes the maximum surface pressure before the casing shoe formation fractures?**

Answer

- D**
- (A) Permeability of formation
 - (B) Porosity of formation.
 - (C) Type of formation fluids
 - (D) Formation strength

67

DS3.68 **We are preparing to circulate Kill Weight Mud (KWM), but the gas influx is starting to move upward. If we do not do anything, what will happen to the pressure inside the gas bubble as it goes higher?**

Answer

- C** (A) Pressure will increase.
- (B) Pressure will decrease.
- (C) Stay approximately the same
- (D) None of the above

68

DS3.69 **With the well open, which of the following operating procedures would cause problems such as Pit Gain, Decreasing Hydrostatic Pressures, Influx displacement of mud within casing annulus and decreasing Bottom Hole Pressure (BHP)?**

Answer

- B** (A) Ballooning
- (B) Gas Migration
- (C) High Equivalent Circulating Density
- (D) High Annular Pressure Loss

69

DS3.70 **What statement best describes kick tolerance?**

Answer

- A** (A) The maximum volume of influx for a given kick intensity that can be taken at a given depth and circulated out without exceeding fracture pressure
- (B) The maximum volume of influx that can be taken at a specific depth without exceeding fracture pressure
- (C) The maximum pressure the cement will hold
- (D) The maximum capacity of the casing

70

DS3.71 **What does pulling pipe out of a weighted pill do to the bottom-hole pressure?**

Answer

- D** (A) Increase
- (B) Stays the same.
- (C) Slowly decreases, then rapidly increases
- (D) Decrease

71

DS3.72 **How is density affected with increasing hydrostatic pressure?**

Answer

- A** (A) It increases fluid density.
- (B) It decreases fluid density.
- (C) Pressure has no effect on density.
- (D) None of the above

72

DS3.73 **What is the right way to check for trapped pressure after finishing the well kill operation?**

Answer

- A** (A) Bleed a small amount of pressure from the choke and shut the well back in. Then watch the pressure gauge. If the well is underbalanced, pressure will climb; if the pressure was trapped it will remain steady.
- (B) Full open choke manifold and bleed off all pressure. Watch for the flow at the pit. If the well is static, the bled pressure was trapped.
- (C) Using recorded SICP, subtract off initial SICP for trapped pressure. Then, subtract trapped pressure from current SIDPP.
- (D) Continue pumping until 500 psi of pressure is added to the CP. Then, bleed off 500 psi plus SIDPP.

73

DS3.74 **What can lead to a larger kick size and a higher Shut-in Casing Pressure (SICP)?**

Answer

- C** (A) Low porosity
- (B) Wrong type of drilling fluid being used
- (C) High formation permeability
- (D) Low formation permeability

74

DS3.75 **Which of these factors can change the Shut-in Casing Pressure (SICP)?**

Answer

- D** (A) Mud pH
- (B) Choke line length
- (C) Mud density
- (D) The kick volume.

75

DS3.76	During a kill start-up operation, the actual Initial Circulating Pressure (ICP_{actual}) is greater than the calculated Initial Circulating Pressure (ICP_{calculated}). Which of the following would not be a reason for this?
Answer	D
	(A) The HCR being in the open position
	(B) The slow circulating rate pressure will be different because the hole is deeper since it was recorded.
	(C) Delayed circulation allowed the gas to migrate
	(D) None of the above
76	
DS3.77	In formation known for ballooning, during a connection, well flowed and shut-in. Stabilized SIDPP = 100 psi, SICP = 200 psi. Pit Gain = 5 bbls.. Operations bled off 1 bbl The current SIDPP = 100 psi, SICP = 225 psi. What is occurring downhole?
Answer	D
	(A) Ballooning formations
	(B) Losing mud to formation
	(C) The well is overbalanced in the static condition
	(D) Formation fluid influx
77	
DS3.78	We are currently drilling a HPHT (high pressure, high temperature) well. Our readings from the BHA show a downhole temperature which is significantly higher than surface temperature. How will the downhole hydrostatic pressure be affected?
Answer	C
	(A) None, the mud in the well is a constant density
	(B) It will increase the effective mud density downhole and cause a higher hydrostatic pressure
	(C) It will decrease the effective mud density downhole and cause a lower hydrostatic pressure
	(D) Temperature will not affect mud density
78	
DS3.79	Prior to installation of an annular packing element, which of the following statement best describes what needs to be checked?
Answer	A
	(A) The type of mud in the well
	(B) The manufacturer of the rams
	(C) The desired hydraulic closing pressure
	(D) The hydraulically operated pistons
79	

DS3.80 **What type of equipment would show a lagging indicator of an increase in formation pressure?**

Answer

- C** (A) Clogged shaker screens
- (B) Stroke counters
- (C) Mud Gas Detector
- (D) Vacuum Degasser

80

DS3.81 **Which following statement best describes the proper procedure to be followed, if the mud pop-off valve opened during the kill operation?**

Answer

- D** (A) Shut down the pump, close the IBOP or Kelly Cock, and close the choke
- (B) Shut down the pump, activate the shear rams, close the choke
- (C) Shut down the pump, close diverter
- (D) Shut down the pump and close the choke

81

DS3.82 **If a non-return valve, such as an Inside BOP (IBOP), is stabbed within the drill string, which of the following statement is true?**

Answer

- C** (A) According to regulations, IBOP's cannot be stored on the rig floor
- (B) It could leak through the open/close key
- (C) It will not allow wireline to be run inside the drill string
- (D) It should be kept in the Rig Managers office during all operations

82

DS3.83 **Concerning drillstring safety valves, what is the required test pressure?**

Answer

- B** (A) The minimum pressure required to maintain a seal
- (B) The same pressure as the BOP
- (C) Pressure must be equal to or greater than 1500 psi
- (D) The same rated working pressure as the Kelly/top drive

83

DS3.84 **If the BOP stack contains a Shear Ram, what information needs to be understood?**

Answer

- B**
- (A) The hang-off capacity
 - (B) The size and strength of tubulars the ram can shear
 - (C) The required opening pressure
 - (D) The total height of the shear ram in the BOP stack

84

DS3.85 **The Drillpipe Pressure (DP) gauge on standpipe manifold reads 450 psi. On the remote panel, the DP reads 400 psi. What gauge should be used to calculate kill mud?**

Answer

- A**
- (A) Neither. Research why there is a difference and replace the defective gauge.
 - (B) Drillpipe Gauge
 - (C) Standpipe Gauge
 - (D) None of the above

85

DS3.86 **Before starting wireline operations, what precautions should have occurred?**

Answer

- D**
- (A) Increase the mud density before starting the wireline operation to compensate for the decrease in ECD
 - (B) Decrease the mud weight before starting the wireline operation to compensate for the decrease in ECD
 - (C) Pump multiple sweeps to clean the hole before the wireline operation begins
 - (D) The drilling fluid should be circulated and conditioned because the well will be static for an extended period of time

86

DS3.87 **If the kick has entered an open well and gas migration occurs, what happens to the Hydrostatic Pressure in the annulus?**

Answer

- D**
- (A) Gas density is increasing, causing displacement of mud out of the hole and a decrease in hydrostatic pressure
 - (B) Gas width is increasing, causing displacement of mud out of the hole and a decrease in hydrostatic pressure
 - (C) Gas height is decreasing, causing displacement of mud out of the hole and a increasing hydrostatic pressure
 - (D) Gas height is increasing, causing displacement of mud out of the hole and a decrease in hydrostatic pressure

87

DS3.88 **What is the best point within the horizontal well to place a weighted pill?**

Answer

- A** (A) In the vertical part of the well.
- (B) From the Kick-off point (KOP) to the End of build (EOB) section of the hole.
- (C) At the beginning of the horizontal section.
- (D) At the toe of the well measured depth.

88

DS3.89 **Which statement best defines how hydrates form?**

Answer

- C** (A) They form when methanol and other gases combine with water at specific pressure and temperature conditions.
- (B) They form when glycol and gases combine with water at specific pressure and temperature conditions.
- (C) They form when methane or other gases combine with water at specific pressure and temperature conditions.
- (D) They form only in subtropical regions of the world

89

DS3.90 **Which statement best describes the objective of the Lube and Bleed Method?**

Answer

- C** (A) To increase mud density in the well and force the influx back into the formation
- (B) To increase surface pressure by controlling gas migration
- (C) To reduce surface pressure by increasing hydrostatic pressure and removing dry gas
- (D) To reduce surface pressure by removing dry gas

90

DS3.91 **How does the gas migration rate (ft/hr) affect the selection of proper bullheading speed?**

Answer

- D** (A) Gas migration has no effect on our choice of pump speed.
- (B) Bullheading never works, use the Volumetric Method
- (C) Gas migration is never a consideration when bullheading.
- (D) None of the above

91

DS3.92 **During bullheading operations, what is the main element being performed?**

Answer

- B**
- (A) Forcing pipe into a hole under pressure using hydraulic jacks
 - (B) Pumping into a shut-in well to force formation fluids back into the formation.
 - (C) Circulating formation fluids out of a well using a choke and circulating system.
 - (D) Using BOP's as primary barrier regardless of mud type

92

DS3.93 **When stripping into the hole, which of the following will maintain a constant bottom hole pressure (BHP), assuming no influx migration?**

Answer

- B**
- (A) Bleed off the drillpipe open ended displacement while stripping each stand.
 - (B) Bleed off the drillpipe closed end displacement while stripping each stand.
 - (C) Bleed off the drillpipe annular displacement while stripping each stand.
 - (D) Strip to bottom without bleeding any gain in casing pressure

93

DS3.94 **While tripping out, well flowed and was shut-in. While preparing to strip back to bottom (350 ft), the crew noted the influx was not migrating as the SICP did not increase. How would we maintain constant BHP while stripping back to bottom?**

Answer

- B**
- (A) Bleed off a volume of mud equal to 75 psi of hydrostatic pressure.
 - (B) Bleed off a volume of mud equal to the closed end displacement of the pipe stripped into the well.
 - (C) Bleed off a volume of mud equal to the metal displacement of pipe stripped into the well.
 - (D) Strip to bottom with open BOPs. Once on bottom, close BOP's, rig up and bullhead down the annulus until SICP reaches zero.

94

DS3.95 **While POOH at 11,000 ft, improper hole fill up prompted a flowcheck which indicated that the well is flowing. The well is 14,000 ft TVD/14,500 ft MD. After shutting in the well what would be the proper kill method?**

Answer

- D**
- (A) Using the volumetric method to kill the well.
 - (B) Start circulating the current mud weight until the kick is out of the well.
 - (C) Strip to bottom with open BOPs. Once on bottom, close BOP's, rig up and bullhead down the annulus until SICP reaches zero.
 - (D) Strip back to bottom using correct stripping procedures, then circulate the kick out using the Driller's Method.

95

DS3.96 **If a kick has entered the wellbore, how can we determine that the kick is below the drill bit requiring stripping back to bottom?**

Answer

- D**
- (A) SICP is greater than SIDPP
 - (B) SIDPP is greater than SICP
 - (C) SICP is equal to Accumulator Pressure
 - (D) SIDPP is equal to SICP

96

DS3.97 **We are tripping out, well was swabbed and shut-in. After stablization, the SICP begins increasing. What well control method should be used to manage gas migration in the well?**

Answer

- C**
- (A) Bullheading Method
 - (B) Lube and Bleed Method
 - (C) Volumetric Method.
 - (D) Reverse Circulating Method

97

DS3.98 **During tripping operations, trip chart indicated a kick has been swabbed into the well. We are currently 1239 feet off bottom. What is the safest next-step?**

Answer

- C**
- (A) Begin the Wait and Weight method immediately
 - (B) Start stripping out of the hole and keeping the hole full
 - (C) Strip to bottom before starting well kill operations
 - (D) In order to facilitate getting to bottom quickly, leave BOP open and strip to bottom, then close BOP

98

DS3.99 **During well control operations, we are displacing the Current Mud Weight (CMW) with Kill Mud Weight (KMW). If the drillstring is a tapered string, which of following will NOT vary as KMW is displaced during the operations?**

Answer

- A**
- (A) BHP
 - (B) ICP
 - (C) Casing Pressure
 - (D) Shut-in Drillpipe Pressure

99

DS3.100 **The well is filled with 12.2 pounds per gallon (ppg) mud, and a 500-foot cement plug has been set and tested. The mud above the plug is replaced with 10.2 ppg brine. If the cement plug fails, which way would the fluid move across the cement plug?**

Answer

- A** (A) Pressure from below would cause fluid to move up.
- (B) Pressure from above would cause fluid to move down.
- (C) Pressure from below would mix with the pressure above resulting in no movement of fluid across the plug.
- (D) Nothing will change

100

DS3.101 **The well is filled with 11 ppg mud, and a 600-foot cement plug has been set and tested. If we replace the mud above the plug with 9.0 ppg brine, what will happen to the pressure difference between the top and bottom of the cement plug?**

Answer

- B** (A) Pressure differential will decrease
- (B) Pressure differential will increase
- (C) Pressure differential is equal to 600 psi
- (D) There is no pressure differential

101

DS3.102 **If pump speed is increased during a kill and Bottom Hole Pressure (BHP) is kept constant, what should happen to the pump pressure?**

Answer

- C** (A) Pump pressure should decrease
- (B) Pump pressure should remain the same.
- (C) Pump pressure should increase.
- (D) None of the above

102

DS3.103 **During the 1st Circulation of the Driller's Method, SCRP of 400 psi at 30 spm. What will happen to the BHP, if we increase the pump speed to 40 SPM while keeping the drillpipe pressure constant?**

Answer

- C** (A) Stay the same
- (B) It will stay the same
- (C) It will decrease
- (D) Both B and C

103

DS3.104 **How can the total pressure in the wellbore affect the density of non-aqueous (oil-based) mud downhole?**

Answer

- D** (A) Well depth will not affect the mud density downhole
- (B) Wellbore pressures will not be affected
- (C) Increasing hydrostatic pressure downhole can increase effective mud density in the well
- (D) None of the above

104

DS3.105 **What occurs in a non-aqueous (oil-based) mud when wellbore temperatures increase?**

Answer

- B** (A) Increases the hydrostatic pressure in the well.
- (B) Decreases hydrostatic pressure in the well.
- (C) Hydrostatic pressure remains the same.
- (D) All the above

105

DS3.106 **Why is it important to keep track of pit volume during well control operations?**

Answer

- B** (A) To know when to adjust the drillpipe pressure
- (B) To check for mud gains or losses.
- (C) To keep the bottomhole pressure constant
- (D) Changing pit volumes require circulation rates to be rapidly adjusted

106

DS3.107 **Why is it necessary to lower the mud in the active circulating pit when circulating out a kick?**

Answer

- B** (A) To store the kick fluid as it is circulated from the well.
- (B) If the kick is gas it will expand and pit level will increase
- (C) If the kick is salt water, it will expand and the pit level will increase.
- (D) Lowering the mud in the circulatin pipe is only needed if we are circulating out a water kick.

107

DS3.108 **When should the well be watched using pit level markers or flow levels and flow rate gauges?**

Answer

- A** (A) Anytime there is a potential for a barrier to fail
- (B) During connections
- (C) When running in the hole
- (D) Only when well is shut-in so we may determine if the BOP is leaking.

108

DS3.109 **What type of mud makes it harder for us to figure out the size of the influx downhole?**

Answer

- D** (A) Salt based mud
- (B) Water based mud
- (C) Bentonite Mud
- (D) Oil based mud

109

DS3.110 **What is the definition of hydrates?**

Answer

- C** (A) Nothing to worry about during a well control operation.
- (B) Hydrates are liquid mixtures of water and gas
- (C) Hydrates are solid, frozen mixtures of water and gas
- (D) Dry cement within a cement slurry

110

DS3.111 **As mud is pumped downhole, how does increasing temperature affect mud density?**

Answer

- D** (A) Increased temperature will increase mud weight downhole
- (B) Mud weight will not be affected by wellbore temperature
- (C) Increasing temperatures will always cause plugging hydrates to form
- (D) Increased temperature will decrease mud weight downhole.

111

DS3.112 **How does increasing pressure affect the density of non-aqueous (oil-based) mud?**

Answer

- A** (A) Increases density
- (B) No effect on density
- (C) Decreases density
- (D) Increasing pressure will always cause plugging hydrates to form

112

DS3.113 **During drilling operations, the Derrickhand calls and says they are going to transfer 45 bbls to the active pit in 10 minutes. After 20 minutes, the Driller notices the pit level has gone up by 60 bbls. What is the safest next-step which should be taken?**

Answer

- C** (A) Call the Derrickhand and see if he left a valve open and added too much.
- (B) Keep drilling, Derrickhand added too much mud and did not know.
- (C) Flow check and then call the Derrickhand to check the added volume.
- (D) Begin mixing a slug pill to ensure an influx does not rise up through the drillpipe

113

DS3.114 **Well is shut-in on a kick. The Mud Engineer determines there is not enough barite to kill the well. While we waiting for new stocks of barite, the shut-in pressures are slowly rising by 75 psi every 30 minutes. What is the best next-step?**

Answer

- A** (A) Use the first circulation of Driller's Method
- (B) Bullhead with current mud weight to push gas formation fluids back into the formation.
- (C) Increase MW by as much as possible and circulate to the Bit to reduced SIDPP
- (D) Use the Wait and Weight method to keep pressures as low as possible while circulating.

114

DS3.115 **A well is shut-in with the bit 200 feet off the bottom, and the influx is estimated to be on the bottom. The Shut-in Drillpipe Pressure (SIDPP) reads 300 psi. What would we expect the Shut-in Casing Pressure (SICP) to be?**

Answer

- B** (A) Higher than SIDPP
- (B) The same as SIDPP
- (C) Lower than SIDPP
- (D) Both A and B

115

DS3.116 **KMWin is equal to KMWout. After pumping total surface to surface strokes, pumps are shut down. Stabilized SIDPP and SICP = 200 psi. We believe this is trapped pressure. How do we confirm this?**

Answer

- A** (A) Bleed off 50 psi at the choke then monitor pressure to see if it is static or builds back-up to 200 psi.
- (B) Bleed off 200 psi, open the well and check for flow.
- (C) Start circulating surface to bit strokes then shut down and re-check the pressures.
- (D) Rig up and bullhead down the annulus until SICP reaches zero.

116

DS3.117 **What conditions affect gas breaking out of solution in Oil-Based Mud?**

Answer

- D** (A) The gas and OBM have the same properties.
- (B) The temperature of the mud as it gets cooler near the surface.
- (C) The pH of the OBM
- (D) The decrease in hydrostatic pressure above the gas when it gets close to the surface.

117

DS3.118 **What is the expected surface indication that a kick has gone into solution in oil-based mud?**

Answer

- A** (A) Nothing. When gas goes into solution, it mixes with oil-based mud and does not create a pit gain.
- (B) A pit gain either equal to or smaller than the volume of the kick.
- (C) The drillpipe pressure will increase.
- (D) When gas is at the choke and the choke has to be slightly closed in order to keep CP constant

118

DS3.119 **During a W&W kill operation the pumps were shut down once kill weight mud reached the bit. The Shut-in Drillpipe Pressure (SIDPP) reads 150 psi and the Shut-in Casing Pressure reads 450 psi. Which of the following procedures should be done?**

Answer

- B** (A) Bleed casing pressure to zero and confirm well is dead.
- (B) Make sure all drillpipe and casing pressures gauges are showing pressure to confirm pressure exists.
- (C) Check and see if there are any trapped pressures by comparing current casing pressure to the previous casing pressure which was present before the pumps were shut down.
- (D) Detach TDS and rig up to pump with High-Pressure, Low-Volume portable pump

119

DS3.120 **What changes to the volume of a gas kick occur when it is successfully circulated out?**

Answer

- C** (A) Gas volume will stay the same.
- (B) Gas volume will decrease
- (C) Gas volume will increase
- (D) Both B and C

120

DS3.121 **What should be the immediate plan if while drilling there is a complete loss of returns?**

Answer

- A** (A) Prepare to top-fill the annulus with water (or lightest fluid available) and record volume pumped.
- (B) Start mixing LCM
- (C) Trip out of hole to last casing shoe to keep downhole tools from getting stuck.
- (D) Pump cement downhole immediately

121

DS3.122 **When the mud has entrained air and/or gas, what device is used to determine accurate density?**

Answer

- B** (A) Mud Balance
- (B) Pressurized Mud Balance
- (C) Hydrometer
- (D) Marsh Funnel

122

DS3.123 **An increase in formation pressure may be predicted by which indicator?**

Answer

- D** (A) Pit Gain
- (B) Increase in ROP
- (C) Slight decrease in WOB
- (D) All the above

123

DS3.124 **Why is it important to accurately measure and record the weight and viscosity of the mud returning to the pits?**

Answer

- B**
- (A) It ensures the well plan's mud program is followed
 - (B) It provides information about the effects of the well on the mud.
 - (C) It is required to be filled out on the drilling reports.
 - (D) It provides estimated mud inventories.

124

DS3.125 **Is mud density in non-aqueous (oil-based mud) affected by increased temperature?**

Answer

- D**
- (A) It increases
 - (B) It lowers viscosity, but has no effect on density
 - (C) Increases viscosity of the mud
 - (D) Decreases density

125

DS3.126 **For the most accurate mud weight what piece of equipment should be used?**

Answer

- C**
- (A) Viscosity funnel.
 - (B) Mud balance
 - (C) Pressure mud balance
 - (D) VG Meter

126

DS3.127 **What happens to mud weight readings if a pressurized Mud Balance is used instead of an atmospheric one?**

Answer

- B**
- (A) Mud weight readings would be lower
 - (B) Mud weight readings would be higher
 - (C) Mud weight readings would be the same.
 - (D) None of the above

127

DS3.128 **Why should we leave extra capacity in the active pits while circulating a kick out?**

Answer

- D** (A) To store the kick fluid as it is circulated from the wellbore
- (B) You might lose returns while circulating the kick from the wellbore
- (C) If the kick is oil, it will expand and the active pit level will increase
- (D) If the kick is gas, it will expand and the active pit level will increase

128

DS3.129 **Which of the following conditions increases the difficulty to detect a kick with a Pit Volume Totalizer (PVT)?**

Answer

- B** (A) When you close annular preventer and divert flow to the choke manifold
- (B) When you allow mud to over flow the flow cleaners or shakers.
- (C) When you divert flow into the trip tank
- (D) When circulating returns cease

129

DS3.130 **The cement has been displaced into the casing followed by Current Mud Weight (CMW). What should occur to the active pit levels during this displacement?**

Answer

- A** (A) Stay the Same
- (B) Increase
- (C) Decrease
- (D) Increase quickly and decrease slowly

130

DS3.131 **Which of the following statements best describes situations where it is more difficult to detect a kick with the Pit Volume Totalizer (PVT)?**

Answer

- A** (A) When you allow mud to overflow the shale shakers
- (B) When you allow mud to by-pass the solids control pits
- (C) When the mud density is increased
- (D) When you only circulate through the choke line

131

DS3.132 **We are drilling a well with a tapered drillstring. With larger pipe in upper portion of the well, what occurs to the volume of trip tank as a slug is pumped into the smaller diameter pipe?**

Answer

- C**
- (A) The trip tank volume will indicate surging as the slug length gets longer
 - (B) The trip tank volume will indicate swabbing as the slug length gets longer
 - (C) There will be no change in trip tank volume until all the slug has been displaced into the small diameter pipe
 - (D) There will be no change in trip tank volume until all the slug has been displaced into the large diameter pipe

132

DS3.133 **We are tripping out of the hole with 4-1/2" drillstring, we are monitoring the trip tank. During the last four (4) stands, the trip tank has increased by 10 barrels. What is the safest next step?**

Answer

- D**
- (A) Continue pulling pipe. 10 barrels is the correct displacement for 4-1/2 inch drill pipe
 - (B) Decrease pipe pulling speed, monitor the well on the active mud pits
 - (C) Increase pipe pulling speed to minimize the potential for swabbing
 - (D) Perform a flow check. This could be a kick that has been swabbed in

133

DS3.134 **During the 1st Circulation of the Driller's Method, what will happen the active pit levels as the gas breaks out of solution?**

Answer

- D**
- (A) Decrease
 - (B) Remain constant
 - (C) Decrease rapidly and increase slowly
 - (D) Increase

134

DS3.135 **After completing the 1st Circulation of the Driller's Method, the pumps are staged down and the well is shut-in. Assuming no safety factor was used, what is one indication the gas influx has been successfully removed from the well?**

Answer

- A**
- (A) The Shut-in Casing Pressure (SICP) has decreased to the initial SIDPP
 - (B) SICP is zero
 - (C) SIDPP is zero
 - (D) The choke position is fully open

135

DS3.136 **Within the Bottom Hole Assembly (BHA), a Pressure While Drilling (PWD) Tool located will provide information which indicates an influx while drilling ahead. Accordingly, what PWD information would indicate an influx?**

Answer

- B**
- (A) Increase in Equivalent Circulating Density (ECD)
 - (B) Decrease in Equivalent Circulating Density (ECD)
 - (C) The Equivalent Circulating Density (ECD) would remain the same
 - (D) Increase in drilling fluid viscosity

136

DS3.137 **Which of the following will be affected by a string washout during a well kill?**

Answer

- B**
- (A) Casing pressure
 - (B) Pump pressure
 - (C) Drillpipe pressure
 - (D) Accumulator Pressure

137

DS3.138 **While drilling at 20 feet per hour, MWD fails. Spot slug and POOH. At surface, the MWD is replaced without changing the original bit. At bottom, ROP is averaging 64 ft/hr with the same WOB. What is our next step?**

Answer

- D**
- (A) Shut-in the well and perform reverse circulation.
 - (B) Shut-in the well and circulate bottoms-up.
 - (C) Continue drilling as everything is ok.
 - (D) Carry out a flow check and consider circulating bottoms up through the choke line.

138

DS3.139 **Why is it important to circulate a full bottoms-up before starting a cement job on a deviated well?**

Answer

- D**
- (A) The time taken to circulate bottoms-up will replace the requirement to flow check the well.
 - (B) You want to ensure we will not induce losses during the cement job.
 - (C) It is an BSEE requirement as defined within the Drilling Permit
 - (D) To ensure a good clean column of mud in the annulus.

139

DS3.140 **What should we do if we experience total losses while drilling with water-based mud?**

Answer

- D** (A) Pump lost circulation material immediately
- (B) Stop drilling, shut the well in and see what happens
- (C) Switch to an Synthetic Oil Base Mud
- (D) Stop drilling and try to top-fill the hole up with water and record the volume.

140

DS3.141 **Shut-in on a kick. The Mud Engineer estimates 3,000 sacks of barite needed for KMW. Only 2,200 sacks are available. Both SIDPP and SICP have increased due to gas migration. The MAASP is 500 psi within current SICP. What's the best kill method to use?**

Answer

- D** (A) Begin to Lube and Bleed
- (B) Wait and Weight Method of Well Control
- (C) Begin Volumetric Method
- (D) Driller's Method of Well Control

141

DS3.142 **A well has been shut-in after a kick, and the pressures on the gauges have stabilized. The SIDPP reads 400 psi, and the SICP reads 850 psi. What is happening in the well?**

Answer

- B** (A) The well is not balanced; bottomhole pressure on the annulus side is lower due to contamination by formation fluids.
- (B) The well is balanced; pressures at the bottom are balanced on both sides of the well
- (C) The well is not balanced; BHP on the annular side is higher than on the drillpipe side of the U-tube.
- (D) None of the above

142

DS3.143 **Drilling at 45 ft/hr, the MWD fails. After POOH, we decide to replace the MWD and run the same bit back into the hole. At TD, we begin drilling with same rotary speed and WOB. We're averaging 75 ft/hr. What action could we take?**

Answer

- B** (A) Decrease WOB to get an average ROP of 18 ft/hr
- (B) Carry out a flow check and consider circulating bottoms up
- (C) Shut-in the well and circulated bottoms up
- (D) Increase the Mud Weight by 0.2 ppg trip margin

143

DS3.144 **What is the major benefit of routine fill-ups while running casing?**

Answer

- B**
- (A) To prevent casing burst due to high internal pressure
 - (B) To provide hydrostatic support to the casing and related components
 - (C) To prevent casing collapse due to high external pressure
 - (D) To prevent casing exceeding designed tensile loads

144

DS3.145 **What does increasing drillpipe pressure and fairly constant casing pressure indicate while using the Driller's Method?**

Answer

- C**
- (A) Plugged choke
 - (B) Washed out bit nozzle
 - (C) Plugged bit nozzle
 - (D) Partially plugged choke

145

DS3.146 **Improper hole fill up on a trip out was noticed. After positive flow check, well shut-in on annular preventer. The bit was 1050 ft off bottom. What should the driller's next task be after recording surface pressures?**

Answer

- A**
- (A) Bump the float
 - (B) Perform a prejob safety meeting
 - (C) Monitor well to allow time to develop well killsheet
 - (D) Strip back to bottom

146

DS3.147 **Abnormally high pore pressures are normally due to what natural occurrence?**

Answer

- D**
- (A) Erosion in shale
 - (B) Formations with no permeability.
 - (C) Repositioning of formations below original sediments
 - (D) Trapped fluid in shale.

147

DS3.148 **What is a brief description of a transition zone?**

Answer

- C** (A) A section of formation where the formation undergoes huge temperature decrease.
- (B) A section of formation where the rock undergoes a permeability change.
- (C) A section of formation where formation pressure is changing.
- (D) None of the above

148

DS3.149 **When drilling through a transition zone, what actions should be taken?**

Answer

- A** (A) Increased crew awareness of the warning signs.
- (B) Pit levels should be reset to zero.
- (C) Reset the automatic driller and decrease WOB
- (D) Slow rate of penetration

149

DS3.150 **When drilling through a transition zone to a formation with high pressure, what changes might we notice in the drilling data?**

Answer

- B** (A) Increase in drillpipe pressure
- (B) Gradual increase in ROP
- (C) Reduced drag on connections
- (D) Immediate increase in torque

150

DS3.151 **When drilling in a transition zone to a formation with abnormal pressure, what changes might we notice in the cutting coming over shakers at bottoms up?**

Answer

- C** (A) Decreased size in the cuttings
- (B) Increase in the density of shale cuttings
- (C) Decrease in the density of shale cuttings
- (D) Both A and C

151

DS3.152 **During a stripping operation through a gas bubble, if casing pressure is held constant while stripping, what happens to Bottomhole Pressure (BHP)?**

Answer

- B**
- (A) It increases.
 - (B) It decreases.
 - (C) It stays the same.
 - (D) It increases rapidly then decreases

152

DS3.153 **While tripping in the hole at 4500 ft, the flow check was positive. Due to weak formations, the soft shut-in procedure is to be used. Which of the following statements best describes a soft shut-in procedure?**

Answer

- A**
- (A) Stab a full opening safety valve, close the safety valve, open HCR valve, close BOP, and close choke, record pressures
 - (B) Close the BOP, open the choke, monitor and record pressure
 - (C) Stab a full opening safety valve, open HCR valve and choke, close BOP, close choke, close safety valve, record pressure
 - (D) Close the BOP, stab a full opening safety valve, close the safety valve, open choke, record pressure

153

DS3.154 **What is the main purpose of the weep hole on a Cameron Type "U" Ram type preventer?**

Answer

- B**
- (A) Indicates the seals on the bonnet are leaking fluid
 - (B) Indicates the piston rod mud seal is leaking fluid
 - (C) To reduce the amount of hydraulic shock to weaker formations
 - (D) Allows for ram function test during all rig operations

154

DS3.155 **After cementing, a new bit and BHA is run to the float shoe. The casing will be displaced with mud before drilling shoe. Why is it important to accurately monitor flow rates and pit levels during this portion of the operation?**

Answer

- D**
- (A) To monitor for indications of liquid cement returning to the pits
 - (B) To monitor for fluid losses as any extra cement is cleaned out of the hole
 - (C) To check and verify that the surface lines are not leaking cement
 - (D) To check and verify that the float shoe is not leaking

155

DS3.156 **A 500-foot-long cement plug is set inside the casing within 13 ppg fluid. Once cured, the cement plug is pressure tested with displaced 9 ppg brine. If the cement plug were to fail, what direction would the fluid move across the cement plug?**

Answer **B**

- (A) The pressure from above the plug would cause fluid to move down
- (B) The pressure from below the plug would cause fluid to move up
- (C) The pressure from above and below the plug would remain constant
- (D) The pressure above the plug would suddenly increase followed by a slight decrease

156

DS3.157 **During a circulation of a kick with water-based mud, what will happen to surface pressures as gas breaks out of solution?**

Answer **A**

- (A) Increase rapidly
- (B) Decrease rapidly
- (C) No change in surface pressures
- (D) Decrease slowly

157

DS3.158 **The Derrickhand tells us there's been a 10-barrel gain in the pits over the last 15 minutes. What is the next safest step we should take?**

Answer **D**

- (A) Inform the Mud Engineer
- (B) Ask the Derrickhand to check for any transfers
- (C) Ask the Derrickhand to direct flow to the trip tank so that a more accurate volume measurement can be made.
- (D) Carry out a flow check

158

Drilling Supervisor Practice Test Day 3

DST3.1 **What is the proper definition of kick tolerance?**

Answer

- C**
- (A) The maximum kick intensity at 1000', which can be successfully shut in and circulated out without exceeding the maximum anticipated surface pressure
 - (B) The minimum gas kick volume, for a given kick intensity at a particular depth, which can successfully be shut in and circulated out without exceeding the well's weak point fracture pressure.
 - (C) The maximum gas kick volume, for a given kick intensity at a particular depth, which can successfully be shut in and circulated out without exceeding the well's weak point fracture pressure.
 - (D) The average gas kick volume, for a given kick intensity at a particular depth, which can successfully be shut in and circulated out without exceeding the well's weak point fracture pressure.

1

DST3.2 **Given a kick tolerance of 30 barrels at a specific depth with a 0.4 ppg kick intensity, what would occur if we recalculated a doubled kick intensity of 0.8 ppg to the kick tolerance?**

Answer

- C**
- (A) Maximum kick size will increase
 - (B) Maximum kick size will remain the same
 - (C) Maximum kick size will decrease
 - (D) None of the above

2

DST3.3 **Best definition of Maximum Allowable Annular Surface Pressure (MAASP)?**

Answer

- B**
- (A) The total pressure applied at the shoe which will cause losses.
 - (B) The total surface pressure which will cause losses to the formation on top of the mud hydrostatic pressure.
 - (C) Another name for Mud Cap pressure
 - (D) Total pressure which can be applied to the well minus the SIDPP.

3

DST3.4 **For a given overbalance required for a heavy pill, how is the volume calculated?**

Answer

- A** (A) Height of pill = Overbalance needed / (Pill Wt – MW)/0.052. The Volume needed = Ht of Pill x Hole Capacity.
- (B) Slug weight x 19.24
- (C) Volume = Hole Capacity - Length
- (D) None of the above

4

DST3.5 **Symptoms of ballooning have happened, and we decided to bleed 10 barrels of mud back into the trip tank. What negative effects could this decision cause?**

Answer

- D** (A) The increase likelihood of lost circulation.
- (B) The intensity of the ballooning is increased.
- (C) If the problem was a kick and not ballooning, the kick size would decrease
- (D) If the problem was a kick and not ballooning, the kick would become larger.

5

DST3.6 **What does the term "Fingerprinting the Well" mean?**

Answer

- A** (A) Recording baseline drilling data that can be analyzed against current data to help identify downhole problems
- (B) Recording baseline drilling data that can be analyzed against current data to help identify losses while drilling.
- (C) Recording baseline circulating data of each solids control device (i.e. flow cleaners, desander, desilter, etc.). Volume information is used to determine total losses to formation.
- (D) Recording baseline tripping data from the trip tank which can be analyzed to help identify potential kick

6

DST3.7 **How can we differentiate between a ballooning formation and a kick?**

Answer

- D** (A) You can't tell the difference between the two until we pump bottoms up
- (B) There will be shut-in pressures for a kick, no pressures for ballooning
- (C) If we bleed-off some shut-in pressure and shut the well back in, the pressure will become variable. It will build sometimes and decrease at other times. This is due to gas compressibility
- (D) If we bleed-off some shut-in pressure then shut the well back in, the pressure will build back up with a kick. If the well is ballooning, the pressure will decrease with each bleed-off and not build back.

7

DST3.8 **You suspect ballooning and decide to bleed 25 bbls of mud back into the trip tank. The well is circulated until bottoms-up. What is the safest way to carry out this operation?**

Answer

- A** (A) Circulate through a fully open choke with the BOP closed and monitor pit level closely
- (B) Circulate full bottoms-up and route returns through the vacuum degasser
- (C) Make sure to bleed casing pressure to zero prior to circulating bottoms-up
- (D) None of the above

8

DST3.9 **If the well is ballooning, which of these actions should be followed?**

Answer

- C** (A) Increase circulating pressure
- (B) Raise the MW
- (C) Fingerprint connections for flowback trends
- (D) Decrease MW

9

DST3.10 **Drilling with 7 bbls/stand loss. At connection, flowed and shut-in with 6 bbls gain. SIDPP = 100 psi, SICP = 110 psi. Bleed #1 = 1 bbl, shut-in. SIDPP = 90 psi, SICP = 100 psi. Bleed #2 = 1 bbl, SIDPP = 30 psi, SICP = 35 psi. What's happening?**

Answer

- C** (A) The formation has fractured at the shoe.
- (B) A kick is in the well
- (C) The well is ballooning.
- (D) None of the above

10

DST3.11 **Mud losses while drilling have been at a constant rate of 5 barrels per hour for the last hour. During a connection the flowback is greater than 5 barrels and mud is still returning. What is the next step?**

Answer

- D** (A) Resume drilling operations.
- (B) Shut-in the well and start bullheading the fluid back into the wellbore.
- (C) Continue drilling and increase pump rate
- (D) Shut-in the well and call supervisor

11

DST3.12 **If a well shows a gain in the mud pits while the pumps are shut down and no gain while the pump is running, what does this indicate?**

Answer

- D** (A) Mud is started down the drillpipe again.
- (B) Mud weight inside the drillstring is creating an overbalance against formation pressure.
- (C) Drillpipe pressure loss is creating an overbalance against formation pressure
- (D) Annular pressure loss is creating an overbalance against formation pressure.

12

DST3.13 **What is a sign that ballooning is taking place?**

Answer

- B** (A) Gradual gains when pumps are on and losses when pumps are off.
- (B) Gradual losses when pumps are on and gains when pumps are off
- (C) No losses while drilling but gains at the connection.
- (D) Observing complete fluid losses when the pumps are off

13

DST3.14 **When the casing is run in the hole with a Non-Return Float assembly and is not being properly filled, and the float assembly fails at 2300 feet causing mud to u-tube inside the casing, what will happen to the Bottom Hole Pressure (BHP)?**

Answer

- C** (A) Bottom hole pressure will stay the same.
- (B) Bottom hole pressure will increase.
- (C) Bottom hole pressure will decrease.
- (D) Bottom hole pressure will increase rapidly, then decrease slowly

14

DST3.15 **After cement has been pumped inside the casing and is now being displaced into position with mud, what should happen to the active pit level during this stage of the operation?**

Answer

- D** (A) Pit level will increase.
- (B) Pit level will decrease.
- (C) Pit level will increase rapidly
- (D) Pit level will stay constant.

15

DST3.16 **What are the consequences of a self-fill float collar (autofill-tube) failing to convert to a check valve when cement is heavier than the displacement mud?**

Answer

- A** (A) Cement could u-tube back up inside the casing when pumps are stopped.
- (B) The autofill-tube self-fill float collars are not designed as check valves and should not be used.
- (C) Cement could u-tube into the casing and cause the casing to separate, since casing collars are a weak point.
- (D) Autofill-tube type float collars never fail

16

DST3.17 **Which of these things would make it more likely for the well to surge while running casing?**

Answer

- C** (A) Larger casing to wellbore clearance.
- (B) Heavy wall cake in well.
- (C) Tighter casing to wellbore clearance.
- (D) Smaller well cake in well

17

DST3.18 **What is a benefit of doing regular fill-ups when running casing into the hole?**

Answer

- C** (A) To prevent casing burst due to high internal pressure.
- (B) To prevent casing collapse due to high internal pressure.
- (C) To provide hydrostatic support to the casing and related components.
- (D) To prevent tensile separation at a tool joint

18

DST3.19 **Centralizers are added to the casing before casing is run into the well to ensure a good casing job. What problems can occur when using centralizers?**

Answer

- B** (A) Creates voids in the cement between the casing and wellbore.
- (B) Increased chances of surging the well while running casing.
- (C) Centralizers may become dislodged and form a bird's nest of steel within the BOP stack.
- (D) Centralizers are not installed and thrown into the reserve pit

19

DST3.20 **What could happen if the casing is not kept full when running it into the hole with a Non-Return Float assembly?**

Answer

- D** (A) Casing could burst
- (B) The string may start exceeding the weight limit of derrick
- (C) Casing could corrode and burst
- (D) Float may fail causing mud to u-tube up inside the casing

20

DST3.21 **Which of the following would not raise the risk of causing a surge in the well while running casing?**

Answer

- B** (A) High gel strength mud
- (B) Lower casing grade
- (C) Faster running speed
- (D) None of the above

21

DST3.22 **What can lead to a kick when running casing into the hole?**

Answer

- C** (A) Keeping the casing full of mud
- (B) Swabbing the formation
- (C) Surging and losses with drop in mud level
- (D) Casing float does not hold when tension loads are placed on the casing.

22

DST3.23 **We need to run back to the float shoe after the cement job on the casing. The drilling prognosis states we are to circulate the hole clean before drilling out of the shoe. Why is it important for us to monitor pit and flow levels during this operation?**

Answer

- D** (A) To ensure equipment is functioning properly before drilling next hole section.
- (B) Ensure the cement is dry.
- (C) To ensure the cement does not get mixed with the drilling fluid
- (D) To check that the float shoe is not leaking

23

DST3.24 **During running casing operations, if the self-fill (auto-fill) float becomes plugged resulting in improper fill-up, what are the well control risks we should be aware of?**

Answer

- A** (A) If the plug suddenly clears the mud level in the annulus will drop.
- (B) Losses may occur due to higher mud level in the annulus
- (C) You will not be able to reverse circulate to kill the well.
- (D) If the plug suddenly clears, the casing will probably collapse

24

DST3.25 **During a casing running operation with a float assembly, if the casing is positioned across the BOP and reaches the rig floor when an influx is detected, which shut-in procedure should we follow?**

Answer

- C** (A) Close blind shears, install casing swage, open choke valve on BOP
- (B) Close annular, install casing swage, open choke valve on the BOP
- (C) Install casing swage, open choke valve on BOP, close annular
- (D) Close annular with full operating pressure, install casing swage, open choke valve on the BOP

25

DST3.26 **While running casing in the well and the self-fill float becomes plugged and the casing is not regularly filled, which statement best describes the impact to well control risks?**

Answer

- B** (A) Surge pressure could potentially breakdown the formation at the bottom of the well
- (B) If the plug suddenly clears the drilling fluid level in the annulus will fall
- (C) You will not be able to reverse circulate to kill the well
- (D) Trip tank could overflow due to increase in returns

26

DST3.27 **What would indicate the self-fill casing float assembly is properly working as casing is run into the hole?**

Answer

- C** (A) Casing pressure will increase by the same amount over time
- (B) Hook load decreases by the buoyed weight of steel run in the hole
- (C) Returns from the well equal the volume of steel run in the hole
- (D) Surge pressure could potentially breakdown the formation at the bottom of the well

27

DST3.28 **What conditions would verify the self-fill float shoe is plugged on a casing string?**

Answer

- D** (A) Returns to trip tank will decrease
- (B) The surge pressure will decrease
- (C) The pump pressure will decrease
- (D) The typical increase in hook load with depth will slow down due to the buoyancy effect

28

DST3.29 **What statement best describes abnormal pressure?**

Answer

- C** (A) Formation pressure which is equal to the hydrostatic pressure caused by a full column of the formation water
- (B) Formation pressure which is less than the hydrostatic pressure caused by a full column of the formation water
- (C) Formation pressure which is greater than the hydrostatic pressure caused by a full column of the formation water
- (D) Formation Fracture pressure

29

DST3.30 **Which statement correctly indicates what value Bottomhole Pressure (BHP) must be at equal to or greater than, during well kill operations?**

Answer

- B** (A) Is less than formation fluid pressure
- (B) At least equal to formation fluid pressure
- (C) Is greater than formation fluid pressure
- (D) At least equal to formation fracture pressure

30

DST3.31 **Well shut-in on kick. Stabilized SIDPP = 300 psi, SICP = 500 psi. Both pressures rise due to gas migration. If the SIDPP is held constant at 300 psi, what will happen to the BHP?**

Answer

- A** (A) Stay the same.
- (B) Increase
- (C) Decrease
- (D) Increase rapidly then decrease slowly

31

DST3.32 **At the end of the final circulation, KWMin equals KMWout coming out the well. After shut-in, SIDPP = SICP = 150 psi. How do we determine if extra pressure is trapped pressure?**

Answer

- A** (A) Bleed off 50 psi at the choke then monitor pressure to see if it is static or builds back up to 100 psi
- (B) Bleed off 200 psi at the choke then monitor pressure to see if it is static or builds back up to 200 psi
- (C) Bleed off 100 psi at the choke then monitor pressure to see if it is static or builds back up to 150 psi
- (D) Bleed off 50 psi at the choke then monitor pressure to see if it is static or builds back up to 1000 psi

32

DST3.33 **How does increasing temperature affect the density of non-aqueous (oil-based) mud?**

Answer

- D** (A) Increases hydrostatic pressure in wellbore
- (B) Hydrostatic pressure will be constant regardless of temperature
- (C) Hydrostatic pressure will only be affected by change in pump pressures.
- (D) Hydrostatic pressure in wellbore will decrease

33

DST3.34 **Which statement best describes the safest action which should take place when drilling a transition zone?**

Answer

- D** (A) Circulate bottoms up at each connection
- (B) Increase mud weight by 0.2 ppg and keep drilling
- (C) Close annular preventer and drill ahead taking returns through chokeline
- (D) Finger print connections for flow back trends.

34

DST3.35 **While mixing Kill Weight Mud (KWM) for the Wait and Weight Method, both Shut-in Drillpipe (SIDPP) and Shut-in Casing Pressures (SICP) begin rising. What is the type of influx?**

Answer

- C** (A) Oil
- (B) Salt Water
- (C) Gas
- (D) Condensate

35

DST3.36 **A well is shut in with the bit 500 feet off the bottom and it's assumed that all the influx is below the bit. The Shut-In Drillpipe Pressure (SIDPP) is 300 psi. What would the Shut-In Casing Pressure (SICP) be?**

Answer **B**

- (A) The SICP should be higher than the SIDPP as the influx is within the annulus between open hole and drillstring
- (B) The same as SIDPP
- (C) The SICP should be lower than the SIDPP.
- (D) SICP cannot be determined

36

DST3.37 **During continuous drilling, what occurs to the maximum allowable volume of gas influx which can be taken and circulated out without breaking down the casing shoe or formation?**

Answer **D**

- (A) Increases
- (B) Stay the same
- (C) Increases rapidly followed by decreasing slowly
- (D) Decreases

37

DST3.38 **As we drill deeper into zones of abnormal pressure, what occurs between thresholds of pore pressure and hydrostatic pressure of the current mud weight?**

Answer **A**

- (A) Reduces the hydrostatic pressure overbalance.
- (B) Increases the hydrostatic pressure
- (C) Hydrostatic pressure remains the same.
- (D) All the above

38

DST3.39 **During circulating operations, the well was shut-in before the pumps were fully shut-off. We suspect that trapped pressure on the system. What effect will this have on the wellbore?**

Answer **C**

- (A) All pressures within the wellbore will have additional underbalance.
- (B) All pressures within the wellbore will be balanced
- (C) All pressures within the wellbore will have additional overbalance.
- (D) None of the above

39

DST3.40 **When staging pumps up to speed (SPUTS), if casing pressure is allowed to exceed Shut-in Casing Pressure (SICP), what is the impact on bottom hole pressure (BHP)?**

Answer

- B**
- (A) Bottomhole pressure will decrease and lead to additional kicks.
 - (B) Bottomhole pressure will increase and may fracture the formation.
 - (C) Bottomhole pressure will remain constant.
 - (D) None of the above

40

DST3.41 **Shut-in on 20 bbls kick. SIDPP = 400 psi, SICP = 600 psi. Start Driller's Method, had pump failure, shut-in. Current SIDPP = 400 psi, SICP = 600 psi. Pressures rise from gas migration. If CP = 600 psi by opening choke, what happens to BHP?**

Answer

- C**
- (A) Increase
 - (B) Stay the same
 - (C) Decrease
 - (D) Slowly decrease then increase dramatically

41

DST3.42 **How is an unknown formation strength determined before drilling below a casing seat?**

Answer

- C**
- (A) EMW Test
 - (B) FIT
 - (C) LOT
 - (D) BOP Test

42

DST3.43 **What happens to the hydrostatic pressure in the annulus, if gas influx migrates in an open well?**

Answer

- D**
- (A) Gas height is staying the same, causing the hydrostatic pressure to remain constant
 - (B) Gas height is increasing, causing displacement of mud out of the hole resulting in an increase in hydrostatic pressure.
 - (C) Within OBM, gas remains in solution
 - (D) Gas height is increasing, causing displacement of the mud out of the hole and a decrease in hydrostatic pressure

43

DST3.44 **During stripping operations, if the Casing Pressure (CP) is held constant when entering the gas influx will result in this happening to Bottom Hole Pressure (BHP)?**

Answer

- C**
- (A) It increases
 - (B) It stays the same.
 - (C) It decreases
 - (D) None of the above

44

DST3.45 **The casing pressure gauge on the remote choke panel shows 400 psi, while the casing pressure gauge on the choke manifold shows 600 psi. What action should be taken?**

Answer

- C**
- (A) Start the kill using 600 psi as the choke manifold CP is the most accurate gauge.
 - (B) Start the kill using 400 psi because the remote choke will be used during the kill.
 - (C) Investigate the reason for the difference.
 - (D) Run to nearest exit

45

DST3.46 **During well control operations, why is casing pressure usually higher than the shut-in drillpipe pressure when the influx is gas?**

Answer

- C**
- (A) The volume of the annulus is greater than the volume of drillpipe.
 - (B) The hydrostatic pressure in the annulus is greater than the hydrostatic pressure in the drillpipe.
 - (C) The hydrostatic pressure in the drillpipe is greater than the hydrostatic pressure in the annulus.
 - (D) The influx is always within the drillpipe

46

DST3.47 **When drilling through a transition zone, what key properties should be monitored?**

Answer

- B**
- (A) Influx gradient
 - (B) Formation fluid pressure
 - (C) Influx density
 - (D) ROP

47

DST3.48 **Which density measuring instrument is used when fluid has entrained air and/or gas?**

Answer

- A** (A) A pressurized balance
- (B) A triple beam mud balance
- (C) A densitometer
- (D) A hydrometer balance

48

DST3.49 **During drilling operations, high connection gas readings are encountered. Unfortunately, a power outage occurs stopping circulation and rotation. Power to hoist is ok. What is the Driller's next safest step?**

Answer

- C** (A) Line up on the active pit and monitor the well
- (B) Line up on the trip tank and monitor the well
- (C) Space out and shut the well in then monitor for pressure build-up in the well
- (D) Activate diverter system

49

DST3.50 **Why is it important for us to strip the pipe back to bottom after swabbing in a kick during a trip out of the hole?**

Answer

- D** (A) To bullhead the influx back into the formation
- (B) To keep the pipe moving.
- (C) To ensure the rig crews master stripping in the hole
- (D) To get below the influx and circulate it out.

50

DST3.51 **What does stripping back to bottom accomplish?**

Answer

- D** (A) Outrunning the kick before it gets to the surface.
- (B) Reduce casing pressure by exchanging hydrostatic pressure for surface pressure.
- (C) Reduce drillpipe pressures by exchanging hydrostatic pressure for surface pressure
- (D) Running pipe into the shut-in well to get the drillstring below the influx, while holding BHP constant.

51

DST3.52 **When pipe stripped to bottom after a swabbed kick, what mud weight should be used to kill the well?**

Answer

- C**
- (A) KMW based upon SICP before stripping
 - (B) KMW based on SIDPP before stripping
 - (C) The original MW
 - (D) The Current Mud Weight

52

DST3.53 **After pulling out 1,500 feet, the well kicked and was shut in. SICP = 250 psi, swabbed volume is 9 bbls, casing capacity is 0.147 bbl/ft. Assuming the swabbed influx is below the bit, what is the SIDPP?**

Answer

- B**
- (A) Higher than SICP
 - (B) Same as the SICP
 - (C) Lower than the SICP
 - (D) Both B and C

53

DST3.54 **During trip, swabbed 5 bbls and shut-in. Successfully stripped to bottom, started Driller's Method 1st Circulation. As we approach circulating bottoms-up, the pit level begins slowly rising as bottoms-up gets closer to the surface. What's next?**

Answer

- A**
- (A) With the well shut-in, continue circulation through the choke using the 1st circulation of the Driller's Method.
 - (B) Continue circulating with well open, but switch on vacuum degasser
 - (C) Shut-in well, record SIDPP and use Wait and Weight method to kill well
 - (D) With the well shut-in, swap to manual choke and continue circulation using the 2nd circulation of the Driller's Method.

54

DST3.55 **A 500-foot cement plug across shoe. Displaced mud above plug with 9.5 ppg brine. The mud below the plug is 11.8 ppg. The top of the cement is at 8,200 feet. If the plug fails and the well is open, what would happen to the well?**

Answer

- B**
- (A) Well will remain static
 - (B) Well will flow
 - (C) Well will go overbalanced and possibly fracture well
 - (D) None of the above

55

DST3.56 **If the Current Mud Weight (CMW) is held constant when entering an abnormal pressure zone, what happens to the overbalance pressure?**

Answer

- A** (A) Overbalance decreases.
- (B) Overbalance increases.
- (C) Overbalance stays the same.
- (D) None of the above

56

DST3.57 **The pump is turned off, the well is flowing, and the mud pit is gaining mud. When the pump is restarted, the gain in the mud pit stops. What is happening?**

Answer

- C** (A) Mud weight inside the drill string is creating an overbalance against formation pressure.
- (B) The swabs on the pump needs to be replaced.
- (C) Annular pressure loss is creating an overbalance against formation pressure.
- (D) Need to line up on active pits

57

DST3.58 **We are drilling with a 8-1/2" bit and an average Rate of Penetration (ROP) of 115 ft/hr. For the last 60 minutes, the pit level has not changed. What would we do?**

Answer

- A** (A) Make a flow check to check why pit volume is stable.
- (B) Continue drilling all is good.
- (C) Contact Mud Engineer to add more volume into the active pits to compensate for additional hole capacity.
- (D) POOH to surface without filling the hole.

58

DST3.59 **The results of a flow check aren't clear because there are only a small amount of returns from the well. What is the best way to find out if the well is truly flowing?**

Answer

- A** (A) *Line-up to the trip tank and monitor for gain or losses.
- (B) Swap chokes to manual choke and perform choke drill again. Manual choke will generate more responsive volumes for gains/losses.
- (C) Close in the well and line up through the MGS and monitor for flow at the shakers.
- (D) Line-up to active pits to monitor for gain or losses

59

DST3.60 **Why is it important to check the viscosity and density (weight) of the drilling mud at the suction pit?**

Answer

- C** (A) It keeps the man in charge of the pits busy mixing products during his tour
- (B) In ensures compliance with company policy
- (C) It provides data on the mud before it is pumped into the wellbore.
- (D) For comparison to the drilling mud density of returns from the well. They should be equal.

60

DST3.61 **During wireline operations, the wireline tools are being pulled from the well. The Operator is notified by the Driller that gain in the trip tank has been recorded. What is occurred?**

Answer

- C** (A) Nothing, it is normal to show a gain in the trip tank due to displacement when pulling wireline tools out of the hole
- (B) The wireline tools have been pulled too fast causing surging
- (C) The wireline tools have been pulled too fast causing swabbing
- (D) The well is most likely ballooning downhole

61

DST3.62 **What will we notice if the pop-off valve on the pump goes off while we're circulating out a kick?**

Answer

- C** (A) Slow drop in drillpipe pressure and an increase in casing pressure
- (B) Rapid drop in drillpipe pressure with no change to casing pressure
- (C) Rapid drop in drillpipe pressure and drop in casing pressure
- (D) None of the above

62

DST3.63 **During well control operations, we have pumped KMW to the bit. After shutting down, pressure are checked. We notice pressure on DP. After bleed for trapped pressure, the DP returned to the same value. What could we do?**

Answer

- C** (A) Bleed casing pressure down to zero
- (B) Bleed drillpipe pressure back down to zero
- (C) Circulate some more strokes, then shut-in and check again.
- (D) Change out faulty gauges.

63

DST3.64 **While drilling shallow hole with diverter stack, the well begins to flow. What is the safest next best step?**

Answer

- B**
- (A) Keep pumps on, close diverter, open downwind vent line, close flowline valve to shakers
 - (B) Keep pumps on, open downwind vent line, close flowline valve to shakers, close diverter
 - (C) Shut down pumps, open downwind vent line, close flowline valve to shakers, close diverter
 - (D) Shut down pumps, close diverter, open downwind vent line, close flowline valve to shakers

64

DST3.65 **When drilling through a transition zone into an abnormally pressured formation, what changes might be observed in the mud when it is circulated to the surface?**

Answer

- D**
- (A) Increase mud weight
 - (B) Mud weight remains the same.
 - (C) Decreasing mud weight
 - (D) All the above

65

DST3.66 **When drilling through a transition zone into an abnormally pressured formation, what changes might be observed in the formation cuttings returned to the shakers?**

Answer

- A**
- (A) Decrease in the density of Shale cuttings.
 - (B) Increase in the density of Shale cuttings
 - (C) Increase in background gas.
 - (D) Decrease in background gas

66

DST3.67 **When drilling through a transition zone into an abnormally pressured formation, what changes might be observed in the drilling fluid when it is circulated to the surface?**

Answer

- B**
- (A) Increasing mud weight.
 - (B) Decreasing mud weight.
 - (C) Increasing background gas.
 - (D) None of the above

67

DST3.68 **When stripping, what is the correct pressure for an annular preventer?**

Answer

- D**
- (A) The minimum pressure that allows the tool joint to go through the packing element with a decrease of 30k lbs. of hook load
 - (B) At least equal to or greater than 1500 psi
 - (C) The normal pressure required to close the annular BOP
 - (D) The minimum pressure required to maintain a seal

68

DST3.69 **When entering a gas kick while stripping back to bottom, what happens to Bottomhole Pressure (BHP)?**

Answer

- C**
- (A) Stays the same.
 - (B) It increases
 - (C) It decreases
 - (D) BHP rapidly decreases the increases slowly

69

DST3.70 **During a routine BOP test, the weep hole is observed to be dripping fluid. What is the safest next step?**

Answer

- C**
- (A) The primary ram shaft seal is leaking fluid, secure the well and replace immediately
 - (B) Immediately inject the emergency plastic packing fluid to temporarily seal the wellbore
 - (C) The piston rod packing elements are worn out, replace immediately.
 - (D) Continue with normal operations, use annular BOP to shut in well

70